

Introduction to Signals and Systems

1.1 INTRODUCTION

In this chapter, we begin our development of the analytical framework for signals and systems by introducing their mathematical descriptions and representations. In the chapters that follow, we build on this foundation in order to develop and describe additional concepts and methods that add considerably both to our understanding of signals and systems and to our ability to analyse and solve problems involving signals and systems.

A *signal* is defined as any physical quantity that varies with time, space, or any other independent variable or variables. Signals are represented mathematically as functions of one or more independent variables. For example, a speech signal can be represented mathematically by acoustic pressure as a function of time, and a picture can be represented by brightness as a function of two spatial variables. In this book, we deal almost exclusively with signals that are functions of time.

Signals may be processed further by *systems*, which may modify them or extract additional information from them. A system is an entity that processes a set of input signals to yield another set of output signals. A system may be made up of physical components, as in electrical, mechanical, or hydraulic systems (hardware realization), or it may be an algorithm that computes an output from an input signal (software realization).

1.2 CLASSIFICATION OF SIGNALS

Depending on the independent variables and the value of the function defining the signal, various types of signals can be defined.

1.2.1 Continuous-time and Discrete-time Signals

A continuous-time or analog signal is defined for a continuum of values of the independent variable time t . A discrete-time signal is defined only at discrete-time n , and consequently for discrete-time signals, the independent variable takes on only at a discrete set of values. The amplitude of the discrete-time signal between two time instants is just not defined. A speech signal as a function of time and atmospheric pressure as a function of altitude are examples of continuous-time signals, whereas the quarterly gross domestic product (GDP), monthly sales of a corporation, and stock market daily averages are examples of discrete-time signals. Illustrations of a continuous-time signal $x(t)$ and a discrete-time

signal $x(n)$ are shown in Fig. 1.1. It is important to note that the discrete-time signal $x(n)$ is defined *only* for integer values of the independent variable.

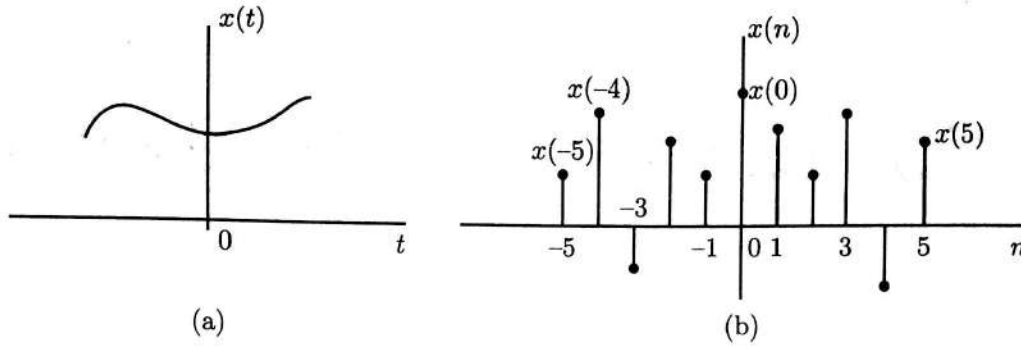


Fig. 1.1 Graphical representation of (a) continuous-time and (b) discrete-time signals

1.2.2 Continuous-valued and Discrete-valued Signals

The value of a continuous-time or discrete-time signal can be continuous or discrete. If a signal takes on all possible values on a finite or an infinite range, it is said to be a continuous-valued signal. Alternatively, if the signal takes on values from a finite set of possible values, it is said to be a discrete-time signal. Usually, these values are equidistant and hence can be expressed as an integer multiple of the distance between two successive values. A discrete-time signal having a set of discrete values is called a *digital signal*. Figure 1.2 shows a digital signal that takes on one of four possible values.

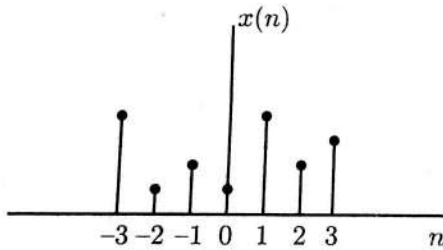


Fig. 1.2 Digital signal with four different amplitude levels

1.2.3 Multichannel and Multidimensional Signals

In some applications, signals are generated by multiple sources or multiple sensors. Such signals, in turn, can be represented in vector form. If $x_k(t)$, $k = 1, 2, 3$, denotes the electrical signal from the k th sensor as a function of time, the set of $k = 3$ signals can be represented by a vector $\mathbf{x}(t)$, where

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix}$$

We refer to such a vector of signals as a *multichannel signal*. In electrocardiography, e.g., 3-lead and 12-lead electrocardiograms are often used in practice, which result in 3-channel and 12-channel signals.

If a signal is a function of a single-independent variable, it is called a *one-dimensional* signal. On the other hand, a signal is called *M-dimensional* if its value is a function of M independent variables.

A picture is an example of a two-dimensional signal since the intensity or brightness $I(x, y)$ at each point is a function of two independent variables. On the other hand, a black-and-white television picture may be represented as $I(x, y, t)$ since the brightness is a function of time. Hence, the TV picture may be treated as a three-dimensional signal. In contrast, a colour TV picture may be described by three intensity functions of the form $I_r(x, y, t)$, $I_g(x, y, t)$, and $I_b(x, y, t)$, corresponding to the brightness of the three principal colours (red, green, and blue) as functions of time. Hence, the colour TV picture is a three-channel, three-dimensional signal, which can be represented by the vector

$$\mathbf{I}(x, y, t) = \begin{pmatrix} I_r(x, y, t) \\ I_g(x, y, t) \\ I_b(x, y, t) \end{pmatrix}$$

In this book we deal with single-channel, one-dimensional real- or complex-valued signals and we refer to them simply as signals.

1.2.4 Deterministic and Random Signals

A signal whose complete physical description is known, either in a mathematical form or in a graphical form, is a *deterministic signal*. The nature and amplitude of such a signal at any time can be predicted. The pattern of such a signal is regular. A signal whose values cannot be predicted precisely but are known only in terms of probabilistic description, such as mean value or mean-square value, is a *random signal*. A typical example of a random signal is thermal noise in an electrical circuit.

3 TRANSFORMATIONS OF THE INDEPENDENT VARIABLE (TIME)

Three types of transformations are possible on the independent variable: (i) time shifting, (ii) time scaling, and (iii) time reversal or folding.

1.3.1 Time Shifting

A signal $x(t)$ may be shifted in time by replacing the independent variable t by $t - t_0$, where t_0 is known as the shifting factor. If $t_0 > 0$, the signal is shifted to the right and the time shift results in a delay of the signal by t_0 units of time. If $t_0 < 0$, the signal is shifted to the left and the time shift results in an advance of the signal by $|t_0|$ units in time. A time shift in continuous time is illustrated in Fig. 1.3, in which we have three signals $x(t)$, $x(t - t_0)$, and $x(t + t_0)$ that are identical in shape but are displaced or shifted relative to each other. We will also encounter time shifts in discrete time, as illustrated in Fig. 1.4.

1.3.2 Time Scaling

A second modification of the independent variable involves replacing t by at , where a is known as scaling factor. If $a > 1$, the scaling results in compression (speeded up), and if

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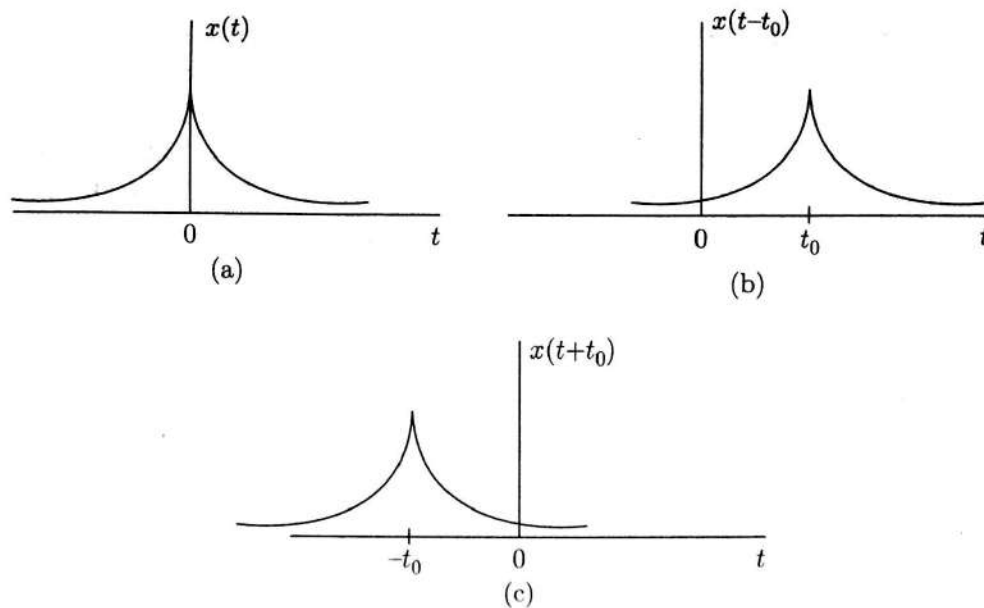


Fig. 1.3 Continuous-time signals related by time shift

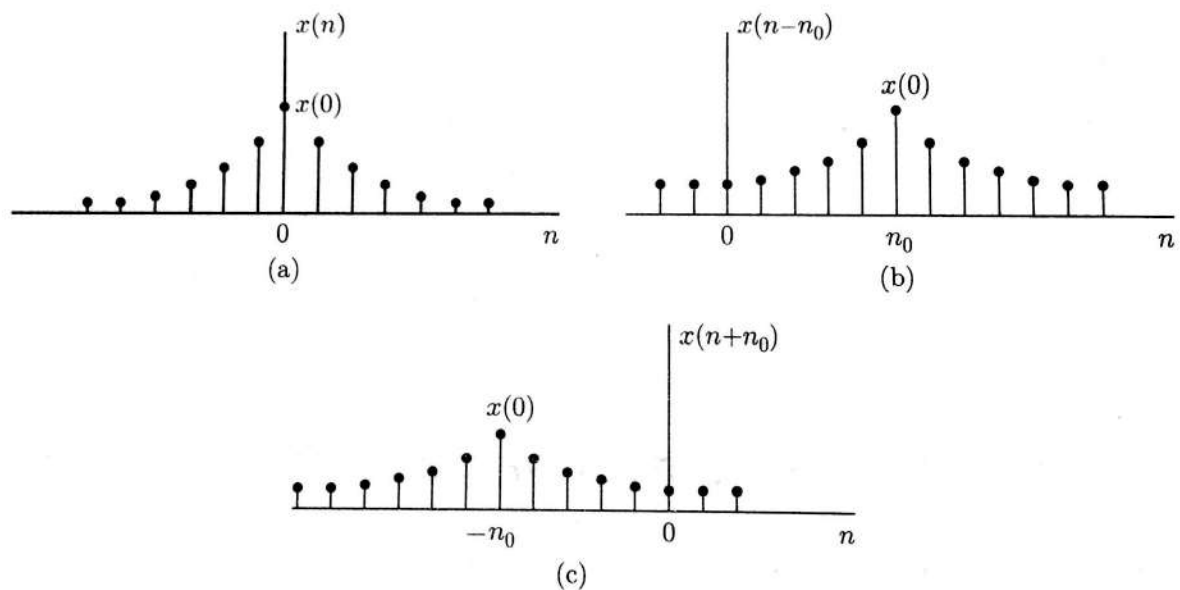


Fig. 1.4 Discrete-time signals related by time shift

$0 < a < 1$, the scaling results in expansion (slowed down). In Fig. 1.5 we have illustrated three signals, $x(t)$, $x(2t)$, and $x(t/2)$, that are related by linear scale changes in the independent variable.

1.3.3 Time Reversal

A third modification of the independent variable involves replacing t by $-t$. The results of this operation is *folding* or *reflection* or *time reversal* of the signal about the time origin $t = 0$. For example, as illustrated in Fig. 1.6, the signal $x(-t)$ is obtained from the signal by a re-election about $t = 0$ (i.e., by reversing the signal). Similarly, as depicted in Fig. 1.7, the signal $x(-n)$ is obtained from the signal $x(n)$ by a re-election about $n = 0$.

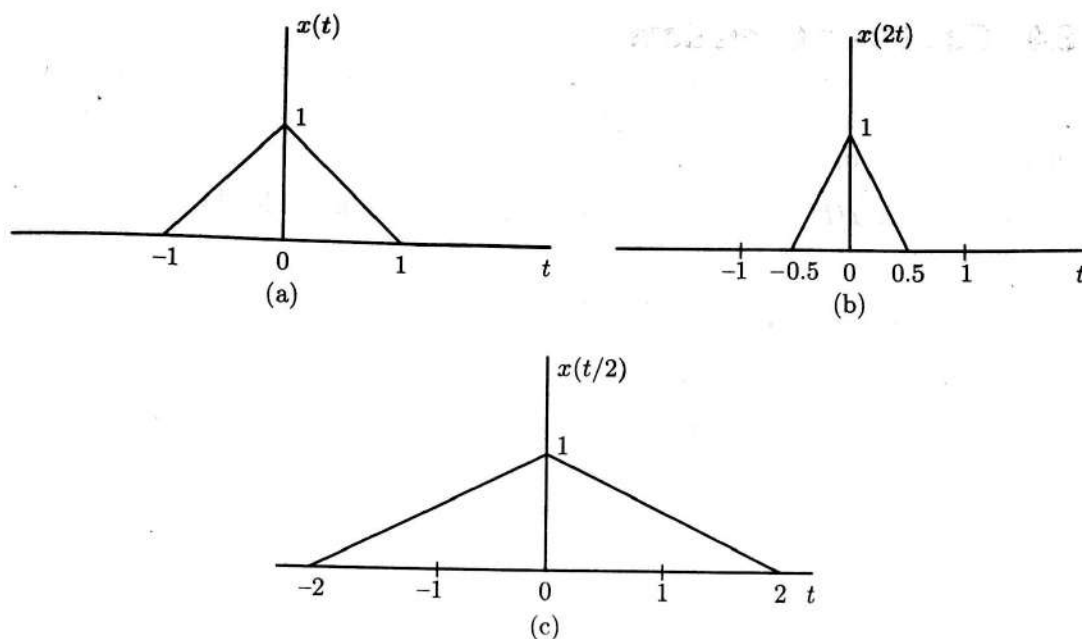


Fig. 1.5 Time-scaling operation

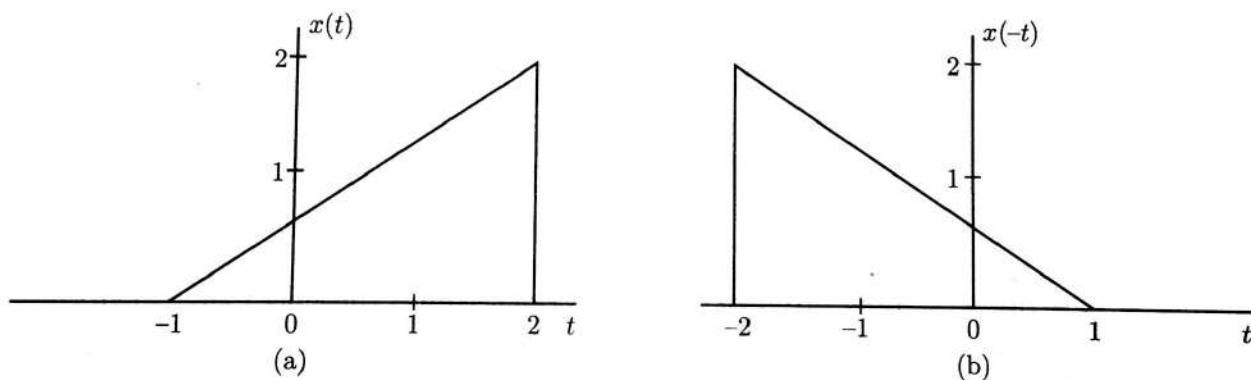


Fig. 1.6 Continuous-time signals related by time-reversal operation

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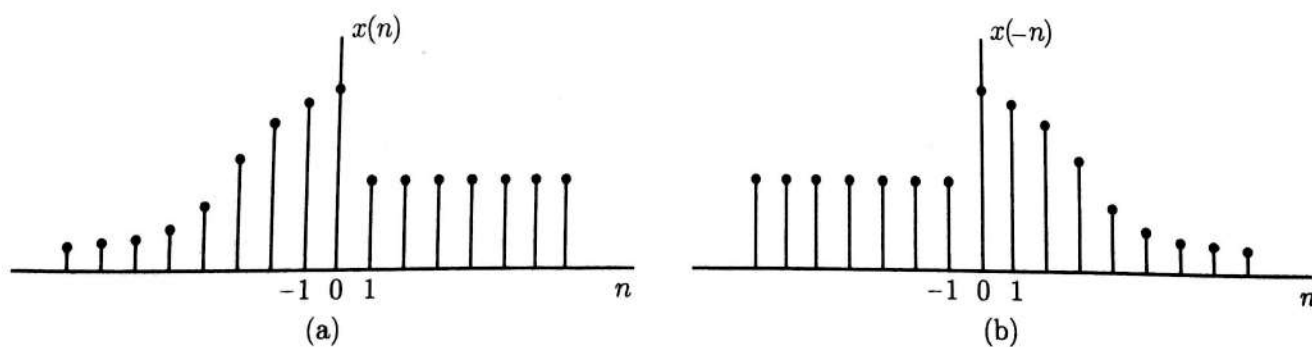


Fig. 1.7 Discrete-time signals related by time-reversal operation

1.3.4 Combined Operations

Certain complex operations require simultaneous use of more than one of the operations just described. The most general operation involving all the three operations is $x(at - b)$, which is realized in two possible sequences of operation:

1. Time-shift $x(t)$ by b to obtain $x(t - b)$. Now time-scale the shifted version $x(t - b)$ by a (i.e., replace t by at) to obtain $x(at - b)$.
2. Time-scale $x(t)$ by a to obtain $x(at)$. Now time-shift $x(at)$ by b/a [i.e., replace t by $(t - b/a)$] to obtain $x(a(t - b/a)) = x(at - b)$. In either case, if a is negative, time scaling involves time reversal.

It is important to note that the operations of folding (time reversal) and time delaying (or advancing) a signal are not commutative. If we denote the time-delay operation by TD and the folding operation by FD , we can write

$$TD_k[x(n)] = x(n - k), \quad k > 0 \quad (1.1)$$

$$FD[x(n)] = x(-n) \quad (1.2)$$

Now

$$TD_k\{FD[x(n)]\} = TD_k[x(-n)] = x(-(n - k)) = x(-n + k) \quad (1.3)$$

whereas

$$FD\{TD_k[x(n)]\} = FD[x(n - k)] = x(-n - k) \quad (1.4)$$

Example 1.1 It is often of interest to determine the effect of transforming the independent variable of a given signal $x(t)$ to obtain a signal of the form $x(at + b)$, where a and b are given numbers.

- (a) If $a = 1$, $b < 0$, we get a time-delayed version of $x(t)$.
- (b) If $a = 1$, $b > 0$, we get a time-advanced version of $x(t)$.
- (c) If $a > 1$, $b = 0$, we get a compressed (speeded up) version of $x(t)$.
- (d) If $0 < a < 1$, $b = 0$, we get an expanded (slowed down) version of $x(t)$.
- (e) If $a = -1$, $b = 0$, we get a time-reversed version of $x(t)$.
- (f) If $a > 1$, $b < 0$, we get a compressed and delayed version of $x(t)$.
- (g) If $0 < a < 1$, $b < 0$, we get a slowed down and advanced version of $x(t)$.
- (h) If $-1 < a < 0$, $b = 0$, we get a time-reversed and slowed down version of $x(t)$.
- (i) If $a < -1$, $b = 0$, we get a time-reversed and speeded up version of $x(t)$.

Example 1.2 A triangular pulse signal $x(t)$ is depicted in Fig. 1.8. Sketch each of the following signals derived from $x(t)$: (a) $x(3t)$, (b) $x(3t + 2)$, and (c) $x(-2t - 1)$.

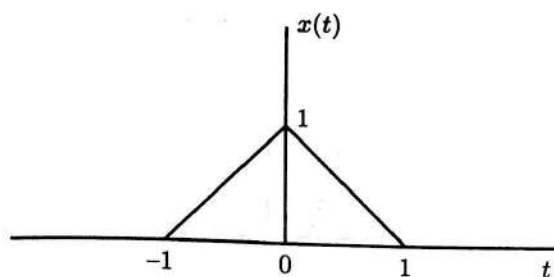


Fig. 1.8 Triangular pulse

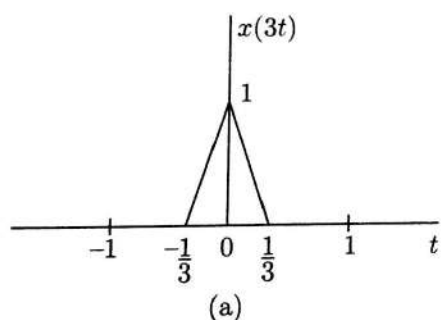
Solution

The mathematical form of $x(t)$ is

$$x(t) = \begin{cases} 1 + t, & -1 \leq t \leq 0 \\ 1 - t, & 0 \leq t \leq 1 \end{cases}$$

- (a) To obtain the mathematical form of $x(3t)$ from $x(t)$, replace t by $3t$ in $x(t)$ [see Fig. 1.9(a)]:

$$x(3t) = \begin{cases} 1 + 3t, & -1 \leq 3t \leq 0 \longrightarrow -\frac{1}{3} \leq t \leq 0 \\ 1 - 3t, & 0 \leq 3t \leq 1 \longrightarrow 0 \leq t \leq \frac{1}{3} \end{cases}$$

Fig. 1.9(a) Signal $x(3t)$

- (b) The mathematical form of $x(3t + 2)$ is

$$x(3t + 2) = \begin{cases} 1 + (3t + 2) = 3 + 3t, & -1 \leq (3t + 2) \leq 0 \longrightarrow -1 \leq t \leq -\frac{2}{3} \\ 1 - (3t + 2) = -1 - 3t, & 0 \leq (3t + 2) \leq 1 \longrightarrow -\frac{2}{3} \leq t \leq -\frac{1}{3} \end{cases}$$

To obtain $x(3t + 2)$ from $x(t)$, the time-shifting and time-scaling operations must be performed in the correct order. Time-shift $x(t)$ by -2 to obtain $x(t - (-2)) = x(t + 2)$. Now, time-scale the shifted version $x(t + 2)$ by 3 (i.e., replace t by $3t$) to obtain $x(3t + 2)$, i.e., $x(t) \rightarrow x(t + 2) \rightarrow x(3t + 2)$ [see Fig. 1.9(b)].

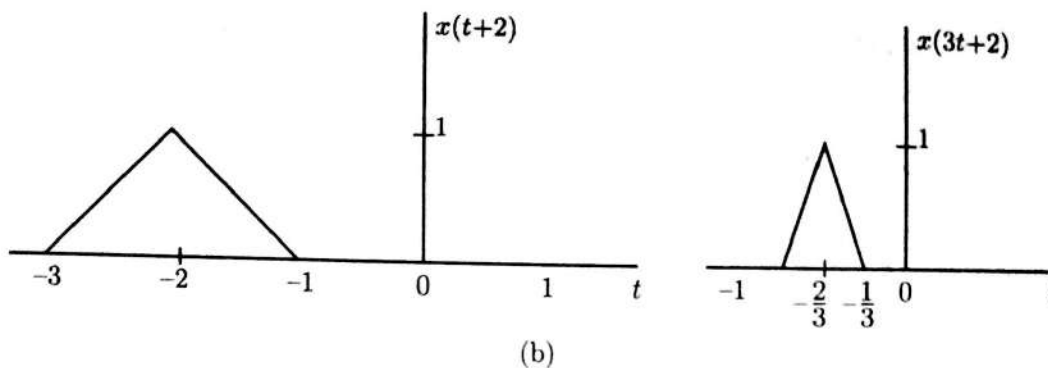


Fig. 1.9(b) Signal $x(t+2)$ and $x(3t+2)$

- (c) To obtain $x(-2t-1)$ from $x(t)$, the time-shifting operation is performed first on $x(t)$ resulting in an intermediate signal $x(t-1)$. Now, time-scale the shifted version $x(t-1)$ by 2 (i.e., replace t by $2t$) to obtain $x(2t-1)$ and after that apply the time-reversal operation [i.e., replace t by $-t$ to obtain $x(-2t-1)$], i.e., $x(t) \rightarrow x(t-1) \rightarrow x(2t-1) \rightarrow x(-2t-1)$ [see Fig. 1.9(c)].

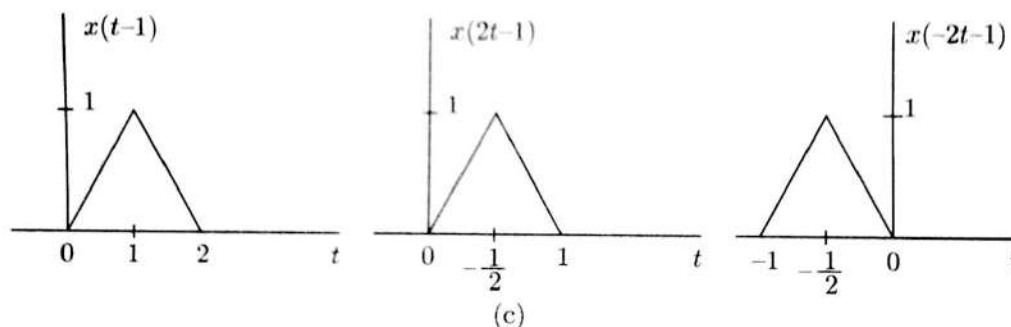


Fig. 1.9(c) Signal $x(t-1)$, $x(2t-1)$, and $x(-2t-1)$

Example 1.3 Let $x(n) = e^{-n/2}u(n)$. Find (a) $2x(5n/3)$, (b) $x(2n)$, and (c) $x(n^2)$.

Solution

(a) With $w(n) = 2x(5n/3)$, we have

$$w(0) = 2x(0) = 2$$

$$w(1) = 2x(5/3) = 0$$

$$w(2) = 2x(10/3) = 0$$

$$w(3) = 2x(5) = 2e^{-5/2}$$

$$w(4) = 2x(20/3) = 0, \text{ etc.}$$

Here we have assumed that $x(n)$ is zero if n is not an integer. It is clear that the general expression for $w(n)$ is

$$w(n) = 2x\left(\frac{5n}{3}\right) = \begin{cases} 2e^{-5n/6}, & n = 0, 3, 6, \text{ etc.} \\ 0, & \text{otherwise} \end{cases}$$

(b) With $y(n) = x(2n)$, we have

$$y(0) = x(0) = 1$$

$$y(1) = x(2) = e^{-1}$$

$$y(2) = x(4) = e^{-2}$$

$$y(3) = x(6) = e^{-3}, \text{ etc.}$$

The general expression for $y(n)$ is therefore

$$y(n) = x(2n) = \begin{cases} e^{-n}, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

(c) With $z(n) = x(n^2)$, we have

$$z(0) = x(0) = 1$$

$$z(1) = x(1) = e^{-1/2}$$

$$z(2) = x(4) = e^{-2}$$

$$z(3) = x(9) = e^{-9/2}, \text{ etc.}$$

The general expression for $z(n)$ is therefore

$$z(n) = x(n^2) = \begin{cases} e^{-n^2/2}, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

The preceding example shows that for discrete-time signals, time scaling does not yield just a stretched or compressed version of the original signal but may give a totally different waveform.

1.4 SINGULARITY FUNCTIONS: UNIT STEP, UNIT IMPULSE, AND UNIT RAMP FUNCTIONS

In the area of signals and systems, step, impulse, and ramp functions play very important roles. They serve as a basis for representing other signals. All these functions are known as singularity functions. Singularity is a point at which a function does not possess a derivative. Each singularity function has a singular point at the origin and is zero elsewhere. Singularity functions are also defined as generalized functions. A *generalized function* is defined by its effect on other functions instead of by its value at every instant of time.

1.4.1 Unit Step Function

- Continuous-time unit step function is defined as

$$u(t) = \begin{cases} 1, & t > 0 \\ \frac{1}{2}, & t = 0 \\ 0, & t < 0 \end{cases} \quad (1.5)$$

and is shown in Fig. 1.10(a)

This signal is important for analytic studies, and it also has many practical applications. Note that the unit step function is continuous for all t , except for $t = 0$ where there is a discontinuity.

- Discrete-time unit step function is defined as

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (1.6)$$

and is shown in Fig. 1.10(b).

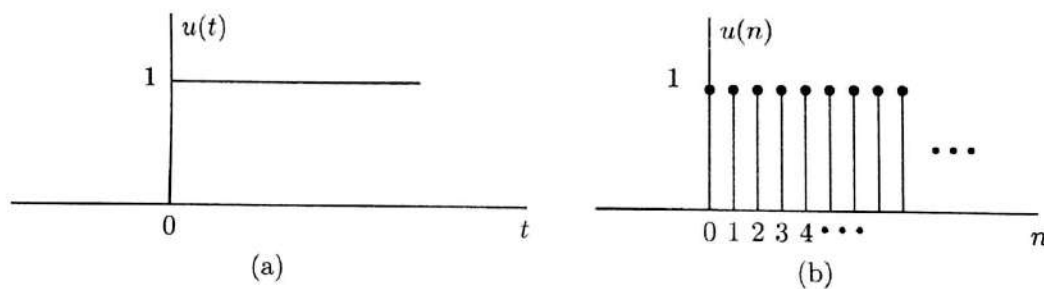


Fig. 1.10 (a) Continuous-time and (b) discrete-time unit step functions

1.4.2 Unit Impulse Function

Continuous-time Unit Impulse Function

Continuous-time unit impulse function, also known as *Dirac delta function* $\delta(t)$, was first defined by P.A.M. Dirac as

$$\delta(t) = 0, \quad t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1, \quad \text{area under unit impulse function is one} \quad (1.7)$$

and is shown in Fig. 1.11(a). The unit impulse can be regarded as a rectangular pulse with a width that has become infinitesimally small, a height that has become infinitely large, and an overall area that has been maintained at unity. The important feature of the unit impulse function is not its shape but the fact that its effective duration (pulse width) approaches zero while its area remains at unity. Pulses such as rectangular, exponential, triangular, and Gaussian types may be used in impulse approximation.

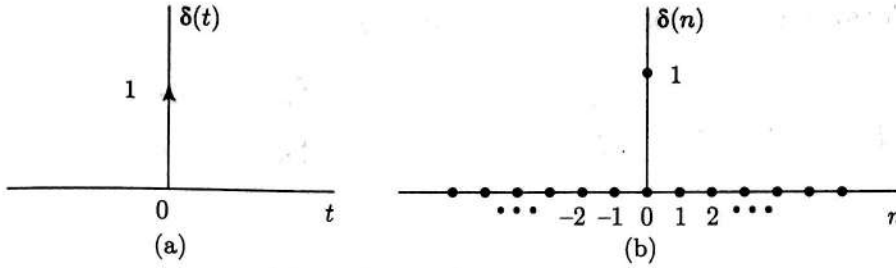


Fig. 1.11 (a) Continuous-time and (b) discrete-time unit impulse functions

Note that the definition of $\delta(t)$ in Eq. (1.7) does not make sense if $\delta(t)$ is an ordinary function. It is meaningful only if $\delta(t)$ is interpreted as a functional, i.e., as a process of assigning the value $x(0)$ to the signal $x(t)$. An ordinary function is specified by its value for all time t . The impulse function is zero everywhere except at $t = 0$, and at this, the only interesting part of its range, it is undefined. These difficulties are resolved by defining the impulse as a generalized function rather than an ordinary function. All the functions derived from the unit impulse function (successive derivatives and integrals) are called *singularity functions*. From Eq. (1.7), it follows that the function $K\delta(t) = 0$ for all $t \neq 0$, and its area is K . Thus, $K\delta(t)$ is an impulse function whose area is K .

Derivative of singularity functions Let $x(t)$ be any test function continuous at $t = 0$. If $g(t)$ is a generalized function, its derivative $dg(t)/dt = g'(t)$ is defined by the following relation:

$$\int_{-\infty}^{\infty} g'(t)x(t)dt = -\int_{-\infty}^{\infty} g(t)x'(t)dt \quad (1.8)$$

where $x'(t)$ is the derivative of $x(t)$.

Properties of continuous-time unit impulse function

1. Sifting property

$$\int_{-\infty}^{\infty} x(t)\delta(t)dt = x(t)|_{t=0} = x(0) \quad (1.9)$$

Proof The multiplication of a continuous-time function $x(t)$ with an impulse located at $t = 0$ results in an impulse, which is located at $t = 0$ and has strength $x(0)$ [the value of $x(t)$ at the location of the impulse].

$$\begin{aligned} \int_{-\infty}^{\infty} x(t)\delta(t)dt &= \int_{-\infty}^{\infty} x(0)\delta(t)dt \\ &= x(0) \int_{-\infty}^{\infty} \delta(t)dt = x(0) \end{aligned}$$

The other form of the sifting property is given by

$$\int_{t_1}^{t_2} x(t)\delta(t-t_0)dt = \int_{t_1}^{t_2} x(t_0)\delta(t-t_0)dt = \begin{cases} x(t_0), & t_1 < t_0 < t_2 \\ 0, & \text{otherwise} \end{cases} \quad (1.10)$$

Proof This can be seen by using the change of variables $\tau = t - t_0$ to obtain

$$\begin{aligned} \int_{t_1}^{t_2} x(t)\delta(t-t_0)dt &= \int_{t_1-t_0}^{t_2-t_0} x(\tau+t_0)\delta(\tau)d\tau \\ &= x(t_0), \quad t_1 < t_0 < t_2 \end{aligned}$$

Note: This function is discontinuous at $t_0 = t_1$ and $t_0 = t_2$. The value of the function at t_1 or t_2 should be given by

$$\int_{t_1}^{t_2} x(t)\delta(t-t_0)dt = \frac{1}{2}x(t_0), \quad t_0 = t_1 \text{ or } t_0 = t_2$$

In general, the property can be written as

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$$

which implies that the signal $x(t)$ can be expressed as a continuous sum of weighted impulses. ■

2. Scaling property

$$\delta(at) = \frac{1}{|a|}\delta(t) \quad (1.11)$$

Proof With a change of variable, $at = \tau$, hence $t = \tau/a$, $dt = (1/a)d\tau$, we obtain the following equations:

- If $a > 0$,

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(at)x(t)dt &= \frac{1}{a} \int_{-\infty}^{\infty} \delta(\tau)x\left(\frac{\tau}{a}\right)d\tau \\ &= \frac{1}{a}x\left(\frac{\tau}{a}\right)\Big|_{\tau=0} = \frac{1}{|a|}x(0) \end{aligned}$$

- If $a < 0$,

$$\begin{aligned}\int_{-\infty}^{\infty} \delta(at)x(t)dt &= -\frac{1}{a} \int_{\infty}^{-\infty} \delta(\tau)x\left(-\frac{\tau}{a}\right)d\tau = \frac{1}{a} \int_{-\infty}^{\infty} \delta(\tau)x\left(-\frac{\tau}{a}\right)d\tau \\ &= \frac{1}{a} x\left(-\frac{\tau}{a}\right)\Big|_{\tau=0} = \frac{1}{|a|} x(0)\end{aligned}$$

Thus for any a ,

$$\int_{-\infty}^{\infty} \delta(at)x(t)dt = \frac{1}{|a|} x(0) = \frac{1}{|a|} \int_{-\infty}^{\infty} \delta(t)x(t)dt = \int_{-\infty}^{\infty} \frac{1}{|a|} \delta(t)x(t)dt$$

for any $x(t)$. From the above equation, we get

$$\delta(at) = \frac{1}{|a|} \delta(t) \quad \blacksquare$$

3. Impulse function is an even function

$$\delta(-t) = \delta(t) \quad (1.12)$$

Proof Substituting $a = -1$ into Eq. (1.11), we obtain

$$\delta(-t) = \frac{1}{|-1|} \delta(t) = \delta(t)$$

which shows that $\delta(t)$ is an even function. ■

4. **Sampling property or multiplication property** Multiplying the signal $x(t)$ by a unit impulse samples the value of the signal at the point at which the impulse is located, i.e.,

$$x(t)\delta(t) = x(t)|_{t=0}\delta(t) = x(0)\delta(t) \quad (1.13)$$

Proof If $x(t)$ is continuous at $t = 0$, then

$$\begin{aligned}\int_{-\infty}^{\infty} [x(t)\delta(t)]\phi(t)dt &= \int_{-\infty}^{\infty} \delta(t)[x(t)\phi(t)]dt \\ &= x(0)\phi(0) = x(0) \int_{-\infty}^{\infty} \delta(t)\phi(t)dt\end{aligned}$$

Therefore,

$$\int_{-\infty}^{\infty} [x(t)\delta(t)]\phi(t)dt = \int_{-\infty}^{\infty} [x(0)\delta(t)]\phi(t)dt$$

for all $\phi(t)$. From the above equation, we get

$$x(t)\delta(t) = x(0)\delta(t) \quad \blacksquare$$

The other form of the multiplication property is given by

$$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0) \quad (1.14)$$

Proof If $x(t)$ is continuous at $t = t_0$, then

$$\begin{aligned} \int_{-\infty}^{\infty} [x(t)\delta(t - t_0)]\phi(t)dt &= \int_{-\infty}^{\infty} \delta(t - t_0)[x(t)\phi(t)]dt \\ &= x(t_0)\phi(t_0) = x(t_0) \int_{-\infty}^{\infty} \delta(t - t_0)\phi(t)dt \end{aligned}$$

Therefore,

$$\int_{-\infty}^{\infty} [x(t)\delta(t - t_0)]\phi(t)dt = \int_{-\infty}^{\infty} [x(t_0)\delta(t - t_0)]\phi(t)dt$$

for all $\phi(t)$. From the above equation, we get

$$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0) \quad \blacksquare$$

5. Differentiation property

$$\int_{-\infty}^{\infty} x(t)\delta'(t)dt = -x'(0) \quad (1.15)$$

Proof The derivative of the singularity function is defined as

$$\int_{-\infty}^{\infty} \delta'(t)x(t)dt = - \int_{-\infty}^{\infty} \delta(t)x'(t)dt = -x'(0) \quad \blacksquare$$

6. Amplitude reversal

$$t\delta'(t) = -\delta(t) \quad (1.16)$$

Proof Consider

$$\begin{aligned} \int_{-\infty}^{\infty} [t\delta'(t)]\phi(t)dt &= \int_{-\infty}^{\infty} \delta'(t)[t\phi(t)]dt = - \int_{-\infty}^{\infty} \delta(t) \frac{d}{dt}[t\phi(t)]dt \\ &= - \int_{-\infty}^{\infty} \delta(t)[\phi(t) + t\phi'(t)]dt = -[\phi(t) + t\phi'(t)]|_{t=0} \\ &= -\phi(0) = - \int_{-\infty}^{\infty} \delta(t)\phi(t)dt \end{aligned}$$

Therefore,

$$\int_{-\infty}^{\infty} [t\delta'(t)]\phi(t)dt = \int_{-\infty}^{\infty} [-\delta(t)]\phi(t)dt$$

From the above equation, we get

$$t\delta'(t) = -\delta(t) \quad \blacksquare$$

Derivative of impulse functions (doublet function) Derivatives of all orders of the impulse function are also singularity functions. The first derivative $d\delta(t)/dt = \delta'(t)$ is referred to as a *doublet function*. A doublet function is defined as

$$\frac{d\delta(t)}{dt} = \delta'(t) = 0, \quad t \neq 0$$

$$\int_{-\infty}^{\infty} \delta'(t)dt = 0, \quad \text{area under unit doublet function is zero} \quad (1.17)$$

$$\int_{-\infty}^{\infty} x(t)\delta'(t)dt = - \int_{-\infty}^{\infty} x'(t)\delta(t)dt = -x'(t)\Big|_{t=0} = -x'(0)$$

where $x(t)$ is any continuous function having a continuous first derivative at $t = 0$.

Example 1.4 Prove the following properties of a doublet function:

$$(a) \int_{t_1}^{t_2} x(t)\delta'(t - t_0)dt = -x'(t_0), \quad t_1 < t_0 < t_2 \quad (1.18)$$

$$(b) x(t)\delta'(t - t_0) = x(t_0)\delta'(t - t_0) - x'(t_0)\delta(t - t_0) \quad (1.19)$$

$$(c) \int_{-\infty}^t \delta'(\tau - t_0)d\tau = \delta(t - t_0) \quad (1.20)$$

$$(d) \delta'(at + b) = \frac{1}{a|a|}\delta'\left(t + \frac{b}{a}\right) \quad (1.21)$$

Solution

(a) The given results can be demonstrated using integration by parts as follows:

$$\begin{aligned} \int_{t_1}^{t_2} x(t)\delta'(t - t_0)dt &= \int_{t_1}^{t_2} x(t)d[\delta'(t - t_0)] \\ &= x(t)\delta(t - t_0)\Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} x'(t)\delta(t - t_0)dt \\ &= 0 - 0 - x(t)\Big|_{t=t_0} = -x'(t_0) \end{aligned}$$

(b) Let $\psi(t)$ be a test signal. Therefore,

$$\begin{aligned}
 \int_{-\infty}^{\infty} [x(t)\delta'(t-t_0)]\psi(t)dt &= - \int_{-\infty}^{\infty} [\psi(t)x(t)]'\delta(t-t_0)dt \\
 &= -[\psi'(t)x(t) + \psi(t)x'(t)]\Big|_{t=t_0} \\
 &= -\psi'(t_0)x(t_0) - \psi(t_0)x'(t_0) \\
 &= x(t_0) \int_{-\infty}^{\infty} \delta'(t-t_0)\psi(t)dt - x'(t_0) \int_{-\infty}^{\infty} \delta(t-t_0)\psi(t)dt \\
 \int_{-\infty}^{\infty} [x(t)\delta'(t-t_0)]\psi(t)dt &= \int_{-\infty}^{\infty} [x(t_0)\delta'(t-t_0) - x'(t_0)\delta(t-t_0)]\psi(t)dt
 \end{aligned}$$

Comparing both sides of the above equation, we get

$$x(t)\delta'(t-t_0) = x(t_0)\delta'(t-t_0) - x'(t_0)\delta(t-t_0)$$

(c) Consider the integral

$$\int_{-\infty}^t \delta'(\tau-t_0)d\tau = \delta(\tau-t_0)\Big|_{-\infty}^t = \delta(t-t_0)$$

(d) We know that

$$\begin{aligned}
 \delta(at+b) &= \delta\left(a\left(t+\frac{b}{a}\right)\right) \\
 &= \frac{1}{|a|}\delta\left(t+\frac{b}{a}\right)
 \end{aligned}$$

Differentiating both sides with respect to t , we get

$$\begin{aligned}
 a\delta'(at+b) &= \frac{1}{|a|}\delta'\left(t+\frac{b}{a}\right) \\
 \delta'(at+b) &= \frac{1}{a|a|}\delta'\left(t+\frac{b}{a}\right)
 \end{aligned}$$

Higher-order derivatives of the impulse function give

$$\int_{-\infty}^{\infty} x(t) \frac{d^k \delta(t)}{dt^k} dt = (-1)^k \frac{d^k x(t)}{dt^k} \Big|_{t=0} \quad (1.22)$$

Doublets and higher-order derivatives of the impulse function are not often used as signal models in system analysis.

Discrete-time Unit Impulse Function

Discrete-time unit impulse function (or unit sample function) is defined as

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \quad (1.23)$$

and is shown in Fig. 1.11(b).

Properties of discrete-time unit impulse function

$$1. \quad \delta(kn) = \delta(n) \quad (1.24)$$

where k is an integer.

Proof By definition,

$$\begin{aligned} \delta(n) &= \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \\ \delta(kn) &= \begin{cases} 1, & kn = 0 \longrightarrow n = 0 \\ 0, & kn \neq 0 \longrightarrow n \neq 0 \end{cases} \\ &= \delta(n) \end{aligned}$$

■

2. Unit impulse function $\delta(n)$ and unit step function $u(n)$ are related by

$$\delta(n) = u(n) - u(n-1) \quad (1.25)$$

Proof From the definition of the unit step function,

$$\begin{aligned} u(n) &= \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \\ &= \delta(n) + \delta(n-1) + \delta(n-2) + \dots \\ &= \sum_{k=0}^{\infty} \delta(n-k) \\ &= \delta(n) + \sum_{k=1}^{\infty} \delta(n-k) \end{aligned}$$

and

$$\begin{aligned}
 u(n-1) &= \begin{cases} 1, & n-1 \geq 0 \longrightarrow n \geq 1 \\ 0, & n-1 < 0 \longrightarrow n < 1 \end{cases} \\
 &= \delta(n-1) + \delta(n-2) + \delta(n-3) + \dots \\
 &= \sum_{k=1}^{\infty} \delta(n-k)
 \end{aligned}$$

From the definition of $u(n)$ and $u(n-1)$,

$$\begin{aligned}
 u(n) - u(n-1) &= \delta(n) + \sum_{k=1}^{\infty} \delta(n-k) - \sum_{k=1}^{\infty} \delta(n-k) \\
 &= \delta(n)
 \end{aligned}$$

3. For any arbitrary sequence $x(n)$,

$$x(n)\delta(n-k) = x(k)\delta(n-k) \quad (1.26)$$

Proof From the definition of discrete impulse function,

$$\delta(n-k) = \begin{cases} 1, & (n-k) = 0 \longrightarrow n = k \\ 0, & (n-k) \neq 0 \longrightarrow n \neq k \end{cases}$$

Therefore,

$$\begin{aligned}
 x(n)\delta(n-k) &= x(n)|_{n=k} \delta(n-k) \\
 &= x(k)\delta(n-k)
 \end{aligned}$$

$$4. \quad \sum_{k=-\infty}^n \delta(k) = \sum_{k=0}^{\infty} \delta(n-k) = u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (1.27)$$

Proof Consider the sum of δ function from $-\infty$ to n

$$= \sum_{k=-\infty}^n \delta(k)$$

By replacing k by $(n-m)$, we can write the above equation as

$$\begin{aligned}
 &= \sum_{m=0}^{\infty} \delta(n-m) \\
 &= \delta(n) + \delta(n-1) + \delta(n-2) + \dots \\
 &= u(n)
 \end{aligned}$$

1.4.3 Unit Ramp Function

The ramp function shown in Fig. 1.12(a) is defined by

$$r(t) = tu(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases} \quad (1.28)$$

The discrete-time version of the ramp function shown in Fig. 1.12(b) is defined by

$$r(n) = nu(n) = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (1.29)$$

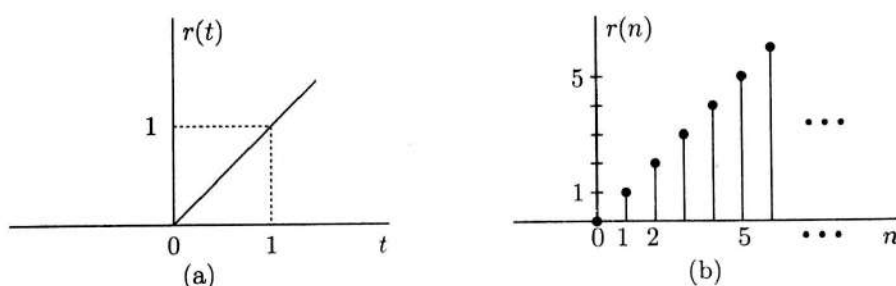


Fig. 1.12 (a) Continuous-time and (b) discrete-time unit ramp functions

Example 1.5 Prove the following relationship between unit impulse function $\delta(t)$, unit step function $u(t)$, and unit ramp function $r(t)$.

$$(a) \quad \frac{du(t)}{dt} = \delta(t) \quad \text{or} \quad u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

$$(b) \quad \frac{dr(t)}{dt} = u(t) \quad \text{or} \quad r(t) = \int_{-\infty}^t u(\tau) d\tau$$

Solution

(a) Let $x(t)$ be any test function continuous at $t = 0$. Now consider the integral

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{du(t)}{dt} x(t) dt &= \int_{-\infty}^{\infty} u'(t) x(t) dt \\ &= u(t)x(t) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} x'(t) u(t) dt \end{aligned}$$

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$$= x(\infty) - 0 - \int_0^{\infty} x'(t) dt$$

$$= x(\infty) - x(t) \Big|_0^{\infty}$$

$$= x(\infty) - [x(\infty) - x(0)]$$

$$= x(0)$$

$$\int_{-\infty}^{\infty} \frac{du(t)}{dt} x(t) dt = \int_{-\infty}^{\infty} \delta(t) x(t) dt$$

Therefore,

$$\frac{du(t)}{dt} = \delta(t)$$

(b) From the definition of unit ramp function, we have

$$r(t) = tu(t)$$

$$\frac{dr(t)}{dt} = \frac{d}{dt}[tu(t)] = t\delta(t) + u(t) = 0 + u(t) = u(t)$$

Example 1.6 Draw the waveforms of the following signals:

- (a) $x_1(t) = u(t+2)$, (b) $x_2(t) = u(t-2)$, (c) $x_3(t) = u(-t)$, (d) $x_4(t) = u(-2t+1)$,
 (e) $x_5(t) = u(-2t-1)$, (f) $x_6(t) = x_1(t) - x_2(t)$, (g) $x_7(t) = u(t) - 2u(t-1) + u(t-2)$,
 (h) $x_8(t) = dx_6(t)/dt$, and (i) $x_9(t) = dx_7(t)/dt$, where $u(t)$ is the unit step signal.

Solution

(a) $u(t+2)$ is a unit step signal shifted towards left by 2 units [Fig. 1.13(a)].

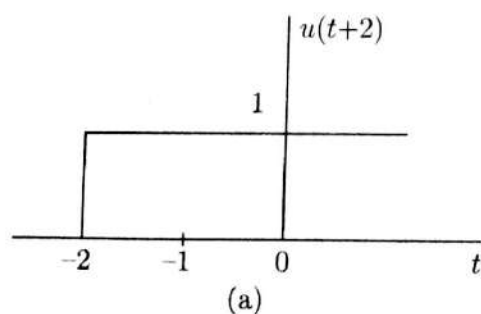


Fig. 1.13(a)

(b) $u(t-2)$ is a unit step signal shifted towards right by 2 units [Fig. 1.13(b)].

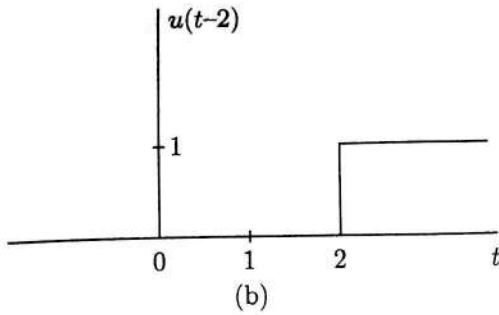


Fig. 1.13(b)

(c) $u(-t)$ is a folded version of $u(t)$ [Fig. 1.13(c)].

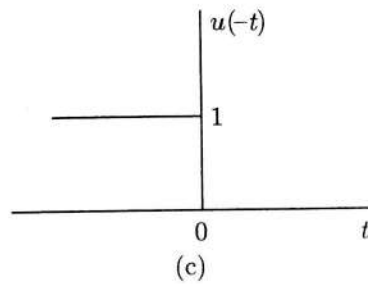


Fig. 1.13(c)

(d) The mathematical form of $u(-2t + 1)$ is

$$u(-2t + 1) = \begin{cases} 1, & (-2t + 1) > 0 \longrightarrow t < \frac{1}{2} \\ 0, & (-2t + 1) < 0 \longrightarrow t > \frac{1}{2} \end{cases}$$

To obtain $u(-2t + 1)$ from $u(t)$, time-shifting, time-scaling, and time-reversal operations must be performed in the correct order, i.e., $u(t) \rightarrow u(t + 1) \rightarrow u(2t + 1) \rightarrow u(-2t + 1)$ [Fig. 1.13(d)].

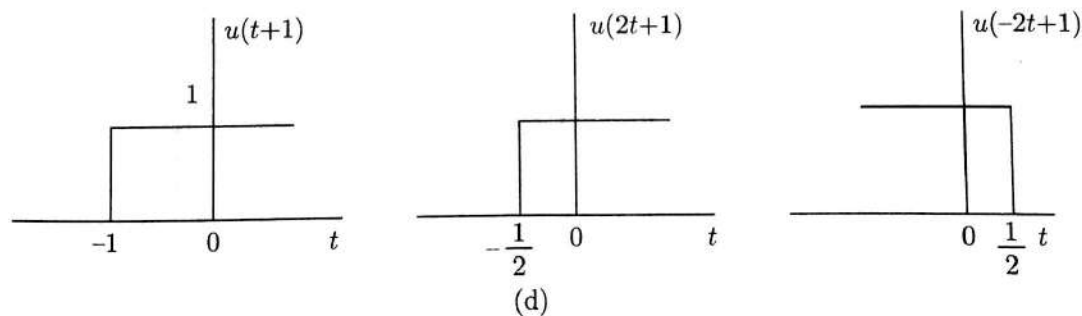


Fig. 1.13(d)

(e) The mathematical form of $u(-2t - 1)$ is

$$u(-2t - 1) = \begin{cases} 1, & (-2t - 1) > 0 \longrightarrow t < -\frac{1}{2} \\ 0, & (-2t - 1) < 0 \longrightarrow t > -\frac{1}{2} \end{cases}$$

To obtain $u(-2t - 1)$ from $u(t)$, time-shifting, time scaling, and time-reversal operations must be performed in the correct order, i.e., $u(t) \rightarrow u(t - 1) \rightarrow u(2t - 1) \rightarrow u(-2t - 1)$ [Fig. 1.13(e)].

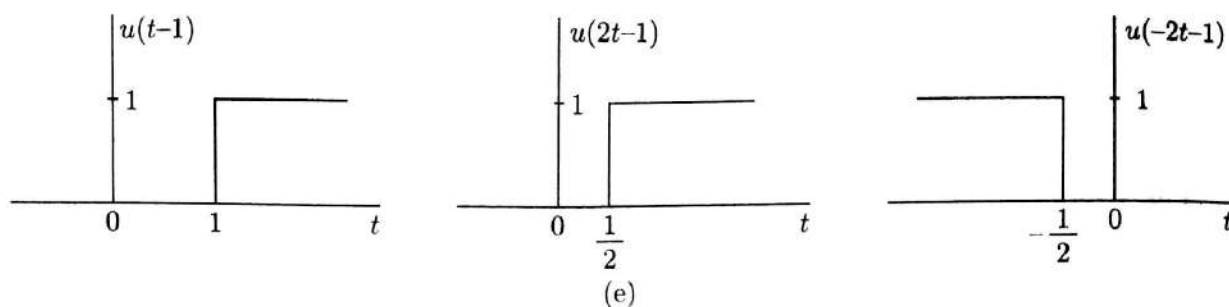


Fig. 1.13(e)

(f) $x_6(t) = x_1(t) - x_2(t) = u(t + 2) - u(t - 2)$ is a gate signal or rectangle signal as shown in Fig. 1.13(f).

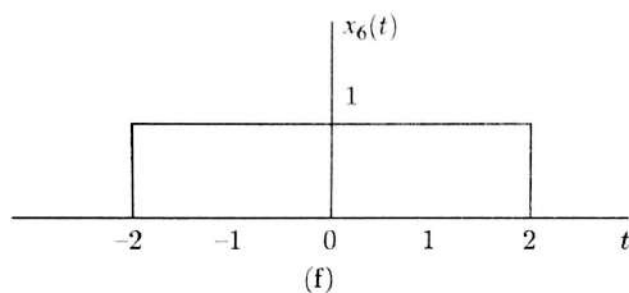


Fig. 1.13(f)

(g) $x_7(t) = u(t) - 2u(t - 1) + u(t - 2)$ is shown in Fig. 1.13(g).

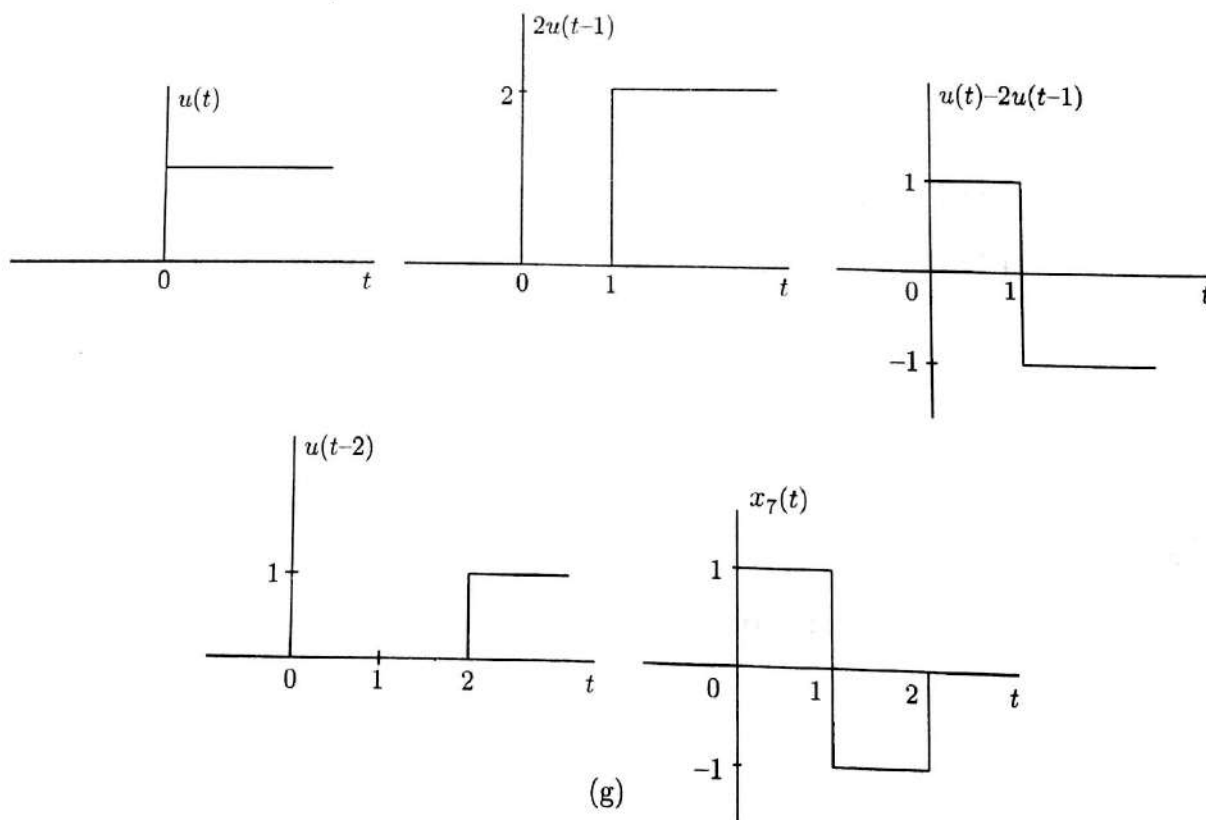


Fig. 1.13(g)

(h) The derivative of $x_6(t)$ is determined as

$$x_8(t) = \frac{dx_6(t)}{dt} = \frac{d}{dt}[x_1(t) - x_2(t)]$$

substituting the values of $x_1(t)$ and $x_2(t)$ from parts (a) and (b), we get [Fig. 1.13(h)]

$$x_8(t) = \frac{d}{dt}u(t+2) - \frac{d}{dt}u(t-2) = \delta(t+2) - \delta(t-2)$$

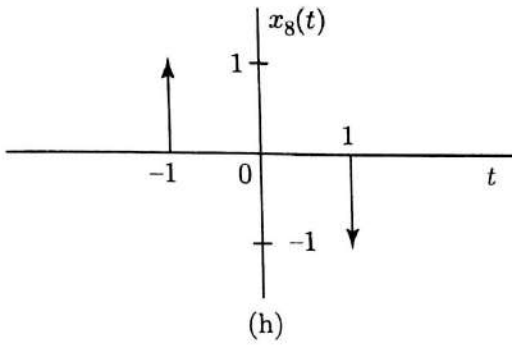


Fig. 1.13(h)

(i) The derivative of $x_7(t)$ is determined as [Fig. 1.13(i)]

$$x_9(t) = \frac{dx_7(t)}{dt} = \frac{d}{dt}[u(t) - 2u(t-1) + u(t-2)] = \delta(t) - 2\delta(t-1) + \delta(t-2)$$

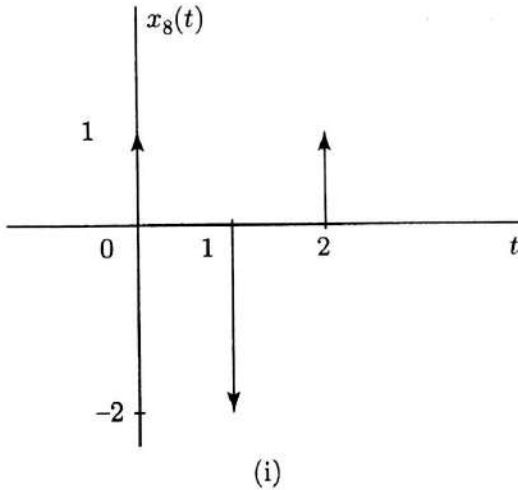


Fig. 1.13(i)

Example 1.7 Draw the waveforms of the following signals:

(a) $x_1(t) = r(t-1)$, (b) $x_2(t) = r(t+1)$, (c) $x_3(t) = r(-t)$, (d) $x_4(t) = r(3t)$,
 (e) $x_5(t) = r(-3t)$, (f) $x_6(t) = r(3t+1)$, (g) $x_7(t) = r(-3t-1)$, (h) $x_8(t) = r(t) - r(t-1)$,
 (i) $x_9(t) = x_8(t) - r(t-1) + r(t-2)$, (j) $x_{10}(t) = dx_8(t)/dt$, and (k) $x_{11}(t) = dx_9(t)/dt$,
 where $r(t)$ is the unit ramp signal.

Solution

(a) $r(t-1)$ is a ramp signal shifted towards right by 1 unit [Fig. 1.14(a)].

(b) $r(t+1)$ is a ramp signal shifted towards left by 1 unit [Fig. 1.14(b)].

(c) $r(-t)$ is a folded version of $r(t)$ [Fig. 1.14(c)].

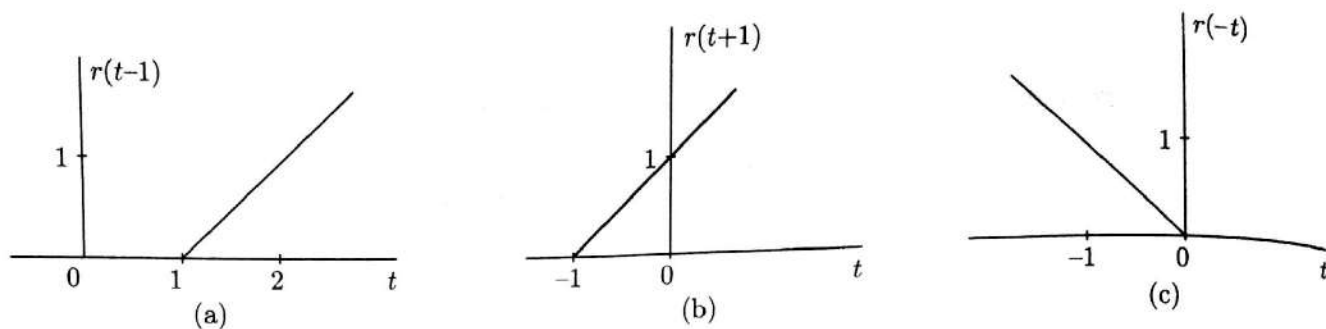


Fig. 1.14(a), (b), (c)

(d) $r(3t)$ is a time-scaled version of $r(t)$ [Fig. 1.14(d)].

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$x_4(t) = r(3t) = \begin{cases} 3t, & 3t \geq 0 \rightarrow t \geq 0 \\ 0, & 3t < 0 \rightarrow t < 0 \end{cases}$$

(e) To obtain $r(-3t)$, apply time-reversal operation on $r(3t)$ [Fig. 1.14(e)].

(f) To obtain $x_6(t) = r(3t + 1)$ from $r(t)$, the time-shifting and time-scaling operations must be performed in the correct order, i.e., $r(t) \rightarrow r(t + 1) \rightarrow r(3t + 1)$ [Fig. 1.14(f)].

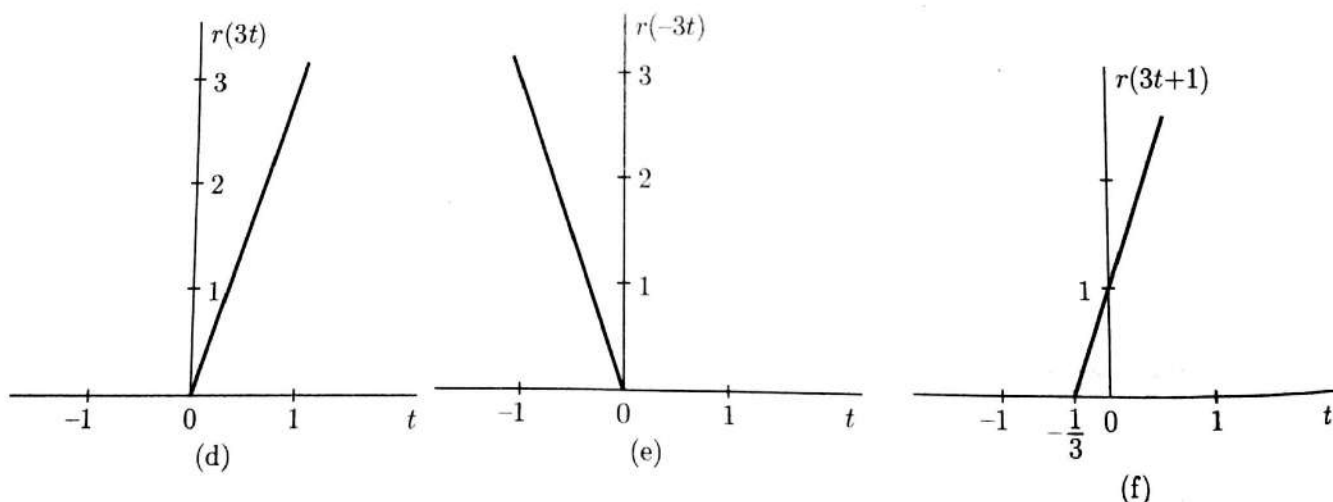


Fig. 1.14(d), (e), (f)

(g) To obtain $x_7(t) = r(-3t - 1)$ from $r(t)$, time-shifting, time-scaling, and time-reversal operations must be performed in the correct order, i.e., $r(t) \rightarrow r(t - 1) \rightarrow r(3t - 1) \rightarrow r(-3t - 1)$ [Fig. 1.14(g)].

(h) $x_8(t) = r(t) - r(t - 1)$ is shown in Fig. 1.14(h).

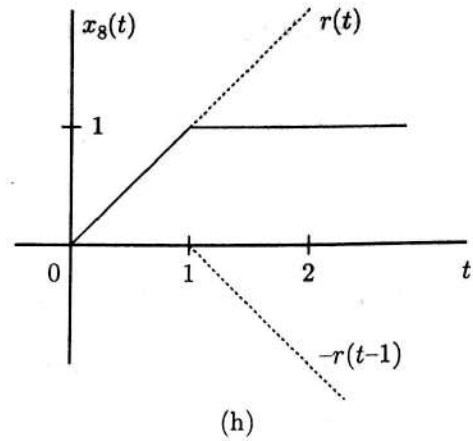
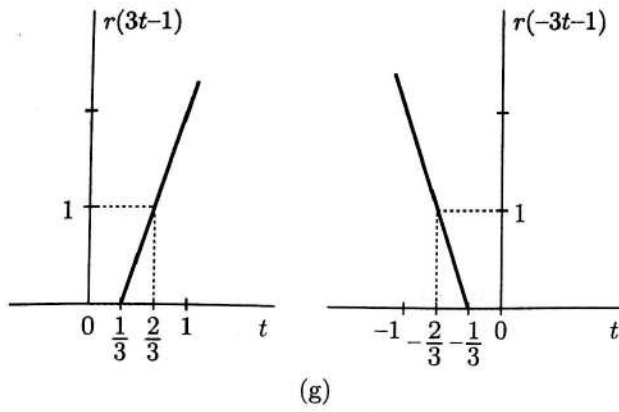


Fig. 1.14(g), (h)

(i) $x_9(t)$ is shown in Fig. 1.14(i).

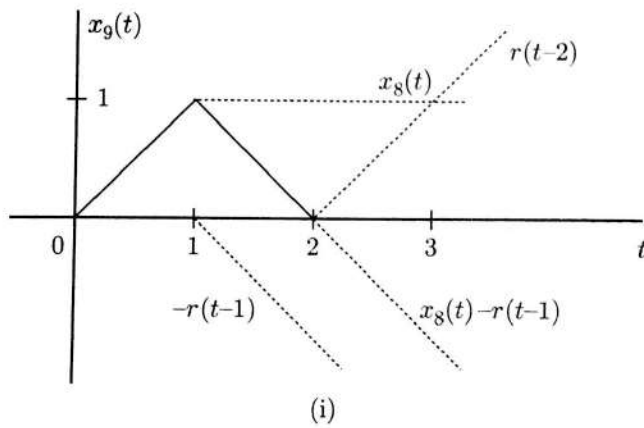


Fig. 1.14(i)

(j) The derivative of $x_8(t)$ is determined as [Fig. 1.14(j)]

$$x_{10}(t) = \frac{dx_8(t)}{dt} = \frac{d}{dt}[r(t) - r(t-1)] = u(t) - u(t-1)$$

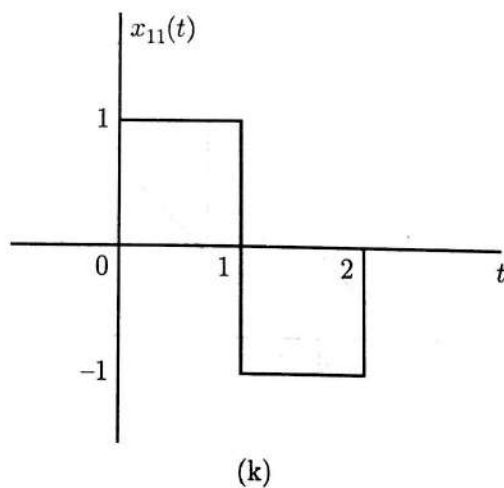
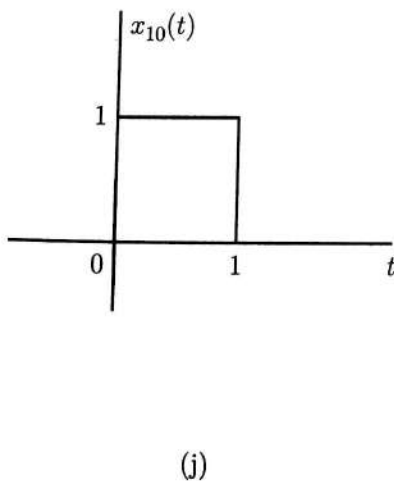


Fig. 1.14(j), (k)

(k) The derivative of $x_{11}(t)$ is determined as [Fig. 1.14(k)]

$$x_{11}(t) = \frac{dx_9(t)}{dt} = \frac{d}{dt}[r(t) - 2r(t-1) + u(t-2)] = u(t) - 2u(t-1) + u(t-2)$$

Example 1.8 Draw the waveforms of the following signals:

- (a) $x_1(t) = r(t) - r(t-1) - u(t-1)$, (b) $x_2(t) = u(t) + r(t) - 2r(t-1) + r(t-2) - u(t-2)$,
 (c) $x_3(t) = -u(t+1) + r(t+1) - r(t-1) - u(t-1)$, (d) $x_4(t) = dx_1(t)/dt$,
 (e) $x_5(t) = u(t+1) - u(t) + x_1(t)$, and (f) $x_6(t) = dx_3(t)/dt$.

Solution

(a) The procedure of obtaining $x_1(t)$ is illustrated in Fig. 1.15(a).

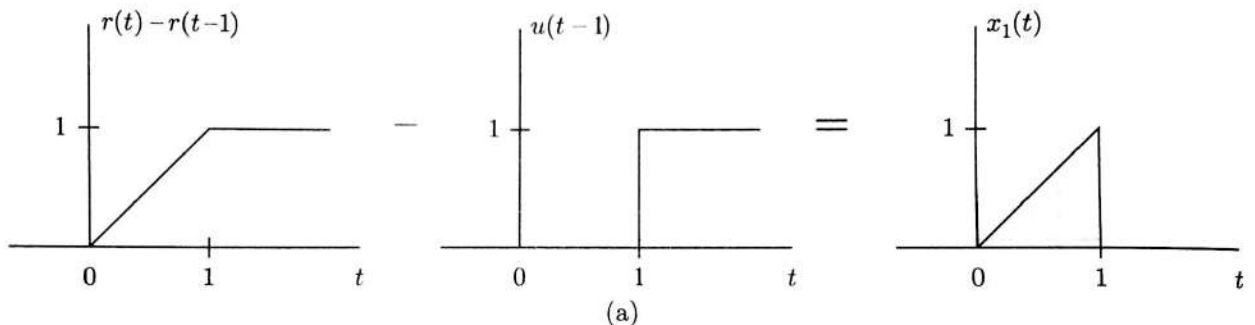


Fig. 1.15(a)

(b) The procedure of obtaining $x_2(t)$ is illustrated in Fig. 1.15(b).

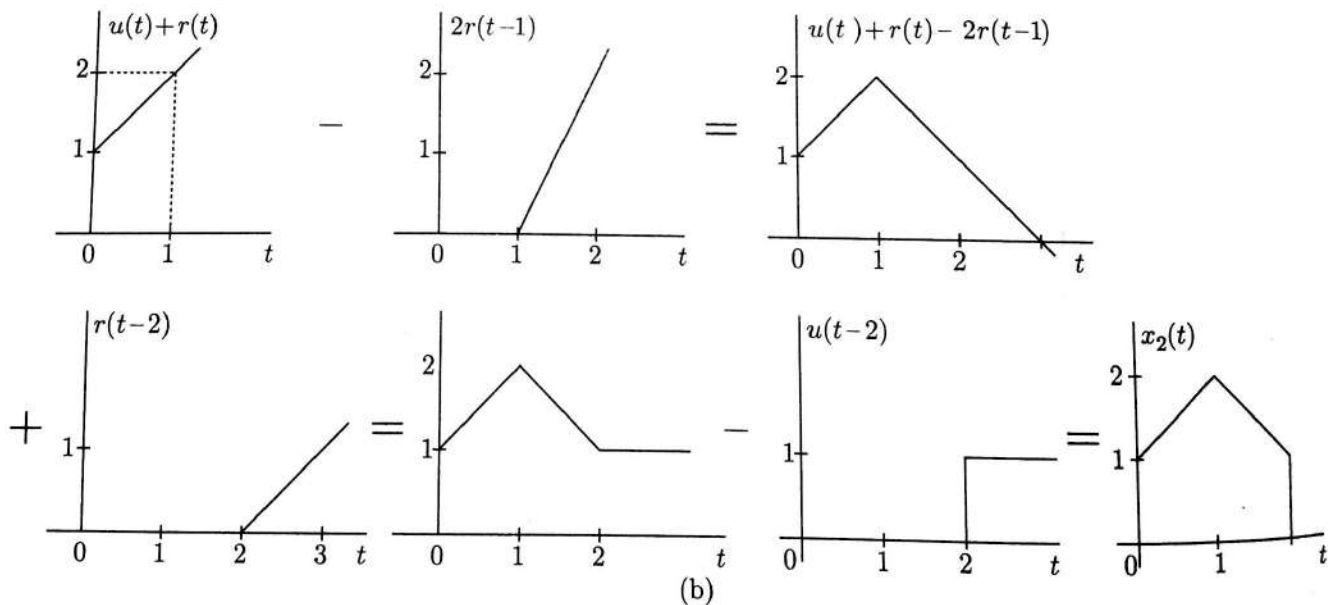


Fig. 1.15(b)

(c) The procedure of obtaining $x_3(t)$ is illustrated in Fig. 1.15(c).

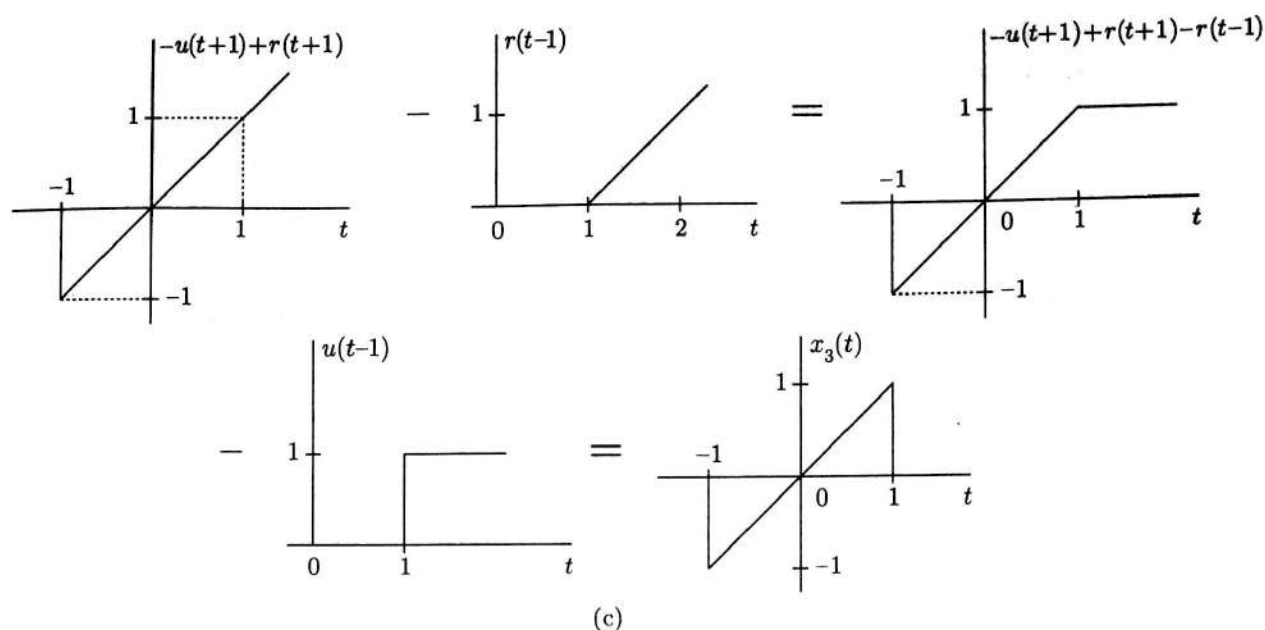


Fig. 1.15(c)

(d) The derivative of $x_1(t)$ is determined as [Fig. 1.15(d)]

$$x_4(t) = \frac{dx_1(t)}{dt} = \frac{d}{dt}[r(t) - r(t-1) - u(t-1)] = u(t) - u(t-1) - \delta(t-1)$$

(e) $x_5(t)$ is shown in Fig. 1.15(e).

(f) The derivative of $x_3(t)$ is determined as [Fig. 1.15(f)]

$$\begin{aligned} x_6(t) &= \frac{dx_3(t)}{dt} = \frac{d}{dt}[-u(t+1) + r(t+1) - r(t-1) - u(t-1)] \\ &= -\delta(t+1) + u(t+1) - u(t-1) - \delta(t-1) \end{aligned}$$

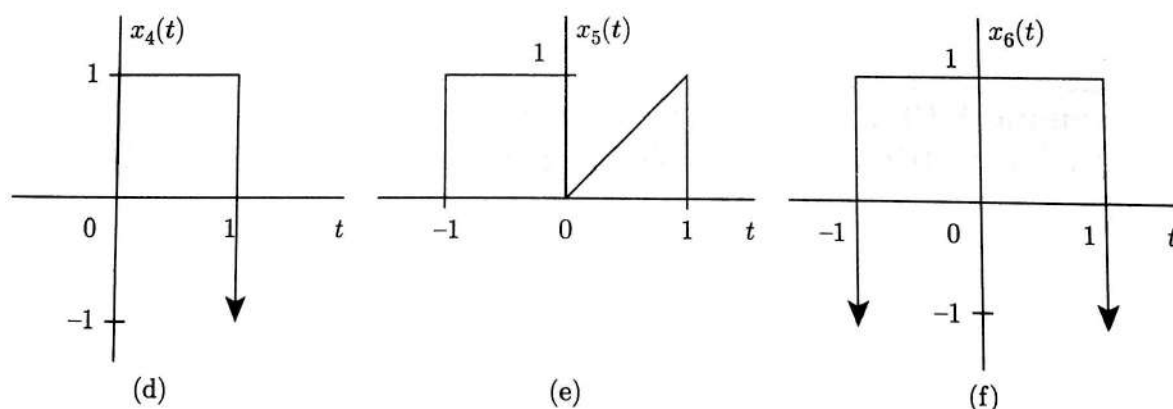


Fig. 1.15(d), (e), (f)

Example 1.9 Evaluate the following integrals:

- (a) $\int_{-2}^1 (t + t^2) \delta(t-3) dt$, (b) $\int_{-2}^4 (t + t^2) \delta(t-3) dt$, (c) $\int_{-\infty}^{\infty} (t^2 + \cos \pi t) \delta(t-1) dt$,
 (d) $\int_0^3 e^{t-2} \delta(2t-4) dt$, and (e) $\int_{-\infty}^{\infty} e^{-t} \delta(2t-2) dt$.

Solution

(a) Since $t = 3$ is not in the interval $-2 < t < 1$, the shifting property [Eq. (1.10)] yields

$$\int_{-2}^1 (t + t^2) \delta(t - 3) dt = 0$$

(b) Since $t = 3$ is within the interval $-2 < t < 4$, the shifting property [Eq. (1.10)] yields

$$\int_{-2}^4 (t + t^2) \delta(t - 3) dt = (t + t^2)|_{t=3} = 3 + 3^2 = 12$$

(c) By using the shifting property [Eq. (1.10)], we obtain

$$\int_{-\infty}^{\infty} (t^2 + \cos \pi t) \delta(t - 1) dt = (t^2 + \cos \pi t)|_{t=1} = 1 + \cos \pi = 1 - 1 = 0$$

(d) By using the scaling property [Eq. (1.11)] and then the shifting property [Eq. (1.10)], we get

$$\int_0^3 e^{t-2} \delta(2t - 4) dt = \int_0^3 e^{t-2} \delta[2(t - 2)] dt = \int_0^3 e^{t-2} \frac{1}{2} \delta(t - 2) dt = \frac{1}{2} e^{t-2} |_{t=2} = \frac{1}{2} e^0 = \frac{1}{2}$$

(e) By using the scaling property [Eq. (1.11)] and then the shifting property [Eq. (1.10)], we get

$$\int_{-\infty}^{\infty} e^{-t} \delta(2t - 2) dt = \int_{-\infty}^{\infty} e^{-t} \delta(2(t - 1)) dt = \int_{-\infty}^{\infty} e^{-t} \frac{1}{2} \delta(t - 1) dt = \frac{1}{2} e^{-t} |_{t=1} = \frac{1}{2} e^{-1} = \frac{1}{2e}$$

Example 1.10 Evaluate the following integrals:

(a) $\int_{-\infty}^{\infty} e^{-2t} \delta'(t) dt$, (b) $\int_{-4}^4 (t - 2)^2 \delta'(-\frac{1}{3}t + \frac{1}{2}) dt$, and (c) $\int_{-4}^1 t e^{-2t} \delta''(t - 1) dt$.

Solution

(a) Consider the given integral

$$\int_{-\infty}^{\infty} e^{-2t} \delta'(t) dt = - \int_{-\infty}^{\infty} \frac{d}{dt} e^{-2t} \delta(t) dt = 2e^{-2t} |_{t=0} = 2$$

(b) Consider the given integral

$$\int_{-4}^4 (t - 2)^2 \delta' \left(-\frac{1}{3}t + \frac{1}{2} \right) dt = \int_{-4}^4 (t - 2)^2 \delta' \left(-\frac{1}{3} \left(t - \frac{3}{2} \right) \right) dt$$

$$\begin{aligned}
&= \int_{-4}^4 3(t-2)^2 \delta' \left(t - \frac{3}{2} \right) dt \\
&= -3 \int_{-4}^4 \frac{d}{dt} (t-2)^2 \delta \left(t - \frac{3}{2} \right) dt \\
&= -6(t-2) \Big|_{t=3/2} = 3
\end{aligned}$$

(c) Consider the given integral

$$\int_{-4}^1 t e^{-2t} \delta''(t-1) dt = (-1)^2 \frac{d^2}{dt^2} (t e^{-2t}) \Big|_{t=1} = (4t-4) e^{-2t} \Big|_{t=1} = 0$$

1.5 PERIODIC AND APERIODIC SIGNALS

A periodic signal has a waveform that repeats over and over, with the time between the repeats defined as the period of the signal. The continuous-time signal $x(t)$ is periodic if and only if

$$x(t+T) = x(t), \quad \text{for all } t \quad (1.30)$$

where T is positive and is the period of the signal. In other words, a periodic signal $x(t)$ remains unchanged when time-shifted by one period. For this reason a periodic signal must start at $t = -\infty$; if it started at some finite instant say $t = 0$, the time-shifted signal $x(t+T)$ would start at $t = -T$ and $x(t+T) \neq x(t)$. Therefore a periodic signal, by definition, must start at $t = -\infty$ and continued forever. It should be noted that no physical signals are actually periodic since they all begin at some time and/or cease to exist at some later time. The fundamental period T_0 of $x(t)$ is the smallest positive value of T for which Eq. (1.30) holds. This definition of fundamental period works, except if $x(t)$ is a constant. In this case the fundamental period is undefined since $x(t)$ is periodic for any choice of T . A signal $x(t)$ that is not periodic will be referred to as an *aperiodic* signal.

The discrete-time signal $x(n)$ is periodic if and only if

$$x(n+N) = x(n), \quad \text{for all } n \quad (1.31)$$

where N is a positive integer and is the period of the signal measured in terms of number of sample spacings (samples/cycle). The smallest value of N for which the definition hold is called the fundamental period of the periodic signal.

Example 1.11 Show that

- the complex exponential signal $x(t) = e^{j\omega_0 t}$ is periodic and its fundamental period is $2\pi/\omega_0$.
- the complex exponential signal $x(n) = e^{j\omega_0 n}$ is periodic only if $2\pi/\omega_0$ is a ratio number.

Solution

(a) By Eq. (1.30), $x(t)$ will be periodic if

$$\begin{aligned} x(t) &= x(t + T) \\ e^{j\omega_0 t} &= e^{j\omega_0(t+T)} \\ &= e^{j\omega_0 t} \underbrace{e^{j\omega_0 T}}_{=1} \end{aligned}$$

Thus, we must have

$$e^{j\omega_0 T} = 1 = e^{j2m\pi}, \quad m = \text{positive integer}$$

Comparing the powers of the exponent, we get

$$\begin{aligned} \omega_0 T &= 2m\pi \\ T &= \frac{2m\pi}{\omega_0} \end{aligned}$$

Thus, the fundamental period (smallest positive value) $T = 2\pi/\omega_0$.

(b) By Eq. (1.31), $x(n)$ will be periodic if

$$\begin{aligned} x(n) &= x(n + N) \\ e^{j\omega_0 n} &= e^{j\omega_0(n+N)} \\ &= e^{j\omega_0 n} \underbrace{e^{j\omega_0 N}}_{=1} \end{aligned}$$

Thus, we must have

$$e^{j\omega_0 N} = 1 = e^{j2m\pi}, \quad m = \text{a positive integer}$$

Comparing the powers of the exponent, we get

$$\begin{aligned} \omega_0 N &= 2m\pi \\ \frac{\omega_0}{2\pi} &= \frac{m}{N} = \text{a rational number} \end{aligned}$$

Thus, $x(n)$ is periodic only if $\omega_0/2\pi$ is a rational number.

1.5.1 Properties of Periodic Signals

Following are the properties of a periodic signal:

1. A periodic signal is an everlasting signal.
2. A periodic signal $x(t)$ can be generated by periodic extension of any segment of $x(t)$ of duration T (the period). As a result we can generate $x(t)$ from any segment of $x(t)$ having a duration of one period by replacing this segment and the reproduction thereof end to end ad infinitum on either side.

3. Area under $x(t)$ over any interval of duration T is the same, i.e., for any real numbers a and b

$$\int_a^{a+T} x(t) dt = \int_b^{b+T} x(t) dt \quad (1.32)$$

This results from the fact that a periodic signal takes the same values at the intervals of T .

4. The sum of M periodic continuous-time signals is not necessarily periodic. It is periodic with period T if and only if the condition

$$\frac{T}{T_i} = n_i, \quad 1 \leq i \leq M \quad (1.33)$$

is satisfied, where T_i is the period of the i th signal in the sum and n_i is an integer. Simultaneous solution for the first signal in the sum paired with each additional signal in the sum gives

$$\frac{T_1}{T_i} = \frac{n_i}{n_1}, \quad 2 \leq i \leq M \quad (1.34)$$

Thus, an equivalent necessary and sufficient condition for a sum of periodic continuous-time signals to be periodic is that all the ratios of the period of the first signal in the sum to the period of another signal in the sum be *rational*, i.e., ratios of two integers.

The following steps can be taken to determine the period of the sum signal, if it is periodic. First convert each ratio from Eq. (1.34) to a ratio of integers. If this conversion is not possible, the sum signal is not periodic. Then factor the greatest common divisor (GCD) from the numerator and denominator of each individual ratio. The least common multiple (LCM) of the denominators of the resulting ratios is the value of n_1 . The period of the sum signal is computed as

$$T = T_1 n_1 \quad (1.35)$$

The preceding development is also applicable to the sum of discrete-time periodic signals if T is replaced by N and each T_i is replaced by N_i .

5. A sum of discrete-time periodic signals is always periodic because the period ratios N_1/N_i are always rational.

Example 1.12 Assume $x_1(t)$ and $x_2(t)$ are periodic signals with periods T_1 and T_2 , respectively. Under what conditions is the sum

$$x(t) = x_1(t) + x_2(t)$$

periodic and what will be the period of $x(t)$ if it is periodic?

Solution

Given that $x_1(t)$ and $x_2(t)$ are periodic with periods T_1 and T_2 , respectively. Thus, $x_1(t)$ and $x_2(t)$ may be written as

$$x_1(t) = x_1(t + T_1) = x_1(t + mT_1)$$

and

$$x_2(t) = x_2(t + T_2) = x_2(t + nT_2)$$

where m and n are integers. Now if T_1 and T_2 are such that

$$T = mT_1 = nT_2$$

Then

$$\begin{aligned} x(t + T) &= x_1(t + T) + x_2(t + T) \\ &= x_1(t + mT_1) + x_2(t + nT_2) \\ &= x_1(t) + x_2(t) \\ x(t + T) &= x(t) \end{aligned}$$

i.e., $x(t)$ is periodic in this case. Therefore, the condition for $x(t)$ to be periodic is

$$\frac{T_1}{T_2} = \frac{n}{m} = \text{a rational number}$$

From the above analysis, it is clear that the smallest common period is the LCM of T_1 and T_2 and is given by the equation

$$mT_1 = nT_2 = T$$

if the integers m and n are relatively prime. On the other hand, if the ratio T_1/T_2 is an irrational number, then the signals $x_1(t)$ and $x_2(t)$ will not have a common period and so $x(t)$ will not be periodic.

Example 1.13 Determine whether the continuous-time signals $x(t)$ and $y(t)$ and the discrete-time signal $z(n)$ are periodic. Compute the period for any that are periodic.

(a) For $x(t)$,

$$x(t) = x_1(t) + x_2(t) + x_3(t)$$

where $x_1(t)$, $x_2(t)$, and $x_3(t)$ have periods of 4, 1.25, and $\sqrt{2}$ s, respectively.

(b) For $y(t)$,

$$y(t) = y_1(t) + y_2(t) + y_3(t)$$

where $y_1(t)$, $y_2(t)$, and $y_3(t)$ have periods of 1.08, 3.6, and 2.025 s, respectively.

(c) For $z(n)$,

$$z(n) = z_1(n) + z_2(n)$$

where $z_1(n)$ and $z_2(n)$ have periods of 90 and 54 s, respectively.

Solution

(a) For $x(t)$,

$$\frac{T_1}{T_2} = \frac{4}{1.25} = \frac{400}{125} = \frac{16}{5} = \text{rational number}$$

$$\frac{T_1}{T_3} = \frac{4}{\sqrt{2}} = \text{irrational number}$$

The signal $x(t)$ is not periodic since T_1/T_3 is not a rational number.

(b) For $y(t)$,

$$\frac{T_1}{T_2} = \frac{1.08}{3.6} = \frac{108}{360} = \frac{3}{10} = \frac{3}{(2)(5)} = \text{rational number}$$

$$\frac{T_1}{T_3} = \frac{1.08}{2.025} = \frac{1080}{2025} = \frac{8}{15} = \frac{8}{(3)(5)} = \text{rational number}$$

The signal $y(t)$ is periodic since the ratios are rational.

$n_1 = \text{LCM of denominators} = (2)(3)(5) = 30$

$$T = T_1 n_1 = (1.08)(30) = 32.4 \text{ s}$$

(c) The discrete-time signal $z(n)$ is periodic since $z_1(n)$ and $z_2(n)$ are periodic.

$$\frac{N_1}{N_2} = \frac{90}{54} = \frac{5}{3} = \text{rational number}$$

$n_1 = \text{LCM of denominators} = 3$

$$N = N_1 n_1 = (90)(3) = 270$$

Example 1.14 Determine whether or not each of the following signals is periodic. If a signal is periodic, specify its fundamental period. (a) $x_1(t) = je^{j10t}$, (b) $x_2(t) = e^{(-1+j)t}$, (c) $x_3(n) = e^{j7\pi n}$, (d) $x_4(n) = 3e^{j3\pi(n+1/2)/5}$, and (e) $x_5(n) = 3e^{j3/5(n+1/2)}$.

Solution

(a) $x_1(t)$ is a periodic complex exponential signal:

$$x_1(t) = je^{j10t} = e^{j(10t+(\pi/2))} = e^{j(\omega_0 t + \phi)}$$

which gives $\omega_0 = 10$. Therefore, the fundamental period of $x_1(t)$ is

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{10} = \frac{\pi}{5}$$

(b) $x_2(t)$ is a complex exponential multiplied by a decaying exponential. Therefore, $x_2(t)$ is not a periodic signal.

$$x_2(t) = e^{(-1+j)t} = e^{j(1+j)t} = e^{-t}e^{jt} = e^{j\omega_0 t}$$

which gives $\omega_0 = 1 + j$, a complex number. The frequency of a signal can never be complex; it must have a real value. So $x_2(t)$ is not periodic.

(c) $x_3(n)$ is a periodic signal:

$$x_3(n) = e^{j7\pi n} = e^{j\omega_0 n}$$

which gives

$$\omega_0 = 7\pi$$

$$2 \times \omega_0 = 7 \times 2\pi$$

$$\frac{\omega_0}{2\pi} = \frac{m}{N} = \frac{7}{2} = \text{rational number}$$

Thus, $x_3(n)$ is periodic with fundamental period $N = 2$.

(d) $x_4(n)$ is periodic:

$$x_4(n) = 3e^{j3\pi(n+1/2)/5} = 3e^{j(3\pi/10)}e^{j(3\pi/5)n} = Ke^{j\omega_0 n}$$

which gives

$$\omega_0 = \frac{3\pi}{5}$$

$$2 \times \omega_0 = \frac{2\pi \times 3}{5}$$

$$\frac{\omega_0}{2\pi} = \frac{m}{N} = \frac{3}{10} = \text{rational number}$$

Thus, $x_4(n)$ is periodic with fundamental period $N = 10$.

(e) $x_5(n)$ is not periodic:

$$x_5(n) = 3e^{j3/5(n+1/2)} = 3e^{j(3/10)}e^{j(3/5)n} = Ke^{j\omega_0 n}$$

which gives

$$\omega_0 = \frac{3}{5}$$

$$\frac{\omega_0}{2\pi} = \frac{3}{10\pi} = \text{irrational number}$$

Since $\omega_0/2\pi$ is an irrational number, $x_5(n)$ is not periodic.

Example 1.15 Determine whether or not each of the following signals is periodic:

(a) $x_1(t) = 2e^{j(t+\pi/4)}u(t)$,

(b) $x_2(n) = u(n) + u(-n)$, and

(c) $x_3(n) = \sum_{k=-\infty}^{\infty} [\delta(n-4k) - \delta(n-1-4k)]$.

Solution

(a) $x_1(t)$ is not a periodic signal because it is zero for $t < 0$, i.e., $x_1(t)$ is not an everlasting signal.

(b) $x_2(n)$, as shown in Fig. 1.16, is not a periodic signal:

$$x_2(n) = u(n) + u(-n)$$

$$x_2(n+N) = u(n+N) + u(-(n+N)), \quad N = \text{an integer}$$

$$= u(n+N) + u(-n-N) \neq x_2(n)$$

$x_2(n)$ is not periodic because $x_2(n) \neq x_2(n+N)$.

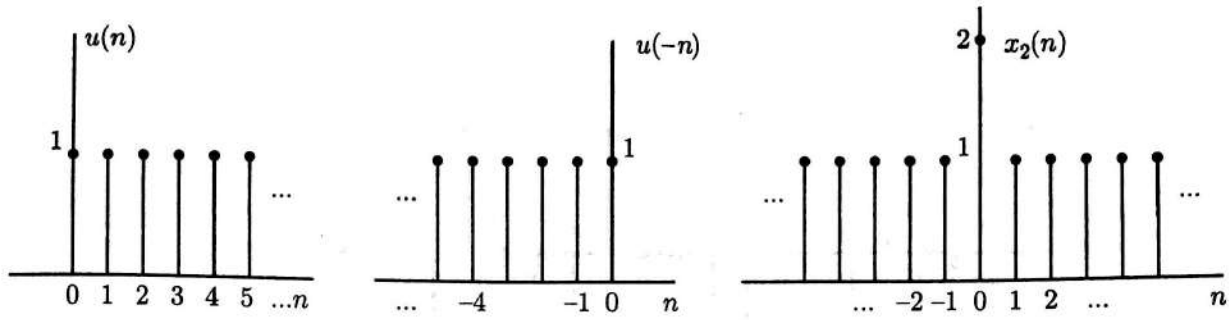


Fig. 1.16 Signal $u(n)$, $u(-n)$, and $x_2(n) = u(n) + u(-n)$

(c) **Method 1:** Given that

$$\begin{aligned}
 x_3(n) &= \sum_{k=-\infty}^{\infty} [\delta(n - 4k) - \delta(n - 1 - 4k)] \\
 x_3(n + N) &= \sum_{k=-\infty}^{\infty} [\delta(n + N - 4k) - \delta(n + N - 1 - 4k)] \\
 &= \sum_{k=-\infty}^{\infty} \left[\delta\left(n - 4\left(k - \frac{N}{4}\right)\right) - \delta\left(n - 1 - 4\left(k - \frac{N}{4}\right)\right) \right]
 \end{aligned}$$

A change of variable is performed by letting $m = k - (N/4)$. Therefore,

$$\begin{aligned}
 x_3(n + N) &= \sum_{m=-\infty}^{\infty} [\delta(n - 4m) - \delta(n - 1 - 4m)] \\
 &= x_3(n)
 \end{aligned}$$

Hence $x_3(n)$ is periodic if $m = k - (N/4)$ is an integer. For m to be an integer, the minimum value of N should be 4.

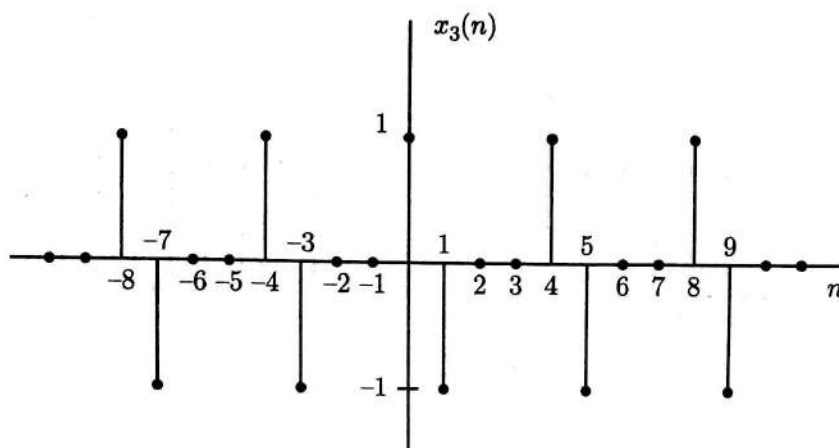
Method 2: $x_3(n)$ is periodic with period $N = 4$.

$$x_3(n) = \sum_{k=-\infty}^{\infty} [\delta(n - 4k) - \delta(n - 1 - 4k)]$$

$$\begin{aligned}
 x_3(n) &= \cdots + \delta(n + 8) - \delta(n + 7) + \delta(n + 4) - \delta(n + 3) + \delta(n) - \delta(n - 1) + \delta(n - 4) \\
 &\quad - \delta(n - 5) + \cdots
 \end{aligned}$$

$$\begin{aligned}
 x_3(n + 4) &= \cdots + \delta(n + 8) - \delta(n + 7) + \delta(n + 4) - \delta(n + 3) + \delta(n) - \delta(n - 1) + \delta(n - 4) \\
 &\quad - \delta(n - 5) + \cdots \\
 &= x_3(n)
 \end{aligned}$$

$x_3(n)$ is periodic with period $N = 4$ (see Fig. 1.17).

Fig. 1.17 Signal $x_3(n)$

1.6 SOME ELEMENTARY SIGNALS

Several elementary signals that occur frequently in applications also serve as a basis for representing other signals. Many of these signals have features that make them particularly useful in solving engineering problems and, therefore, of importance in our subsequent studies.

1.6.1 Real Exponential Signals

- **Continuous-time exponential signal** A real exponential signal, in its most general form, is written as

$$x(t) = Ce^{at} \quad (1.36)$$

where both C and a are real parameters. The parameter C is the amplitude of the exponential signal measured at time $t = 0$. Depending on whether the other parameter a is positive or negative, we may identify two special cases:

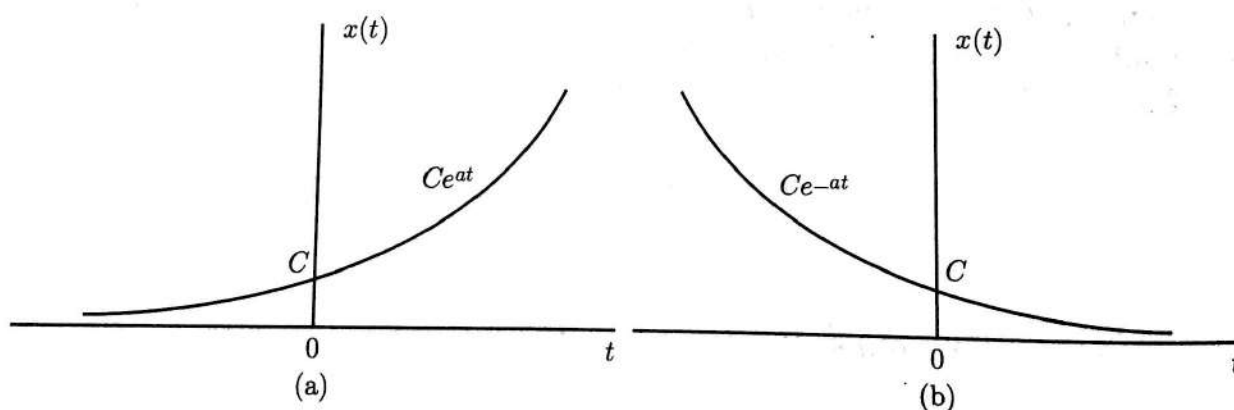


Fig. 1.18 (a) Growing exponential and (b) decaying exponential

Growing exponential If a is positive ($a > 0$), then as t increases $x(t)$ is a growing exponential—a form that is used in describing many different physical processes, including chain reactions in atomic explosions and complex chemical reactions.

Decaying exponential If a is negative ($a < 0$), then $x(t)$ is a decaying exponential—a signal that is also used to describe a wide variety of phenomenon, including the process of radioactive decay and the responses of RC circuits and damped mechanical systems.

The two forms of an exponential signal are illustrated in Fig. 1.18.

- **Discrete-time exponential signal** In discrete time, the complex exponential signal or sequence is defined as

$$x(n) = Cr^n \quad (1.37)$$

If C and r are real, then $x(n)$ is known as real exponential signal.

This could alternately be expressed as

$$x(n) = Ce^{\alpha n} \quad (1.38)$$

where

$$r = e^{\alpha}$$

As illustrated in Fig. 1.19, if $|r| > 1$, the magnitude of the signal grows exponentially with n , while if $|r| < 1$, we have a decaying exponential. Furthermore, if r is positive, all the values of Cr^n are of the same sign, but if r is negative, then the sign of $x(n)$ alternates.

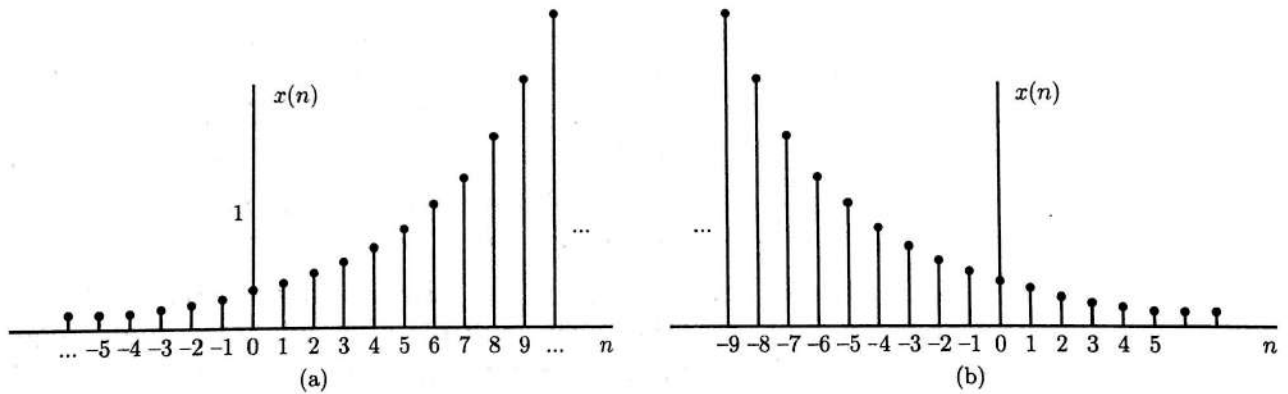


Fig. 1.19 (a) Growing exponential and (b) decaying exponential

1.6.2 Complex Exponential Signals

The mathematical forms of complex exponential signals are the same as those shown in Eqs (1.36) and (1.37), with the following differences: In the continuous-time case, in Eq. (1.36), parameter C or parameter a or both assume complex values. Similarly, in the discrete-time case, in Eq. (1.37), parameter C or parameter r or both assume complex values. Two commonly encountered examples of complex exponential signals are $e^{j\omega t}$ and $e^{j\omega n}$.

Consider

$$x(t) = e^{j\omega_0 t}$$

An important property of this signal is that it is periodic. To verify this, we recall from Eq. (1.30) that $x(t)$ will be periodic with period T if

$$x(t) = x(t + T)$$

$$\begin{aligned}
 e^{j\omega_0 t} &= e^{j\omega_0(t+T)} \\
 &= e^{j\omega_0 t} \underbrace{e^{j\omega_0 T}}_{=1}
 \end{aligned}$$

Thus, it follows that for periodicity, we must have

$$e^{j\omega_0 T} = 1 = e^{j2\pi m} \quad (1.39)$$

where $m = \pm 1, \pm 2, \dots$ is an integer. If $\omega_0 = 0$, then $x(t) = 1$, which is periodic for any value of T . If $\omega_0 \neq 0$, then the fundamental period T_0 of $x(t)$, i.e., the smallest positive value of T for which Eq. (1.39) holds, is

$$T_0 = \frac{2\pi}{|\omega_0|} \quad (1.40)$$

Thus, the signals $e^{j\omega_0 t}$ and $e^{-j\omega_0 t}$ have the same fundamental period. Note that (1) the larger the magnitude of ω_0 , the higher is the rate of oscillation in the signal and (2) $e^{j\omega_0 t}$ is periodic for any value of ω_0 .

Consider the discrete-time exponential signals $e^{j\omega_0 n}$ and $e^{j(\omega_0 + 2\pi)n}$:

$$e^{j(\omega_0 + 2\pi)n} = e^{j2\pi n} e^{j\omega_0 n} = e^{j\omega_0 n} \quad (1.41)$$

From Eq. (1.41), we see that

1. the exponential at frequency $\omega_0 + 2\pi$ is the same as that at frequency ω_0 , whereas the continuous-time exponential signals $e^{j\omega_0 t}$ are all distinct for distinct values of ω_0 . In discrete-time, these signals are not distinct, as the signal with frequency ω_0 is identical to the signals with frequencies $\omega_0 \pm 2\pi$, $\omega_0 \pm 4\pi$, and so on.
2. in considering discrete-time complex exponentials, we need to consider only a frequency interval of length 2π to choose ω_0 from.
3. the low frequency (i.e., slowly varying) discrete-time exponentials have values of ω_0 near 0, 2π , and any other even multiple of π , while the high frequencies (corresponding to rapid variations) are located near $\omega = \pm\pi$ and other odd multiples of π .

Note in particular that for $\omega_0 = \pi$ or any other odd multiple of π ,

$$e^{j\pi n} = (e^{j\pi})^n = (-1)^n \quad (1.42)$$

so that this signal oscillates rapidly, changing sign at each point in time.

Consider the periodicity of discrete-time complex exponential signal $e^{j\omega_0 n}$. In order for the signal $e^{j\omega_0 n}$ to be periodic with period $N > 0$, we must have

$$\begin{aligned}
 e^{j\omega_0 n} &= e^{j\omega_0(n+N)} \\
 &= e^{j\omega_0 n} \underbrace{e^{j\omega_0 N}}_{=1}
 \end{aligned}$$

Thus, it follows that for periodicity, we must have

$$e^{j\omega_0 N} = 1 = e^{j2\pi m}, \quad m = \text{a positive integer}$$

Equating the powers of the exponential signal, we have

$$\omega_0 N = 2\pi m$$

or equivalently,

$$\frac{\omega_0}{2\pi} = \frac{m}{N} = \text{a rational number} \quad (1.43)$$

According to Eq. (1.43), the signal $e^{j\omega_0 n}$ is periodic if $\omega_0/2\pi$ is a rational number and is not periodic otherwise. Or, we can say that for $e^{j\omega_0 n}$ to be periodic, both m and N must be integers.

1.6.3 Signum Function

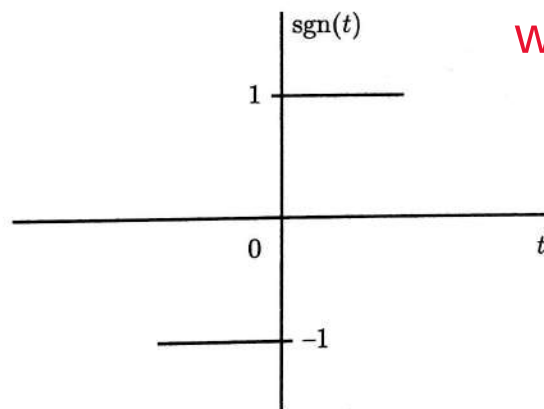
The signum function (written sgn) shown in Fig. 1.20 is defined by

$$\text{sgn}(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases} \quad (1.44)$$

The signum function can be expressed in terms of the unit step function as

$$\text{sgn}(t) = -1 + 2u(t) \quad (1.45)$$

The signum function is one of the most often used signals in communication and control theory.



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Fig. 1.20 Signum function

1.6.4 Sampling Function

A sampling function $\text{Sa}(t)$ is defined as

$$\text{Sa}(t) = \frac{\sin t}{t} \quad (1.46)$$

Since the denominator is an increasing function of t and the numerator is bounded, i.e., $|\sin t| \leq 1$, $\text{Sa}(t)$ is simply a damped sine wave. Figure 1.21(a) shows that $\text{Sa}(t)$ is an even function of t having its peak at $t = 0$ and zero-crossing at $t = \pm n\pi$. The value of the function at $t = 0$ is established by using L'Hôpital's rule. A closely related function

is $\text{sinc } t$, which is defined by

$$\text{sinc } t = \frac{\sin \pi t}{\pi t} = \text{Sa}(\pi t) \quad (1.47)$$

and is shown in Fig. 1.21(b). Note that $\text{sinc } t$ is a compressed version of $\text{Sa}(t)$; the compression factor is π .

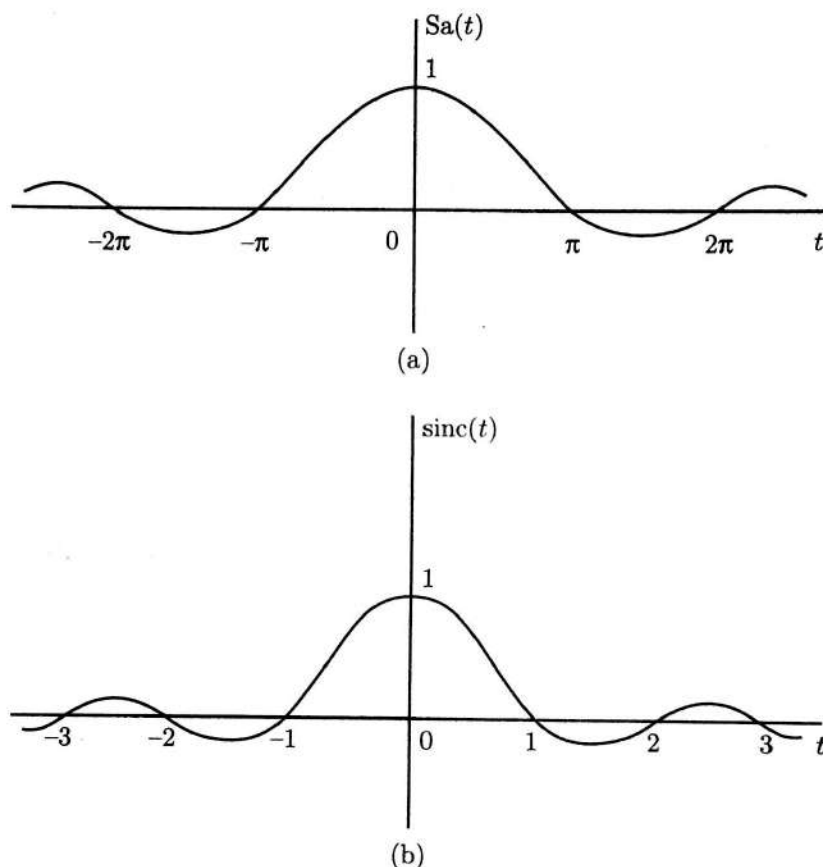


Fig. 1.21 The sampling function

1.7 ENERGY AND POWER SIGNALS

The energy of a continuous-time and a discrete-time signal is defined as follows:

Continuous-time signal

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt \quad (1.48)$$

Discrete-time signal

$$E_x = \sum_{n=-\infty}^{+\infty} |x(n)|^2 \quad (1.49)$$

The power of a continuous-time and a discrete-time signal is defined as follows:

Continuous-time signal

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt \quad (1.50)$$

or, it may be defined as

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt \quad (1.51)$$

Discrete-time signal

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^{+N} |x(n)|^2 \quad (1.52)$$

If $0 < E_x < \infty$, i.e., if E_x is finite and $P_x = 0$, then $x(t)$ [or $x(n)$] is referred to as an *energy* signal. If $E_x = \infty$, but $0 < P_x < \infty$, i.e., P_x is finite, then $x(t)$ [or $x(n)$] is referred to as a *power* signal.

Observe that power is the time average of energy. Since the averaging is over an infinitely large interval, a signal with finite energy has zero power, and a signal with finite power has infinite energy. Therefore, a signal cannot both be an energy signal and a power signal. On the other hand, there are signals that are neither energy nor power signals.

Note: (i) All practical signals have finite energies and are therefore energy signals. It is impossible to generate a true power signal in practice because such a signal has infinite duration and infinite energy.

(ii) All finite periodic signals are power signals; however, not all power signals are periodic, e.g., unit step signal.

Example 1.16 For an energy signal $x(t)$ with energy E_x , show that

- (a) the energy of the signals $x_1(t) = -x(t)$, $x_2(t) = x(-t)$, and $x_3(t) = x(t - T)$ is E_x .
- (b) the energy of $x_1(t) = x(at)$ as well as $x_2(t) = x(at - b)$ is E_x/a .
- (c) the energy of $x_1(t) = ax(t)$ is $a^2 E_x$.

Solution

By definition

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

- (a) (i) The energy contained in the signal $x_1(t) = -x(t)$ is given by

$$E_{x_1} = \int_{-\infty}^{\infty} |x_1(t)|^2 dt = \int_{-\infty}^{\infty} |-x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt = E_x$$

(ii) The energy contained in the signal $x_2(t) = x(-t)$ is given by

$$E_{x_2} = \int_{-\infty}^{\infty} |x_2(t)|^2 dt = \int_{-\infty}^{\infty} |x(-t)|^2 dt$$

A change of variables is performed by letting $-t = \tau$, which also yields $dt = -d\tau$, $\tau = \infty$ as $t = -\infty$, and $\tau = -\infty$ as $t = \infty$. Therefore,

$$E_{x_2} = - \int_{\infty}^{-\infty} |x(\tau)|^2 d\tau = \int_{-\infty}^{\infty} |x(\tau)|^2 d\tau = E_x$$

(iii) The energy contained in the signal $x_3(t) = x(t - T)$ is given by

$$E_{x_3} = \int_{-\infty}^{\infty} |x_3(t)|^2 dt = \int_{-\infty}^{\infty} |x(t - T)|^2 dt$$

A change of variables is performed by letting $(t - T) = \tau$, which also yields $dt = d\tau$, $\tau = -\infty$ as $t = -\infty$, and $\tau = \infty$ as $t = \infty$. Therefore,

$$E_{x_3} = \int_{-\infty}^{\infty} |x(\tau)|^2 d\tau = E_x$$

(b) (i) The energy contained in the signal $x_1(t) = x(at)$ is given by

$$E_{x_1} = \int_{-\infty}^{\infty} |x_1(t)|^2 dt = \int_{-\infty}^{\infty} |x(at)|^2 dt$$

A change of variables is performed by letting $at = \tau$, which also yields $dt = (1/a)d\tau$, $\tau = -\infty$ as $t = -\infty$, and $\tau = \infty$ as $t = \infty$. Therefore,

$$E_{x_1} = \frac{1}{a} \int_{-\infty}^{\infty} |x(\tau)|^2 d\tau = \frac{E_x}{a}$$

(ii) The energy contained in the signal $x_2(t) = x(at - b)$ is given by

$$E_{x_2} = \int_{-\infty}^{\infty} |x_2(t)|^2 dt = \int_{-\infty}^{\infty} |x(at - b)|^2 dt$$

A change of variables is performed by letting $(at - b) = \tau$, which also yields $dt = (1/a)d\tau$, $\tau = -\infty$ as $t = -\infty$, and $\tau = \infty$ as $t = \infty$. Therefore,

$$E_{x_2} = \frac{1}{a} \int_{-\infty}^{\infty} |x(\tau)|^2 d\tau = \frac{E_x}{a}$$

(c) The energy contained in the signal $x_1(t) = ax(t)$ is given by

$$E_{x_1} = \int_{-\infty}^{\infty} |x_1(t)|^2 dt = \int_{-\infty}^{\infty} |ax(t)|^2 dt = a^2 \int_{-\infty}^{\infty} |x(t)|^2 dt = a^2 E_x$$

Note: The time inversion and time shifting does not affect signal energy. On the other hand, time compression of a signal ($a > 1$) reduces the energy and time expansion of a signal ($a < 1$) increases the energy.

Example 1.17 Show that all finite periodic signals are power signals. Or, show that if $x(t)$ is periodic with fundamental period T , then the power P_x defined by Eq. (1.50) is the same as the average power $x(t)$ over any interval of length T , i.e.,

$$P_x = \frac{1}{T} \int_0^T |x(t)|^2 dt$$

Solution

Method 1: Power contained in a signal $x(t)$ is defined as

$$P_x = \lim_{n \rightarrow \infty} \frac{1}{2nT} \int_{-nT}^{nT} |x(t)|^2 dt$$

If $x(t)$ is periodic with period T , then the integral in the above equation is the same over any interval of length T . Therefore,

$$\begin{aligned} P_x &= \lim_{n \rightarrow \infty} \frac{1}{2nT} \left[\int_{-nT}^{(-n+1)T} |x(t)|^2 dt + \cdots + \int_{-T}^0 |x(t)|^2 dt + \int_0^T |x(t)|^2 dt + \cdots \right. \\ &\quad \left. + \int_{(n-1)T}^{nT} |x(t)|^2 dt \right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{2nT} \left[\sum_{k=-n}^{n-1} \int_{kT}^{(k+1)T} |x(t)|^2 dt \right] \\ P_x &= \lim_{n \rightarrow \infty} \frac{1}{2nT} \left[\sum_{k=-n}^{n-1} \int_0^T |x(t+kT)|^2 dt \right] \end{aligned}$$

Since $x(t)$ is periodic, i.e., $x(t) = x(t + kT)$, we get

$$\begin{aligned}
 P_x &= \lim_{n \rightarrow \infty} \frac{1}{2nT} \left[\sum_{k=-n}^{n-1} \int_0^T |x(t)|^2 dt \right] \\
 &= \lim_{n \rightarrow \infty} \frac{1}{2nT} \left[2n \int_0^T |x(t)|^2 dt \right] \\
 P_x &= \frac{1}{T} \int_0^T |x(t)|^2 dt < \infty
 \end{aligned}$$

Since the power contained in a periodic signal is finite, all finite periodic signals are power signals.

Method 2: The signal energy in one period of a periodic signal with period T is

$$E_{x1} = \int_0^T |x(t)|^2 dt$$

Since all periods are alike, the energy in n periods is

$$E_{xn} = n \int_0^T |x(t)|^2 dt$$

The average signal power for all time (i.e., over all periods) is the signal power

$$P_x = \lim_{n \rightarrow \infty} \frac{E_{xn}}{nT} = \frac{1}{T} \int_0^T |x(t)|^2 dt$$

We see that the signal power of a periodic signal is equal to the average signal power in any period of the signal. If the signal energy in one period of a periodic signal is finite, then the total energy of the periodic signal is infinite, because it contains an infinite number of periods, and the signal power of the periodic signal is finite. Therefore, the signal is a power signal.

Example 1.18 Determine the values of P_x and E_x for each of the following signals:
 (a) $x_1(t) = e^{-2t}u(t)$, (b) $x_2(t) = e^{j(2t+\pi/4)}$, (c) $x_3(t) = A \cos(t)$, (d) $x_1(n) = (1/2)^n u(n)$,
 (e) $x_2(n) = e^{j((\pi/2)n+\pi/8)}$, and (f) $x_3(n) = \cos((\pi/4)n)$.

Solution

(a) Given that

$$x_1(t) = e^{-2t}u(t)$$

$$|x_1(t)|^2 = e^{-4t}u(t)$$

Therefore,

$$E_x = \int_{-\infty}^{\infty} |x_1(t)|^2 dt = \int_0^{\infty} e^{-4t} dt = \frac{1}{4}$$

$$\begin{aligned} P_x &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x_1(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} e^{-4t} u(t) dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{T/2} e^{-4t} dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \left. \frac{e^{-4t}}{-4} \right|_0^{T/2} = \lim_{T \rightarrow \infty} \frac{1}{T} [e^{-4T} - 1] = 0 \end{aligned}$$

Since $P_x = 0$ and $0 < E_x < \infty$, $x_1(t)$ is an energy signal.

(b) Given that

$$x_2(t) = e^{j(2t+\pi/4)}$$

$$|x_2(t)|^2 = 1$$

Therefore,

$$E_x = \int_{-\infty}^{\infty} (1) dt = \infty$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x_2(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} dt = \lim_{T \rightarrow \infty} \frac{1}{T} T = 1$$

Since $E_x = \infty$ and $0 < P_x < \infty$, $x_2(t)$ is a power signal.

(c) Given that

$$x_3(t) = A \cos(t)$$

$$|x_3(t)|^2 = A^2 \cos^2(t)$$

Therefore,

$$E_x = \int_{-\infty}^{\infty} A^2 \cos^2(t) dt = \int_{-\infty}^{\infty} \frac{A^2}{2} [1 + \cos(2t)] dt = \infty$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} A^2 \cos^2(t) dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \frac{A^2}{2} [1 + \cos(2t)] dt = \frac{A^2}{2} + 0 = \frac{A^2}{2}$$

Since $E_x = \infty$ and $0 < P_x < \infty$, $x_3(t)$ is a power signal.

(d) Given that

$$x_1(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$|x_1(n)|^2 = \left(\frac{1}{4}\right)^n u(n)$$

Therefore,

$$E_x = \sum_{n=-\infty}^{\infty} |x_1(n)|^2 = \sum_{n=-\infty}^{\infty} \left(\frac{1}{4}\right)^n u(n) = \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n = \frac{1}{1 - (1/4)} = \frac{4}{3}$$

$$\begin{aligned} P_x &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x_1(n)|^2 = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N \left(\frac{1}{4}\right)^n u(n) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N \left(\frac{1}{4}\right)^n \\ &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \left(\frac{1 - (1/4)^{N+1}}{1 - (1/4)} \right) = 0 \end{aligned}$$

Since $0 < E_x < \infty$ and $P_x = 0$, $x_1(n)$ is an energy signal.

(e) Given that

$$x_2(n) = e^{j((\pi/2)n + \pi/8)}$$

$$|x_2(n)|^2 = 1$$

Therefore,

$$E_x = \sum_{n=-\infty}^{\infty} |x_2(n)|^2 = \sum_{n=-\infty}^{\infty} 1 = \infty$$

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x_2(n)|^2 = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N 1 = \lim_{N \rightarrow \infty} \frac{1}{2N+1} 2N+1 = 1$$

Since $E_x = \infty$ and $0 < P_x < \infty$, $x_2(n)$ is a power signal.

(f) Given that

$$x_3(n) = \cos\left(\frac{\pi}{4}n\right)$$

$$|x_3(n)|^2 = \cos^2\left(\frac{\pi}{4}n\right) = \frac{1 + \cos((\pi/2)n)}{2}$$

Therefore,

$$E_x = \sum_{n=-\infty}^{\infty} |x_3(n)|^2 = \sum_{n=-\infty}^{\infty} \frac{1 + \cos((\pi/2)n)}{2} = \sum_{n=-\infty}^{\infty} \frac{1}{2} + \sum_{n=-\infty}^{\infty} \frac{\cos((\pi/2)n)}{2} = \infty$$

$$\begin{aligned} P_x &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x_3(n)|^2 = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N \frac{1 + \cos(\pi/2)n}{2} \\ &= \left(\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N \frac{1}{2} \right) + \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N \frac{1}{2} \cos(\pi/2)n \\ &= \left(\lim_{N \rightarrow \infty} \frac{1}{2} \frac{1}{(2N+1)} (2N+1) \right) + 0 = \frac{1}{2} \end{aligned}$$

Since $E_x = \infty$ and $0 < P_x < \infty$, $x_3(n)$ is a power signal.

Example 1.19 Determine whether the following signals are power or energy signals or neither.

- (a) $x(t) = A \sin t$, $-\infty < t < \infty$
- (b) $x(t) = A[u(t+a) - u(t-a)]$, $a > 0$
- (c) $x(t) = e^{-a|t|}$, $a > 0$
- (d) $x(t) = u(t)$
- (e) $x(t) = r(t) = tu(t)$

Solution

(a) The given signal $x(t) = A \sin t$ is a periodic signal with period $T = 2\pi$ and hence it is a power signal:

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} A^2 \sin^2 t dt = \frac{A^2}{2\pi} \int_{-\pi}^{\pi} \frac{1}{2} (1 - \cos 2t) dt = \frac{A^2}{2} < \infty$$

(b) The given signal $x(t) = A[u(t+a) - u(t-a)]$, where $a > 0$, is a finite duration signal and hence it is an energy signal:

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-a}^a A^2 dt = 2aA^2$$

(c) Consider the given signal

$$x(t) = e^{-a|t|} = \begin{cases} e^{-at}, & t > 0 \\ e^{at}, & t < 0 \end{cases}$$

The energy contained in the signal $x(t)$ is given by

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} e^{-2a|t|} dt = \int_{-\infty}^0 e^{2at} dt + \int_0^{\infty} e^{-2at} dt = \frac{1}{a} < \infty$$

Thus, $x(t)$ is an energy signal.

(d) Consider the given signal

$$x(t) = u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

The energy contained in the signal $x(t)$ is given by

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |u(t)|^2 dt = \int_{-\infty}^{\infty} 1^2 dt = \infty$$

The power contained in the signal $x(t)$ is given by

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |u(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{T/2} 1^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \frac{T}{2} = \frac{1}{2}$$

Thus, the unit step function $u(t)$ is a power signal.

(e) Consider the given signal

$$x(t) = r(t) = tu(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

The energy contained in the signal $x(t)$ is given by

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^{\infty} t^2 dt = \infty$$

The power contained in the signal $x(t) = r(t)$ is given by

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{T/2} t^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \frac{(T/2)^3}{3} = \lim_{T \rightarrow \infty} \frac{T^2}{24} = \infty$$

Thus, $x(t) = r(t)$, i.e., ramp signal is neither an energy signal nor a power signal.

Example 1.20 Consider the continuous-time signal

$$x(t) = \delta(t + 2) - \delta(t - 2)$$

Calculate the value of E_y for the following signal:

$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

Solution

Given that

$$\begin{aligned} y(t) &= \int_{-\infty}^t x(\tau) d\tau \\ &= \int_{-\infty}^t (\delta(\tau + 2) - \delta(\tau - 2)) d\tau \\ &= \int_{-\infty}^t \delta(\tau + 2) d\tau - \int_{-\infty}^t \delta(\tau - 2) d\tau \\ &= u(t + 2) - u(t - 2) = \begin{cases} 1, & -2 < t < 2 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

Therefore,

$$E_y = \int_{-\infty}^{\infty} |y(t)|^2 dt = \int_{-2}^2 1 dt = 4$$

1.8 EVEN AND ODD SIGNALS

A signal $x(t)$ or $x(n)$ is referred to as an *even* signal if it is identical with its reflection about the origin. In continuous time a signal is even if

$$x(t) = x(-t) \quad (1.53)$$

while a discrete-time signal is even if

$$x(n) = x(-n) \quad (1.54)$$

An even signal has the same value at instants t (or n) and $-t$ (or $-n$) for all values of t (or n). Clearly, an even signal is symmetrical about the vertical axis.

A signal is referred to as odd if

$$x(t) = -x(-t) \quad (1.55)$$

or

$$x(n) = -x(-n) \quad (1.56)$$

The value of an odd signal at the instant t (or n) is the negative of its value at the instant $-t$ (or $-n$). Therefore, an odd signal is antisymmetric about the vertical axis. An odd signal must necessarily be 0 at $t = 0$ or $n = 0$ since Eqs (1.55) and (1.56) require that $x(0) = -x(0)$.

1.8.1 Even and Odd Components of a Signal

Every signal can be expressed as a sum of even and odd components.

Continuous-time signal

$$x(t) = \mathcal{E}\{x(t)\} + \mathcal{O}\{x(t)\} = x_e(t) + x_o(t) \quad (1.57)$$

where $\mathcal{E}\{x(t)\}$ and $\mathcal{O}\{x(t)\}$ are an even and odd components of $x(t)$, respectively.

Similarly, for **discrete-time signal**

$$x(n) = \mathcal{E}\{x(n)\} + \mathcal{O}\{x(n)\} = x_e(n) + x_o(n) \quad (1.58)$$

The even part of $x(t)$ is given by

$$\mathcal{E}\{x(t)\} = x_e(t) = \frac{x(t) + x(-t)}{2} \quad (1.59)$$

The even part of $x(n)$ is given by

$$\mathcal{E}\{x(n)\} = x_e(n) = \frac{x(n) + x(-n)}{2} \quad (1.60)$$

The odd part of $x(t)$ is given by

$$\mathcal{O}\{x(t)\} = x_o(t) = \frac{x(t) - x(-t)}{2} \quad (1.61)$$

The odd part of $x(n)$ is given by

$$\mathcal{O}\{x(n)\} = x_o(n) = \frac{x(n) - x(-n)}{2} \quad (1.62)$$

1.8.2 Properties of Continuous-time Even and Odd Signals

Following are the properties of continuous-time even and odd signals:

1. If $x_1(t)$ is an odd signal $[x_1(t) = -x_1(-t)]$ and $x_2(t)$ is an even signal $[x_2(t) = x_2(-t)]$, then

(a) $x_3(t) = x_1(t)x_2(t)$ is an odd signal, i.e.,

even signal $\times$ odd signal = odd signal

Proof Consider the signal $x_3(t)$:

$$x_3(t) = x_1(t)x_2(t)$$

$$x_3(-t) = x_1(-t)x_2(-t) = -x_1(t)x_2(t) = -x_3(t) \quad \blacksquare$$

(b) $x_4(t) = x_1(t)x_1(t)$ is an even signal, i.e.,

odd signal $\times$ odd signal = even signal

Proof Consider the signal $x_4(t)$:

$$x_4(t) = x_1(t)x_1(t)$$

$$x_4(-t) = x_1(-t)x_1(-t) = -x_1(t) \times -x_1(t) = x_1(t)x_1(t) = x_4(t) \quad \blacksquare$$

(c) $x_5(t) = x_2(t)x_2(t)$ is an even signal, i.e.,

even signal $\times$ even signal = even signal

Proof Consider the signal $x_5(t)$:

$$x_5(t) = x_2(t)x_2(t)$$

$$x_5(-t) = x_2(-t)x_2(-t) = x_2(t)x_2(t) = x_5(t) \quad \blacksquare$$

2. If $x(t)$ is an even signal, then

$$\int_{-\infty}^{\infty} x(t)dt = 2 \int_0^{\infty} x(t)dt \quad (1.63)$$

Proof Consider the LHS of Eq. (1.63):

$$\begin{aligned} \int_{-\infty}^{\infty} x(t)dt &= \int_{-\infty}^0 x(t)dt + \int_0^{\infty} x(t)dt \\ &= \int_0^{\infty} x(-t)dt + \int_0^{\infty} x(t)dt \end{aligned}$$

Since $x(t)$ is an even signal, i.e., $[x(t) = x(-t)]$, we have

$$\int_{-\infty}^{\infty} x(t)dt = \int_0^{\infty} x(t)dt + \int_0^{\infty} x(t)dt = 2 \int_0^{\infty} x(t)dt \quad \blacksquare$$

3. If $x(t)$ is an odd signal, then

$$\int_{-\infty}^{\infty} x(t) dt = 0 \quad (1.64)$$

Proof Consider the LHS of Eq. (1.64):

$$\begin{aligned} \int_{-\infty}^{\infty} x(t) dt &= \int_{-\infty}^0 x(t) dt + \int_0^{\infty} x(t) dt \\ &= \int_0^{\infty} x(-t) dt + \int_0^{\infty} x(t) dt \end{aligned}$$

Since $x(t)$ is an odd signal, i.e., $[x(t) = -x(-t)]$, we have

$$\int_{-\infty}^{\infty} x(t) dt = - \int_0^{\infty} x(t) dt + \int_0^{\infty} x(t) dt = 0$$

4. If $x(t)$ is an even signal, then its derivative is an odd signal.

Proof Since $x(t)$ is an even signal, we have

$$\begin{aligned} x(t) &= x(-t) \\ \frac{dx(t)}{dt} &= - \frac{dx(-t)}{dt} \end{aligned}$$

5. If $x(t)$ is an odd signal, then its derivative is an even signal.

Proof Since $x(t)$ is an odd signal, we have

$$\begin{aligned} x(t) &= -x(-t) \\ \frac{dx(t)}{dt} &= \frac{dx(-t)}{dt} \end{aligned}$$

6. If $x(t)$ is an arbitrary signal with its even and odd parts denoted by $x_e(t) = \mathcal{E}\{x(t)\}$ and $x_o(t) = \mathcal{O}\{x(t)\}$, respectively, then

$$\int_{-\infty}^{\infty} x^2(t) dt = \int_{-\infty}^{\infty} x_e^2(t) dt + \int_{-\infty}^{\infty} x_o^2(t) dt \quad (1.65)$$

Proof Consider the LHS of Eq. (1.65),

$$\begin{aligned}
 \int_{-\infty}^{\infty} x^2(t) dt &= \int_{-\infty}^{\infty} [x_e(t) + x_o(t)]^2 dt \\
 &= \int_{-\infty}^{\infty} x_e^2(t) dt + \int_{-\infty}^{\infty} x_o^2(t) dt + 2 \int_{-\infty}^{\infty} x_e(t) x_o(t) dt \\
 &= \int_{-\infty}^{\infty} x_e^2(t) dt + \int_{-\infty}^{\infty} x_o^2(t) dt + 2 \left[\int_{-\infty}^0 x_e(t) x_o(t) dt + \int_0^{\infty} x_e(t) x_o(t) dt \right] \\
 &= \int_{-\infty}^{\infty} x_e^2(t) dt + \int_{-\infty}^{\infty} x_o^2(t) dt + 2 \left[\int_0^{\infty} x_e(-t) x_o(-t) dt + \int_0^{\infty} x_e(t) x_o(t) dt \right] \\
 &= \int_{-\infty}^{\infty} x_e^2(t) dt + \int_{-\infty}^{\infty} x_o^2(t) dt + 2 \left[- \int_0^{\infty} x_e(t) x_o(t) dt + \int_0^{\infty} x_e(t) x_o(t) dt \right] \\
 \int_{-\infty}^{\infty} x^2(t) dt &= \int_{-\infty}^{\infty} x_e^2(t) dt + \int_{-\infty}^{\infty} x_o^2(t) dt + 0 = \int_{-\infty}^{\infty} x_e^2(t) dt + \int_{-\infty}^{\infty} x_o^2(t) dt
 \end{aligned}$$

1.8.3 Properties of Discrete-time Even and Odd Signals

Following are the properties of discrete-time even and odd signals

1. Although part (a) in the previous section has been stated in terms of continuous-time signal, the analogous property is also valid in discrete-time.
2. If $x(n)$ is an even signal, then

$$\sum_{n=-\infty}^{\infty} x(n) = x(0) + 2 \sum_{n=1}^{\infty} x(n) \quad (1.66)$$

Proof Consider the LHS of Eq. (1.66)

$$\begin{aligned}
 \sum_{n=-\infty}^{\infty} x(n) &= \sum_{n=-\infty}^{-1} x(n) + \sum_{n=0}^{\infty} x(n) \\
 &= \sum_{n=1}^{\infty} x(-n) + \sum_{n=0}^{\infty} x(n)
 \end{aligned}$$

Since $x(n)$ is an even signal, i.e., $[x(n) = x(-n)]$, we have

$$\sum_{n=-\infty}^{\infty} x(n) = \sum_{n=1}^{\infty} x(n) + x(0) + \sum_{n=1}^{\infty} x(n) = x(0) + 2 \sum_{n=1}^{\infty} x(n)$$

3. If $x(n)$ is an odd signal, then

$$\sum_{n=-\infty}^{\infty} x(n) = 0 \quad (1.67)$$

Proof Consider the LHS of Eq. (1.67)

$$\begin{aligned} \sum_{n=-\infty}^{\infty} x(n) &= \sum_{n=-\infty}^{-1} x(n) + \sum_{n=0}^{\infty} x(n) \\ &= \sum_{n=1}^{\infty} x(-n) + \sum_{n=0}^{\infty} x(n) \end{aligned}$$

Since $x(n)$ is an odd signal, i.e., $[x(n) = -x(-n)]$, $x(0) = 0$, we have

$$\sum_{n=-\infty}^{\infty} x(n) = -\sum_{n=1}^{\infty} x(n) + x(0) + \sum_{n=1}^{\infty} x(n) = 0$$

4. If $x(n)$ is an arbitrary signal with its even and odd parts denoted by $x_e(n) = \mathcal{E}\{x(n)\}$ and $x_o(n) = \mathcal{O}\{x(n)\}$, respectively, then

$$\sum_{n=-\infty}^{\infty} x^2(n) = \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_o^2(n) \quad (1.68)$$

Proof Consider the LHS of Eq. (1.68), and the fact that $x(n) = x_e(n) + x_o(n)$, we have

$$\begin{aligned} \sum_{n=-\infty}^{\infty} x^2(n) &= \sum_{n=-\infty}^{\infty} [x_e(n) + x_o(n)]^2 \\ &= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_o^2(n) + 2 \sum_{n=-\infty}^{\infty} x_e(n)x_o(n) \\ &= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_o^2(n) + 2 \left[\sum_{n=-\infty}^{-1} x_e(n)x_o(n) + \sum_{n=0}^{\infty} x_e(n)x_o(n) \right] \\ &= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_o^2(n) + 2 \left[\sum_{n=1}^{\infty} x_e(-n)x_o(-n) + \sum_{n=0}^{\infty} x_e(n)x_o(n) \right] \end{aligned}$$

Since $x_e(n) = x_e(-n)$, $x_o(n) = -x_o(-n)$, and $x_o(0) = 0$, we have

$$\begin{aligned} \sum_{n=-\infty}^{\infty} x^2(n) &= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_o^2(n) + 2 \left[-\sum_{n=1}^{\infty} x_e(n)x_o(n) + x_e(0)x_o(0) + \sum_{n=1}^{\infty} x_e(n)x_o(n) \right] \\ &= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_o^2(n) + 0 = \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_o^2(n) \end{aligned}$$

Example 1.21 Verify Eqs (1.59) and (1.61).

Solution

From Eq. (1.57), we have

$$x(t) = x_e(t) + x_o(t)$$

$$x(-t) = x_e(-t) + x_o(-t) = x_e(t) - x_o(-t)$$

Thus, the addition of the above two equations gives Eq. (1.59), i.e.,

$$x(t) + x(-t) = 2x_e(t)$$

$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

and the subtraction gives Eq. (1.61), i.e.,

$$x(t) - x(-t) = 2x_o(t)$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

Example 1.22 Determine the even and odd parts of the unit step signal $u(n)$.

Solution

The unit step signal is defined as

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

The even and odd parts of $u(n)$ are

$$\mathcal{E}\{u(n)\} = \frac{u(n) + u(-n)}{2} \quad \text{and} \quad \mathcal{O}\{u(n)\} = \frac{u(n) - u(-n)}{2}$$

The even and odd parts of $u(n)$ as shown in Fig. 1.22 can be determined as follows:

n	...	-3	-2	-1	0	1	2	3	...
$u(n)$	0	0	0	0	1	1	1	1	1
$u(-n)$	1	1	1	1	1	0	0	0	0
$\mathcal{E}\{u(n)\}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\mathcal{O}\{u(n)\}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

Clearly, the even part is

$$\mathcal{E}\{u(n)\} = u_e(n) = \begin{cases} \frac{1}{2}, & n > 0 \\ 1, & n = 0 \\ \frac{1}{2}, & n < 0 \end{cases}$$

and the odd part is

$$\mathcal{O}\{u(n)\} = u_o(n) = \begin{cases} \frac{1}{2}, & n > 0 \\ 0, & n = 0 \\ -\frac{1}{2}, & n < 0 \end{cases}$$

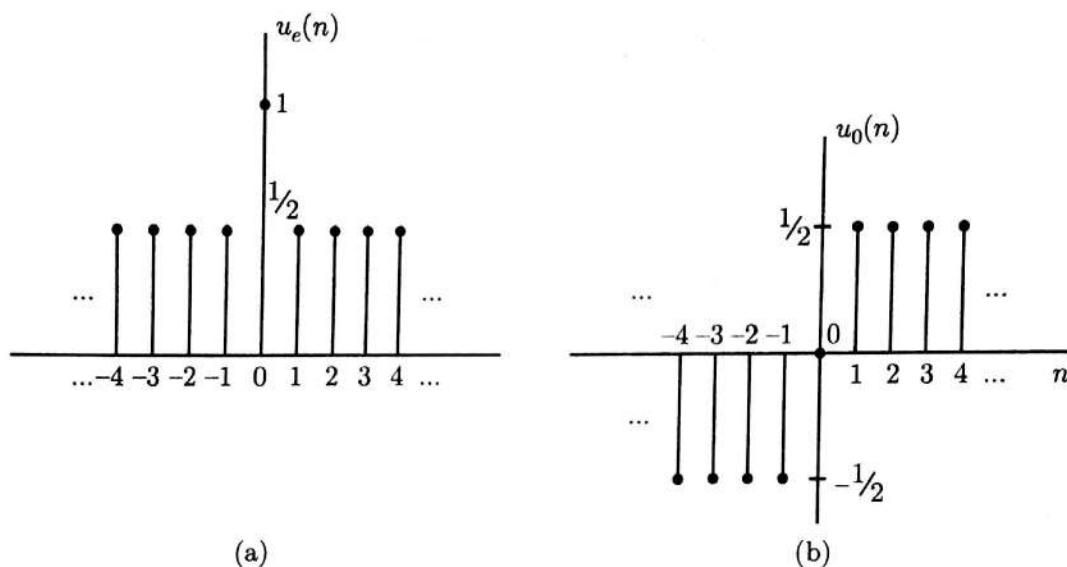


Fig. 1.22 (a) Even and (b) odd parts of $u(n)$

Example 1.23 Find the even and odd components of the signal $x(t) = e^{-2t} \cos(t)$.

Solution

Replacing t with $-t$ in the expression for $x(t)$ yields

$$x(-t) = e^{2t} \cos(-t) = e^{2t} \cos(t)$$

Hence, by applying Eqs (1.59) and (1.61) to the problem at hand, we get

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)] = \frac{1}{2}[e^{-2t} \cos t + e^{2t} \cos t] = \cos t \left[\frac{e^{-2t} + e^{2t}}{2} \right] = \cos t \cosh(2t)$$

and

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)] = \frac{1}{2}[e^{-2t} \cos t - e^{2t} \cos t] = \cos t \left[\frac{e^{-2t} - e^{2t}}{2} \right] = -\cos t \sinh(2t)$$

where $\cosh(2t)$ and $\sinh(2t)$, respectively, denote the hyperbolic cosine and sine of time t .

Example 1.24 Find the even and odd components of the following signals:

$$(a) x(t) = u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases} \quad \text{and} \quad (b) x(t) = \begin{cases} Ae^{-\alpha t}, & t > 0 \\ 0, & t < 0 \end{cases}$$

Solution

(a) Replacing t with $-t$ in the expression for $x(t)$ yields

$$x(-t) = \begin{cases} 1, & -t > 0 \rightarrow t < 0 \\ 0, & -t < 0 \rightarrow t > 0 \end{cases}$$

Hence, by applying Eqs (1.59) and (1.61) to the problem at hand, we get

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)] = \frac{1}{2}, \quad \text{all } t \text{ except } t = 0$$

and

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)] = \begin{cases} \frac{1}{2}, & t > 0 \\ -\frac{1}{2}, & t < 0 \end{cases}$$

The only problem here is the value of these functions at $t = 0$. If we define $x(0) = 1/2$ (the definition here is consistent with our definition of the signal at a point of discontinuity), then

$$x_e(0) = \frac{1}{2} \quad \text{and} \quad x_o(0) = 0$$

Signals $x(t) = u(t)$, for $x_e(t)$ and $x_o(t)$, are plotted in Fig. 1.23.

(b) Replacing t with $-t$ in the expression for $x(t)$ yields

$$x(-t) = \begin{cases} Ae^{\alpha t}, & -t > 0 \rightarrow t < 0 \\ 0, & -t < 0 \rightarrow t > 0 \end{cases}$$

Hence, by applying Eqs (1.59) and (1.61) to the problem at hand, we get

$$\begin{aligned} x_e(t) &= \frac{1}{2}[x(t) + x(-t)] = \begin{cases} \frac{1}{2}Ae^{-\alpha t}, & t > 0 \\ \frac{1}{2}Ae^{\alpha t}, & t < 0 \end{cases} \\ &= \frac{1}{2}Ae^{-\alpha|t|} \end{aligned}$$

and

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)] = \begin{cases} \frac{1}{2}Ae^{-\alpha t}, & t > 0 \\ -\frac{1}{2}Ae^{\alpha t}, & t < 0 \end{cases}$$

Signals $x(t)$, for $x_e(t)$ and $x_o(t)$, are plotted in Fig. 1.24.

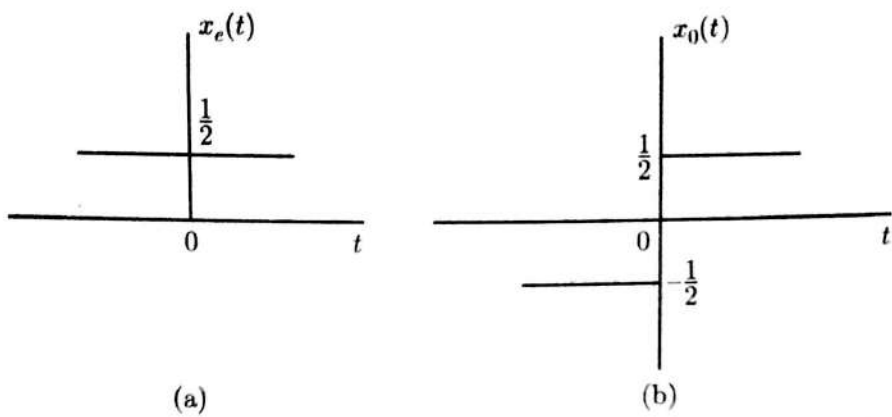


Fig. 1.23 (a) Even and (b) odd parts of $u(t)$

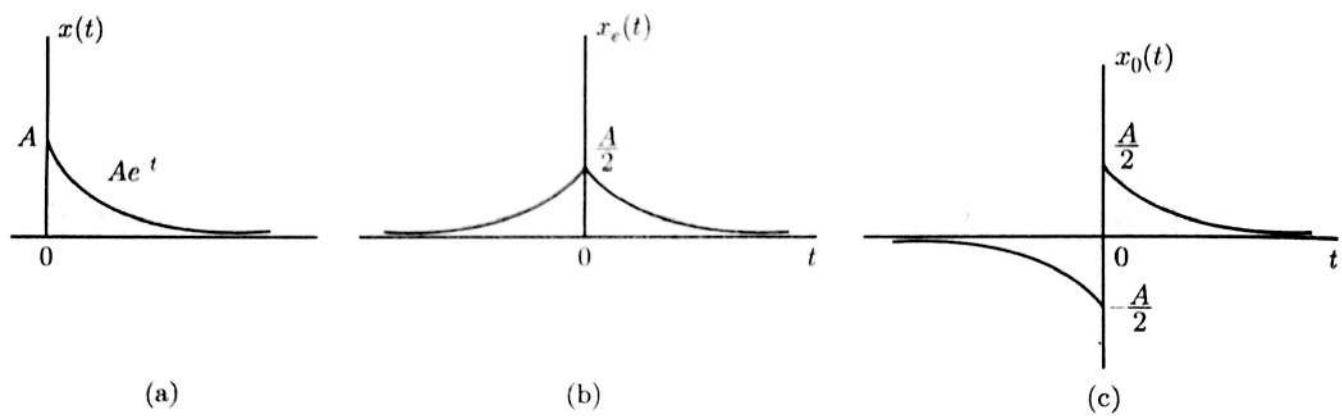


Fig. 1.24 (a) The signal $Ae^{-\alpha t}$, (b) its even part, and (c) its odd part

1.9 CAUSAL, ANTICAUSAL, AND NONCAUSAL SIGNALS

A signal that does not start before $t = 0$ is a causal signal. In other words, $x(t)$ is a causal signal if

$$x(t) = 0, \quad t < 0 \tag{1.69}$$

A signal that starts before $t = 0$ is a noncausal signal. The signal is anticausal if

$$x(t) = 0, \quad t \geq 0 \tag{1.70}$$

Any signal that does not contain any singularities (a delta function or its derivatives) at $t = 0$ can be written as the sum of a causal part $x^+(t)$ and anticausal part $x^-(t)$, i.e.,

$$x(t) = x^+(t) + x^-(t) \tag{1.71}$$

For example, the exponential $x(t) = e^{-at}$ can be written as

$$x(t) = e^{-at}u(t) + e^{-at}u(-t)$$

where the first term represents causal part of $x(t)$ and the second term represents the anticausal part of $x(t)$. Note that multiplying the signal by the unit step ensures that the resulting signal is causal.

1.10 CONTINUOUS-TIME AND DISCRETE-TIME SYSTEMS

Every physical system is broadly characterized by its ability to accept an input such as voltage, current, force, pressure, displacement, etc., and to produce an output in response to this input. For example, a radar receiver is an electronic system whose input is the reflection of an electromagnetic signal from the target and whose output is a video signal displayed on the radar screen. Similarly, a robot is a system whose input is an electric control signal and whose output is a motion or action on the part of the robot. A third example is a filter whose input is a signal corrupted by noise and interference and whose output is the desired signal. In brief, a system can be viewed as a process that results in transforming input signals into output signals.

We are interested in both continuous-time and discrete-time systems. A continuous-time system is one in which continuous-time input signals are transformed into continuous-time output signals. Such a system is represented pictorially as shown in Fig. 1.25(a), where $x(t)$ is the input and $y(t)$ is the output. Alternatively, we will often represent the input-output relation of a continuous-time system by the notation

$$x(t) \rightarrow y(t) \quad (1.72)$$

Similarly, a discrete-time system, i.e., a system that transforms discrete-time inputs into discrete-time outputs, is depicted in Fig. 1.25(b) and will sometimes be represented symbolically as

$$x(n) \rightarrow y(n) \quad (1.73)$$

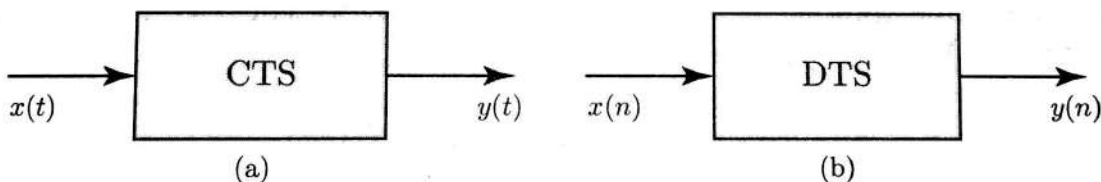


Fig. 1.25 (a) Continuous-time system (CTS) and (b) discrete-time system (DTS)

1.11 BASIC SYSTEM PROPERTIES

Our intent in this section is to lend additional substance to the concept of systems by discussing their classification according to the way the system interacts with the input signal. This interaction, which defines the model for the system, can be linear or nonlinear, time invariant or time varying, memoryless or with memory, static or dynamic, causal or noncausal, and stable or unstable.

1.11.1 Linear and Nonlinear Systems

When the system is linear, the superposition principle can be applied. Superposition simply implies that the response resulting from several input signals can be computed as the sum of the responses resulting from each input signal acting alone. Mathematically, the superposition principle can be stated as follows: Let $y_1(t)$ be the response of a continuous-time system to an input $x_1(t)$ and let $y_2(t)$ be the response corresponding to

an input $x_2(t)$. Then the system is linear (follows the principle of superposition) if

- the response to $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$ and
- the response to $\alpha x_1(t)$ is $\alpha y_1(t)$, where α is any arbitrary constant.

The first property is referred to as the *additivity* property; the second is referred to as the *scaling* or *homogeneity* property. These two properties defining a linear system can be combined into a single statement as

- continuous time: $\alpha x_1(t) + \beta x_2(t) \rightarrow \alpha y_1(t) + \beta y_2(t)$
- discrete time: $\alpha x_1(n) + \beta x_2(n) \rightarrow \alpha y_1(n) + \beta y_2(n)$

Here α and β are any complex constants. When a system violates either the principle of superposition or the property of homogeneity, the system is said to be *nonlinear*.

A direct consequence of the superposition property is that, for linear systems, an input that is zero for all time results in an output that is zero for all time. For example, if $x(n) \rightarrow y(n)$, then the homogeneity property tells us that

$$0 = 0.x(n) \rightarrow 0.y(n) = 0 \quad (1.74)$$

Example 1.25 For each of the following input–output relationships, determine whether the corresponding system is linear:

(a) $y(t) = tx(t)$, (b) $y(t) = x^2(t)$, (c) $y(n) = \mathcal{R}e\{x(n)\}$, and (d) $y(n) = 2x(n) - 3$.

Solution

(a) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$x_1(t) \rightarrow y_1(t) = tx_1(t)$$

$$x_2(t) \rightarrow y_2(t) = tx_2(t)$$

$$x_3(t) \rightarrow y_3(t) = tx_3(t)$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$, i.e.,

$$x_3(t) = ax_1(t) + bx_2(t)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(t) = ay_1(t) + by_2(t)$$

Consider the LHS of the above equation

$$y_3(t) = tx_3(t) = t[ax_1(t) + bx_2(t)] = atx_1(t) + btx_2(t) = ay_1(t) + by_2(t) = \text{RHS}$$

Since LHS = RHS, we conclude that the system is linear.

(b) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$x_1(t) \rightarrow y_1(t) = x_1^2(t)$$

$$x_2(t) \rightarrow y_2(t) = x_2^2(t)$$

$$x_3(t) \rightarrow y_3(t) = x_3^2(t)$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$, i.e.,

$$x_3(t) = ax_1(t) + bx_2(t)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(t) = ay_1(t) + by_2(t)$$

Consider the LHS of the above equation

$$\begin{aligned} y_3(t) &= x_3^2(t) = [ax_1(t) + bx_2(t)]^2 = a^2x_1^2(t) + b^2x_2^2(t) + 2abx_1(t)x_2(t) \\ &= a^2y_1(t) + b^2y_2(t) + 2abx_1(t)x_2(t) \neq \text{RHS} \end{aligned}$$

Since $\text{LHS} \neq \text{RHS}$, we conclude that the system is nonlinear.

(c) When checking the linearity of a system, it is important to remember that the system must satisfy both the additivity and homogeneity properties and that the signals as well as any scaling constants are allowed to be complex. Consider the system specified by

$$y(n) = \mathcal{R}e\{x(n)\}$$

Now consider two arbitrary inputs $x_1(n)$ and $x_2(n)$

$$x_1(n) \rightarrow y_1(n) = \mathcal{R}e\{x_1(n)\}$$

$$x_2(n) \rightarrow y_2(n) = \mathcal{R}e\{x_2(n)\}$$

$$x_3(n) \rightarrow y_3(n) = \mathcal{R}e\{x_3(n)\}$$

Let $x_3(n)$ be a linear combination of $x_1(n)$ and $x_2(n)$, i.e.,

$$x_3(n) = x_1(n) + x_2(n)$$

$$\text{and so } y_3(n) = \mathcal{R}e\{x_1(n) + x_2(n)\} = \mathcal{R}e\{x_1(n)\} + \mathcal{R}e\{x_2(n)\} = y_1(n) + y_2(n)$$

This system is additive; however, it does not satisfy the homogeneity property, as we now demonstrate.

Let

$$x_1(n) = r(n) + js(n)$$

be an arbitrary complex input with real and imaginary parts $r(n)$ and $s(n)$, respectively, so that the corresponding output is

$$y_1(n) = r(n)$$

Now consider scaling $x_1(n)$ by a complex number, e.g., $a = j$, i.e., consider the input

$$x_2(n) = jx_1(n) = j[r(n) + js(n)] = -s(n) + jr(n)$$

The output corresponding to $x_2(n)$ is

$$y_2(n) = \mathcal{R}e\{x_2(n)\} = -s(n)$$

which is not equal to the scaled version of $y_1(n)$

$$ay_1(n) = jr(n)$$

We conclude that the system violates the homogeneity property and hence is not linear.

(d) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(n)$ and $x_2(n)$.

$$x_1(n) \rightarrow y_1(n) = 2x_1(n) - 3$$

$$x_2(n) \rightarrow y_2(n) = 2x_2(n) - 3$$

$$x_3(n) \rightarrow y_3(n) = 2x_3(n) - 3$$

Let $x_3(n)$ be a linear combination of $x_1(n)$ and $x_2(n)$, i.e.,

$$x_3(n) = ax_1(n) + bx_2(n)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(n) = ay_1(n) + by_2(n)$$

Consider the LHS of the above equation

$$y_3(n) = 2x_3(n) - 3 = 2[ax_1(n) + bx_2(n)] - 3 = a2x_1(n) + b2x_2(n) - 3 \neq \text{RHS}$$

Since $\text{LHS} \neq \text{RHS}$, we conclude that the system is nonlinear. Alternatively, since $y(n) = 3$ if $x(n) = 0$, we see that the system violates the 'zero-in/zero-out' property of linear systems given in Eq. (1.74).

Example 1.26 For the system described by the following equations, with the input $x(t)$ and output $y(t)$, determine which of the systems are linear and which are nonlinear:

$$(a) \quad \frac{dy(t)}{dt} + 3y(t) = x(t) \quad (1.75)$$

$$(b) \quad \frac{dy(t)}{dt} + 2y(t) = x^2(t) \quad (1.76)$$

$$(c) \quad \frac{d^2y(t)}{dt^2} + 2y(t) = x(t) \quad (1.77)$$

$$(d) \quad \frac{dy(t)}{dt} + 3y(t) + 4 = x(t) \quad (1.78)$$

Solution

(a) Let the system response to the inputs $x_1(t)$ and $x_2(t)$ be $y_1(t)$ and $y_2(t)$, respectively. Then

$$\frac{dy_1(t)}{dt} + 3y_1(t) = x_1(t) \quad \text{and} \quad \frac{dy_2(t)}{dt} + 3y_2(t) = x_2(t)$$

By multiplying the first equation by a , the second with b , and then adding both, we get

$$\frac{d}{dt}[ay_1(t) + by_2(t)] + 3[ay_1(t) + by_2(t)] = ax_1(t) + bx_2(t)$$

But this equation, is the system equation, i.e., Eq. (1.75) with $x(t) = ax_1(t) + bx_2(t)$ and $y(t) = ay_1(t) + by_2(t)$. Therefore, when the input is $ax_1(t) + bx_2(t)$, the system response is $ay_1(t) + by_2(t)$. Consequently, the system is linear.

(b) Let the system response to the inputs $x_1(t)$ and $x_2(t)$ be $y_1(t)$ and $y_2(t)$, respectively. Then

$$\frac{dy_1(t)}{dt} + 2y_1(t) = x_1^2(t) \quad \text{and} \quad \frac{dy_2(t)}{dt} + 2y_2(t) = x_2^2(t)$$

By multiplying the first equation by a , the second with b , and then adding both, we get

$$\frac{d}{dt}[ay_1(t) + by_2(t)] + 2[ay_1(t) + by_2(t)] = ax_1^2(t) + bx_2^2(t) \quad (i)$$

But this equation is not the system equation, i.e., Eq. (1.76) with $x(t) = ax_1(t) + bx_2(t)$ and $y(t) = ay_1(t) + by_2(t)$ that is,

$$\begin{aligned} \frac{d}{dt}[ay_1(t) + by_2(t)] + 2[ay_1(t) + by_2(t)] &= [ax_1(t) + bx_2(t)]^2 \\ &= a^2x_1^2(t) + b^2x_2^2(t) + 2abx_1(t)x_2(t) \end{aligned} \quad (ii)$$

Since Eq. (i) is not equal to Eq. (ii), the system is nonlinear.

(c) Let the system response to the inputs $x_1(t)$ and $x_2(t)$ be $y_1(t)$ and $y_2(t)$, respectively. Then

$$\frac{d^2y_1(t)}{dt^2} + 2y_1(t) = x_1(t) \quad \text{and} \quad \frac{d^2y_2(t)}{dt^2} + 2y_2(t) = x_2(t)$$

By multiplying the first equation by a , the second with b , and then adding both, we get

$$\frac{d^2}{dt^2}[ay_1(t) + by_2(t)] + 2[ay_1(t) + by_2(t)] = ax_1(t) + bx_2(t)$$

But this equation is the system equation, i.e., Eq. (1.77) with $x(t) = ax_1(t) + bx_2(t)$ and $y(t) = ay_1(t) + by_2(t)$. Therefore, when the input $x(t) = ax_1(t) + bx_2(t)$, the system response is $y(t) = ay_1(t) + by_2(t)$. Consequently, the system is linear.

(d) Let the system response to the inputs $x_1(t)$ and $x_2(t)$ be $y_1(t)$ and $y_2(t)$, respectively. Then

$$\frac{dy_1(t)}{dt} + 3y_1(t) + 4 = x_1(t) \quad \text{and} \quad \frac{dy_2(t)}{dt} + 3y_2(t) + 4 = x_2(t)$$

By multiplying the first equation by a , the second with b , and then adding both, we get

$$\frac{d}{dt}[ay_1(t) + by_2(t)] + 3[ay_1(t) + by_2(t)] + 4(a + b) = ax_1(t) + bx_2(t) \quad (\text{iii})$$

But this equation is not the system equation, i.e., Eq. (1.78) with $x(t) = ax_1(t) + bx_2(t)$ and $y(t) = ay_1(t) + by_2(t)$ that is,

$$\frac{d}{dt}[ay_1(t) + by_2(t)] + 3[ay_1(t) + by_2(t)] + 4 = ax_1(t) + bx_2(t) \quad (\text{iv})$$

Since Eq. (iii) is not equal to Eq. (iv), the system is nonlinear.

1.11.2 Time-varying and Time-invariant Systems

Conceptually, a system is time invariant if the behaviour and characteristics of the system are fixed over time. In other words, a system is said to be time invariant if a time shift in the input signal causes an identical time shift in the output signal. For example, if $y(t)$ is the output of a continuous-time system corresponding to the input $x(t)$, a time-invariant system will have $y(t - t_0)$ as the output when $x(t - t_0)$ is the input, i.e.,

$$\text{if } x(t) \longrightarrow y(t), \quad \text{then } x(t - t_0) \longrightarrow y(t - t_0)$$

Similarly, in discrete time with $y(n)$ the output corresponding to the input $x(n)$, a time-invariant system will have $y(n - n_0)$ as the output when $x(n - n_0)$ is the input, i.e.,

$$\text{if } x(n) \longrightarrow y(n), \quad \text{then } x(n - n_0) \longrightarrow y(n - n_0)$$

The procedure for testing whether a system is time invariant is summarized in the following steps:

1. Let $y_1(t)$ be the output corresponding to $x_1(t)$.
2. Consider a second input $x_2(t)$ obtained by shifting $x_1(t)$

$$x_2(t) = x_1(t - t_0)$$

and find the output $y_2(t)$ corresponding to the input $x_2(t)$.

3. From Step 1, find $y_1(t - t_0)$ and compare with $y_2(t)$.
4. If $y_2(t) = y_1(t - t_0)$, then the system is time invariant; otherwise it is a time-varying system.

Example 1.27 For each of the following input-output relationships, determine whether the corresponding system is time-invariant:

- (a) $y(n) = nx(n)$, (b) $y(n) = x(-n)$, (c) $y(t) = \sin[x(t)]$, and (d) $y(t) = x(2t)$.

Solution

(a) To determine whether a system is time-invariant or not, we consider an arbitrary input $x_1(n)$:

$$x_1(n) \longrightarrow y_1(n) = nx_1(n)$$

$$x_2(n) \longrightarrow y_2(n) = nx_2(n)$$

Let $x_2(n)$ be obtained by shifting $x_1(n)$ in time, i.e.,

$$x_2(n) = x_1(n - n_0)$$

If the system is time-invariant, then

$$y_2(n) = y_1(n - n_0)$$

Considering the LHS of the above equation, we obtain

$$y_2(n) = nx_2(n) = nx_1(n - n_0)$$

Now, consider

$$y_1(n) = nx_1(n)$$

$$y_1(n - n_0) = (n - n_0)x_1(n - n_0)$$

Clearly, $y_2(n) \neq y_1(n - n_0)$, and hence, this system is not time-invariant.

(b) To determine whether a system is time-invariant or not, we consider an arbitrary input $x_1(n)$:

$$x_1(n) \longrightarrow y_1(n) = x_1(-n)$$

$$x_2(n) \longrightarrow y_2(n) = x_2(-n)$$

Let $x_2(n)$ be obtained by shifting $x_1(n)$ in time, i.e.,

$$x_2(n) = x_1(n - n_0)$$

If the system is time-invariant, then

$$y_2(n) = y_1(n - n_0)$$

Considering the LHS of the above equation, we obtain

$$y_2(n) = x_2(-n) = nx_1(-n - n_0)$$

Now, consider

$$y_1(n) = x_1(-n)$$

$$y_1(n - n_0) = x_1(-(n - n_0)) = x_1(-n + n_0)$$

Clearly, $y_2(n) \neq y_1(n - n_0)$, and hence, this system is not time-invariant.

(c) To determine whether a system is time-invariant or not, we consider an arbitrary input $x_1(t)$:

$$x_1(t) \longrightarrow y_1(t) = \sin[x_1(t)]$$

$$x_2(t) \longrightarrow y_2(t) = \sin[x_2(t)]$$

Let $x_2(t)$ be obtained by shifting $x_1(t)$ in time, i.e.,

$$x_2(t) = x_1(t - t_0)$$

If the system is time-invariant, then

$$y_2(t) = y_1(t - t_0)$$

Considering the LHS of the above equation, we obtain

$$y_2(t) = \sin[x_2(t)] = \sin[x_1(t - t_0)]$$

Now, consider

$$y_1(t) = \sin[x_1(t)]$$

$$y_1(t - t_0) = \sin[x_1(t - t_0)]$$

Clearly, $y_2(t) = y_1(t - t_0)$, and hence, this system is time-invariant.

(d) To determine whether a system is time-invariant or not, we consider an arbitrary input $x_1(t)$.

$$x_1(t) \longrightarrow y_1(t) = x_1(2t)$$

$$x_2(t) \longrightarrow y_2(t) = x_2(2t)$$

Let $x_2(t)$ be obtained by shifting $x_1(t)$ in time, that is

$$x_2(t) = x_1(t - t_0)$$

If the system is time-invariant, then

$$y_2(t) = y_1(t - t_0)$$

Considering the LHS of the above equation, we obtain

$$y_2(t) = x_2(2t) = x_1(2t - t_0)$$

Now, consider

$$y_1(t) = x_1(2t)$$

$$y_1(t - t_0) = x_1(2(t - t_0)) = x_1(2t - 2t_0)$$

Clearly, $y_2(t) \neq y_1(t - t_0)$, and hence, this system is not time-invariant.

1.11.3 Causal Systems

A system is causal, or nonanticipatory (also known as physically realizable), if the output at any time t_0 or n_0 depends only on the values of the input at the present time and in the past, i.e., for $t \leq t_0$ or $n \leq n_0$. Equivalently, if two inputs to a causal system are identical up to some time t_0 or n_0 , the corresponding outputs must also be equal up to this time since a causal system cannot predict if the two inputs will be different after t_0 or n_0 (in the future). Mathematically, if

$$x_1(n) = x_2(n), \quad n \leq n_0$$

and the system is causal, then

$$y_1(n) = y_2(n), \quad n \leq n_0$$

All memoryless systems are causal systems since the output responds only to the current value of the input.

Example 1.28 For each of the following input-output relationships, determine whether the corresponding system is causal: (a) $y(n) = nx(n)$, (b) $y(n) = x(-n)$, (c) $y(t) = \sin(x(t))$, (d) $y(t) = x(2t)$, and (e) $y(t) = x(t)\cos(t+1)$.

Solution

(a) Consider the output $y(n)$ at a positive time n_0 :

$$y(n)|_{n=n_0} = y(n_0) = nx(n)|_{n=n_0} = n_0x(n_0)$$

Now, consider the output $y(n)$ at a negative time $-n_0$:

$$y(n)|_{n=-n_0} = y(-n_0) = nx(n)|_{n=-n_0} = -n_0x(-n_0)$$

In both the above cases the present output depends upon the present input, and hence, the system is causal.

(b) Consider the output $y(n)$ at a positive time n_0 :

$$y(n)|_{n=n_0} = y(n_0) = x(-n)|_{n=n_0} = x(-n_0)$$

Now, consider the output $y(n)$ at a negative time $-n_0$:

$$y(n)|_{n=-n_0} = y(-n_0) = x(-n)|_{n=-n_0} = x(n_0)$$

In the first case, the present output depends upon the past input but in the second case the present output depends upon the future inputs, and hence, the system is noncausal.

(c) In this system the current value of the input $x(t)$ influences the current value of the output $y(t)$, and so the system is causal.

(d) Consider the output $y(t)$ at a positive time t_0 :

$$y(t)|_{t=t_0} = y(t_0) = x(2t)|_{t=t_0} = x(2t_0)$$

Now, consider the output $y(t)$ at a negative time $-t_0$:

$$y(t)|_{t=-t_0} = y(-t_0) = x(2t)|_{t=-t_0} = x(-2t_0)$$

In first case, the present output depends upon the future input but in second case the present output depends upon the past input, and so the system is noncausal.

(e) In this system, the output at any time t_0 equals the input at that same time multiplied by a number that varies with time. Thus, only the current value of the input $x(t)$ influences the current value of the output $y(t)$, and we conclude that this system is causal.

1.11.4 Stable Systems

Stability is another important system property. Informally, a stable system is one in which small inputs lead to responses that do not diverge. A system is said to be *bounded-input, bounded-output* (BIBO) stable if and only if every bounded input results in a bounded output. The output of such a system does not diverge if the input does not diverge.

Signal $x(t)$ is said to be bounded if its magnitude does not grow without bound, i.e.,

$$|x(t)| \leq B_x < \infty, \quad \text{for all } t$$

A system is BIBO stable if, for any bounded input $x(t)$, the response $y(t)$ is also bounded. That is,

$$|x(t)| \leq B_x < \infty \text{ implies } |y(t)| \leq B_y < \infty \quad (1.79)$$

Example 1.29 For each of the following input-output relationships, determine whether the corresponding system is stable: (a) $y(t) = tx(t)$, (b) $y(t) = e^{x(t)}$, (c) $y(n) = 1/3[x(n) + x(n-1) + x(n-2)]$, (d) $y(n) = r^n x(n)$, where $r > 1$, and (e) $y(n) = \sum_{k=0}^{\infty} \rho^k x(n-k)$.

Solution

(a) Assume that

$$|x(t)| \leq B_x < \infty, \quad \text{for all } t$$

Using the given input-output relation, we have

$$y(t) = tx(t)$$

and so we may write

$$|y(t)| = |tx(t)| = |t||x(t)| = |t|B_x$$

As $t \rightarrow \infty$ the output $|y(t)| \rightarrow \infty$, which is unbounded, since no matter what finite constant we pick, $|y(t)|$ will exceed that constant for some time t . The condition that the input signal is bounded is not sufficient to guarantee a bounded output signal, and so the system is unstable. To prove stability, we need to establish that all bounded inputs produce a bounded output.

(b) Assume that

$$|x(t)| \leq B_x < \infty, \quad \text{for all } t$$

or

$$-B_x \leq x(t) \leq B_x, \quad \text{for all } t$$

Using the given input-output relation, we have

$$y(t) = e^{x(t)}$$

and so we may write

$$e^{-B_x} \leq y(t) \leq e^{B_x}$$

We conclude that in this system if any input is bounded by an arbitrary positive number B_x , the corresponding output is guaranteed to be bounded by e^{B_x} . Thus, the given system is stable.

(c) Assume that

$$|x(n)| \leq B_x < \infty, \quad \text{for all } n$$

Using the input-output relation, we have

$$\begin{aligned} y(n) &= \frac{1}{3} [x(n) + x(n-1) + x(n-2)] \\ |y(n)| &= \left| \frac{1}{3} [x(n) + x(n-1) + x(n-2)] \right| \\ &\leq \frac{1}{3} [|x(n)| + |x(n-1)| + |x(n-2)|] \\ &\leq \frac{1}{3} [B_x + B_x + B_x] = B_x \end{aligned}$$

Hence, the absolute value of the output signal $y(n)$ is always less than the maximum absolute value of the input signal $x(n)$ for all n , which shows that the moving-average system is stable.

(d) Assume that the input signal $x(n)$ satisfies the condition

$$|x(n)| \leq B_x < \infty, \quad \text{for all } n$$

We then find that

$$|y(n)| = |r^n x(n)| = |r^n| |x(n)|$$

With $r > 1$, the multiplying factor r^n diverges for increasing n . Accordingly, the condition that the input signal is bounded is not sufficient to guarantee a bounded output signal, and so the system is unstable.

(e) Assume that the input signal $x(n)$ satisfies the condition

$$|x(n)| \leq B_x < \infty, \quad \text{for all } n$$

We then find that

$$\begin{aligned} |y(n)| &= \left| \sum_{k=0}^{\infty} \rho^k x(n-k) \right| \\ &\leq \sum_{k=0}^{\infty} |\rho^k| |x(n-k)| \\ &\leq \sum_{k=0}^{\infty} |\rho^k| B_x \end{aligned}$$

Case I When $|\rho| \geq 1$, the output will be unbounded $[y(n) \rightarrow \infty]$ and the system is unstable.

Case II when $|\rho| < 1$,

$$|y(n)| = B_x \frac{1}{1 - \rho}$$

with $|\rho| < 1$, the output will be bounded and the system is stable.

1.11.5 Systems With and Without Memory

A system is said to possess *memory* (with memory system or dynamic system) if its output signal depends on the past or future values of the input signal. The temporal extent of the past or future values, on which the output depends, defines how far the memory of the system extends into the past or future. In contrast, a system is said to be *memoryless* (or static or instantaneous) if its output signal depends only on the present value of the input signal. For example, the system specified by the relationship

$$y(n) = 2x(n) + 3x^2(n) \quad (1.80)$$

is memoryless, as the value of $y(n)$ at any particular time n_0 depends only on the value of $x(n)$ at that time. Similarly, a resistor is a memoryless system: with the input $x(t)$ taken as the current and with the voltage taken as the output $y(t)$, the input-output relationship of a resistor is

$$y(t) = Rx(t) \quad (1.81)$$

where R is the resistance.

An example of a discrete-time system with memory is an *accumulator* or *summer*

$$y(n) = \sum_{k=-\infty}^n x(k) \quad (1.82)$$

and a second example is a delay

$$y(n) = x(n - 1) \quad (1.83)$$

A capacitor is an example of a continuous-time system with memory. With input taken as the current and output as the voltage, the input-output relationship in the case of the capacitor is

$$y(t) = \frac{1}{C} \int_{-\infty}^t x(\tau) d\tau \quad (1.84)$$

where C is the capacitance. It is obvious that the output at any time t depends on the entire history of the input.

1.11.6 Invertibility and Inverse Systems

A system is invertible if by observing the output, we can determine its input. That is, we can construct an inverse system that when cascaded with the given system, as illustrated in Fig. 1.26, yields an output equal to the original input of the given system. In other

words, the inverse system 'undoes' what the given system does to input. So the effect of the given system can be eliminated by cascading it with its inverse system. When several different inputs result in the same output, it is impossible to obtain the input from the output, and the system is *noninvertible*. Therefore, for an invertible system, it is essential that every input has a unique output so that there is a one-to-one mapping between an input and the corresponding output.

A rectifier, specified by an equation $y(t) = |x(t)|$, is noninvertible because the rectification operation cannot be undone.

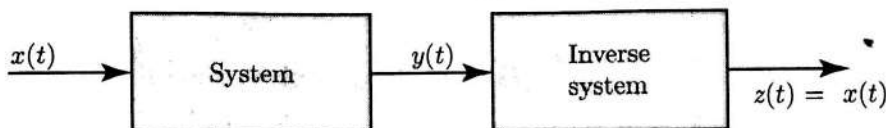


Fig. 1.26 Invertibility and inverse system

SOLVED EXAMPLES

Example 1.30 Determine the fundamental period of the following signals:

- (a) $x(t) = 2 \cos(10t + 1) - \sin(4t - 1)$
 (b) $x(n) = 1 + e^{j4\pi n/7} - e^{j2\pi n/5}$

Solution

- (a) $x(t) = 2 \cos(10t + 1) - \sin(4t - 1)$

Comparing with

$$x(t) = A_1 \cos(\omega_1 t + \phi_1) - A_2 \cos(\omega_2 t + \phi_2)$$

we get

$$\omega_1 = 10, \quad \omega_2 = 4$$

$$T_1 = \frac{2\pi}{10}, \quad T_2 = \frac{2\pi}{4}$$

$$\frac{T_1}{T_2} = \frac{2}{5} = \text{a rational number}$$

Therefore, $x(t)$ is periodic with fundamental period $T = 5T_1 = 2T_2 = \pi$.

- (b) $x(n) = 1 + e^{j4\pi n/7} - e^{j2\pi n/5}$

The first term on the RHS is a constant. The second and third terms on the RHS has frequencies $\omega_1 = 4\pi/7$ and $\omega_2 = 2\pi/5$, respectively.

$$\omega_1 = \frac{4\pi}{7}, \quad \omega_2 = \frac{2\pi}{5}$$

$$7 = \frac{2\pi}{\omega_1} 2, \quad 5 = \frac{2\pi}{\omega_2}$$

$$N_1 = \frac{2\pi}{\omega_1} m_1, \quad N_2 = \frac{2\pi}{\omega_2} m_2$$

$$\frac{N_1}{N_2} = \frac{7}{5} = \text{a rational number}$$

Therefore, $x(n)$ is periodic with fundamental period $N = 5N_1 = 7N_2 = 35$.

Example 1.31 Determine whether or not each of the following signals is periodic. If the signal is periodic, determine its fundamental period.

(a) $x(t) = \left[\cos \left(2t - \frac{\pi}{3} \right) \right]^2$

(b) $x(t) = \mathcal{E}\{\cos(4\pi t) u(t)\}$

(c) $x(t) = \mathcal{E}\{\sin(4\pi t) u(t)\}$

(d) $x(t) = e^{-2t} u(t)$

(e) $x(n) = \cos \left(\frac{\pi}{8} n^2 \right)$

(f) $x(n) = \cos \left(\frac{\pi}{2} n \right) \cos \left(\frac{\pi}{4} n \right)$

Solution

- (a) Consider the given signal

$$\begin{aligned} x(t) &= \left[\cos \left(2t - \frac{\pi}{3} \right) \right]^2 \\ &= \frac{1}{2} \left[1 + \cos \left(4t - \frac{2\pi}{3} \right) \right] \end{aligned}$$

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The first term on the RHS is a constant and the frequency of the second term is $\omega = 4$. Therefore, $x(t)$ is a periodic signal with time period

$$T = \frac{2\pi}{4} = \frac{\pi}{2}$$

(b) Consider the given signal

$$\begin{aligned} x(t) &= \mathcal{E}\{\cos(4\pi t) u(t)\} \\ &= \frac{1}{2} [\cos(4\pi t) u(t) + \cos(-4\pi t) u(-t)] \\ &= \frac{1}{2} [\cos(4\pi t) u(t) + \cos(4\pi t) u(-t)] \\ &= \frac{1}{2} \cos(4\pi t) = \frac{1}{2} \cos(\omega t) \end{aligned}$$

Therefore,

$$\begin{aligned} \omega &= 4\pi \\ T &= \frac{2\pi}{4\pi} = \frac{1}{2} \end{aligned}$$

Thus, $x(t)$ is a periodic signal with period $T = 1/2$ (Fig. 1.27).

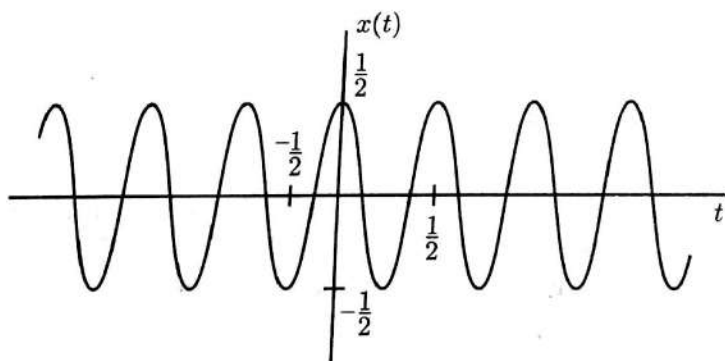


Fig. 1.27

(c) Consider the given signal

$$\begin{aligned} x(t) &= \mathcal{E}\{\sin(4\pi t) u(t)\} \\ &= \frac{1}{2} [\sin(4\pi t) u(t) + \sin(-4\pi t) u(-t)] \\ &= \frac{1}{2} [\sin(4\pi t) u(t) - \sin(4\pi t) u(-t)] \\ &= \frac{1}{2} \sin(4\pi t) [u(t) - u(-t)] \end{aligned}$$

$x(t)$ is not a periodic signal (as shown in Fig. 1.28).

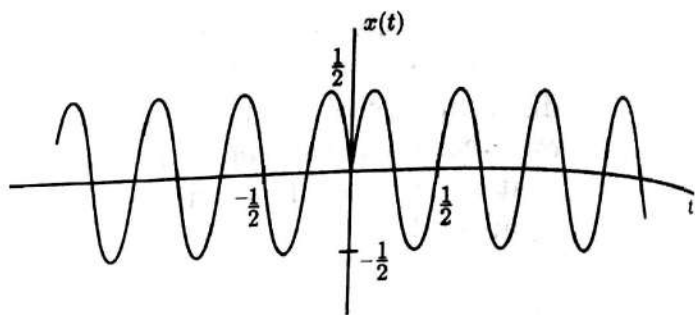


Fig. 1.28

(d) Consider the given signal

$$x(t) = e^{-2t} u(t)$$

$x(t)$ is not a periodic signal because it is zero for $t < 0$ (Fig. 1.29).

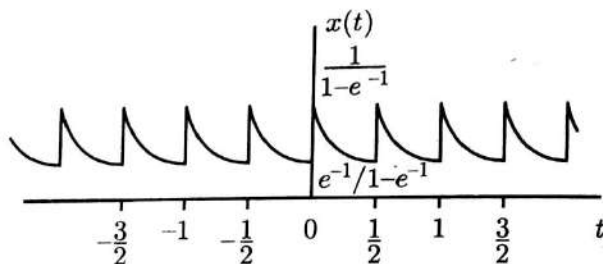


Fig. 1.29

(e) Consider the given signal

$$\begin{aligned} x(n) &= \cos\left(\frac{\pi}{8}n^2\right) \\ x(n+N) &= \cos\left(\frac{\pi}{8}(n+N)^2\right) \end{aligned}$$

for $x(n)$ to be periodic

$$\begin{aligned} x(n+N) &= x(n) \\ \cos\left(\frac{\pi}{8}(n+N)^2\right) &= \cos\left(\frac{\pi}{8}n^2\right) \\ \frac{\pi}{8}(n+N)^2 &= \frac{\pi}{8}n^2 + 2k\pi \\ \frac{\pi}{8}(N^2 + 2nN) &= 2k\pi \\ N^2 + 2Nn &= 16k \end{aligned}$$

where n and k are integers. By inspection, $N = 8, 16, 32, \dots$ will satisfy the above equation. Consequently, $x(n)$ is periodic with period $N = 8$.

(f) Consider the given signal

$$x(n) = \cos\left(\frac{\pi}{2}n\right) \cos\left(\frac{\pi}{4}n\right)$$

$$= \frac{1}{2} \left(\cos\left(\frac{3\pi}{4}n\right) + \cos\left(\frac{\pi}{4}n\right) \right)$$

$$\omega_1 = \frac{3\pi}{4}, \quad \omega_2 = \frac{\pi}{4}$$

$$8 = \frac{2\pi}{\omega_1} 3, \quad 8 = \frac{2\pi}{\omega_2}$$

$$N_1 = \frac{2\pi}{\omega_1} m_1, \quad N_2 = \frac{2\pi}{\omega_2} m_2$$

Both $N_1 = 8$ and $N_2 = 8$ are equal integers. $x(n)$ is periodic with period $N = 8$.

Example 1.32 Consider the discrete-time signal

$$x(n) = 1 - \sum_{k=3}^{\infty} \delta(n-1-k)$$

Determine the values of the integers M and n_0 so that $x(n)$ may be expressed as

$$x(n) = u(Mn - n_0)$$

Solution

Consider the given signal

$$x(n) = 1 - \sum_{k=3}^{\infty} \delta(n-1-k)$$

$$= \sum_{k=-\infty}^{\infty} \delta(n-k) - \sum_{k=4}^{\infty} \delta(n-k)$$

$$= \sum_{k=-\infty}^3 \delta(n-k) + \sum_{k=4}^{\infty} \delta(n-k) - \sum_{k=4}^{\infty} \delta(n-k)$$

$$= \sum_{k=-\infty}^3 \delta(n-k)$$

$$= u(-n+3) = u(Mn - n_0)$$

This implies that $M = -1$ and $n_0 = -3$. The signal $x(n)$ is shown in Fig. 1.30. $x(n)$ can be obtained by shifting $u(n)$ by 3 towards the left and then flipping (time-reversal) the shifted signal.

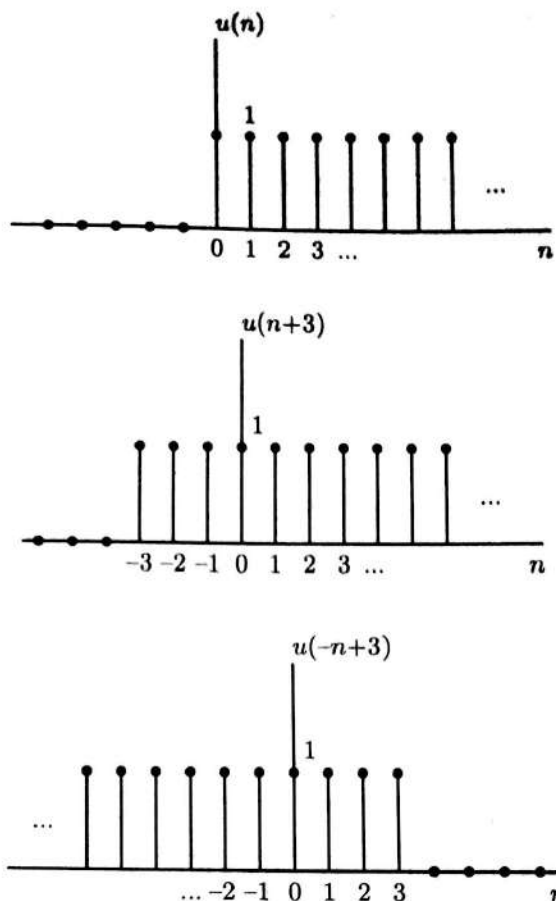


Fig. 1.30

Example 1.33 Consider a periodic signal

$$x(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ -2, & 1 < t < 2 \end{cases}$$

with period $T = 2$. The derivative of this signal is related to the 'impulse train'

$$g(t) = \sum_{k=-\infty}^{\infty} \delta(t-2k)$$

with period $T = 2$. It can be shown that

$$\frac{dx(t)}{dt} = A_1 g(t-t_1) + A_2 g(t-t_2)$$

Determine the values of A_1 , t_1 , A_2 , and t_2 .

Solution

Consider the given signal

$$x(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ -2, & 1 < t < 2 \end{cases}$$

$$x(t) = \dots + 3u(t+2) - 3u(t+1) + 3u(t) \\ - 3u(t-1) + 3u(t-2) - \dots$$

$$\frac{dx(t)}{dt} = \dots + 3\delta(t+2) - 3\delta(t+1) + 3\delta(t) \\ - 3\delta(t-1) + 3\delta(t-2) - \dots$$

and $g(t) = \dots + \delta(t+2) + \delta(t) + \delta(t-2) + \dots$

$$3g(t) = \dots + 3\delta(t+2) + 3\delta(t) + 3\delta(t-2) + \dots$$

$$3g(t-1) = \dots + 3\delta(t+1) + 3\delta(t-1) \\ + 3\delta(t-3) + \dots$$

Thus,

$$\frac{dx(t)}{dt} = 3g(t) + 3g(t-1)$$

Comparing with

$$\frac{dx(t)}{dt} = A_1 g(t-t_1) + A_2 g(t-t_2)$$

we get $A_1 = 3$, $t_1 = 0$, $A_2 = 3$, and $t_2 = 1$. The signal $x(t)$, its derivative $dx(t)/dt$, and $g(t)$ are shown in Fig. 1.31.

Example 1.34 Consider a discrete-time system with input $x(n)$ and output $y(n)$ related by

$$y(n) = \sum_{k=n-n_0}^{n+n_0} x(k)$$

where n_0 is a finite positive integer.

(a) Is this system linear?

(b) Is this system time-invariant?

(c) If $x(n)$ is known to be bounded by a finite integer B_x [i.e., $|x(n)| < B_x$ for all n], it can be shown that $y(n)$ is bounded by a finite number C . We conclude that the given system is stable. Express C in terms of B_x and n_0 .

Solution

(a) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(n)$ and $x_2(n)$.

$$x_1(n) \rightarrow y_1(n) = \sum_{k=n-n_0}^{n+n_0} x_1(k)$$

$$x_2(n) \rightarrow y_2(n) = \sum_{k=n-n_0}^{n+n_0} x_2(k)$$

$$x_3(n) \rightarrow y_3(n) = \sum_{k=n-n_0}^{n+n_0} x_3(k)$$

Let $x_3(n)$ be a linear combination of $x_1(n)$ and $x_2(n)$. That is,

$$x_3(n) = ax_1(n) + bx_2(n)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(n) = ay_1(n) + by_2(n)$$

Consider the LHS of the above equation

$$y_3(n) = \sum_{k=n-n_0}^{n+n_0} x_3(k)$$

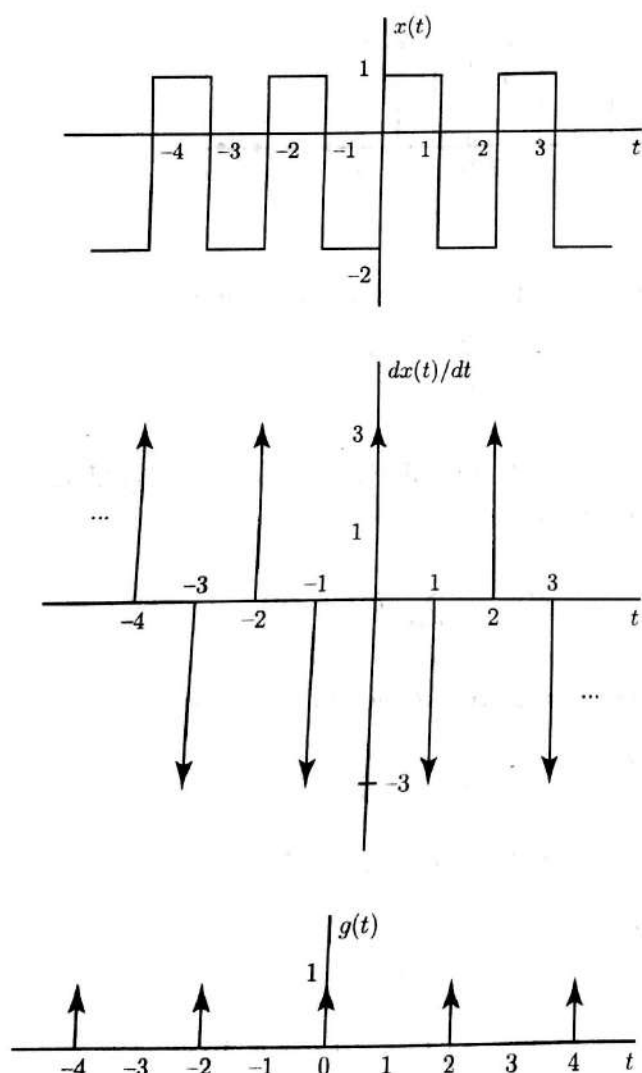


Fig. 1.31

$$\begin{aligned}
&= \sum_{k=n-n_0}^{n+n_0} (ax_1(k) + bx_2(k)) \\
&= a \sum_{k=n-n_0}^{n+n_0} x_1(k) + b \sum_{k=n-n_0}^{n+n_0} x_2(k) \\
&= ay_1(n) + by_2(n) = \text{RHS}
\end{aligned}$$

Since LHS = RHS, we conclude that the system is linear.

(b) Let $x_1(n)$ be an arbitrary input to the system.

$$\begin{aligned}
x_1(n) &\longrightarrow y_1(n) = \sum_{k=n-n_0}^{n+n_0} x_1(k) \\
x_2(n) &\longrightarrow y_2(n) = \sum_{k=n-n_0}^{n+n_0} x_2(k)
\end{aligned}$$

Let $x_2(n)$ be obtained by shifting $x_1(n)$ in time, i.e.,

$$x_2(n) = x_1(n - m)$$

If the system is time-invariant, then

$$y_2(n) = y_1(n - m)$$

Consider the LHS of the above equation,

$$\begin{aligned}
y_2(n) &= \sum_{k=n-n_0}^{n+n_0} x_2(k) \\
&= \sum_{k=n-n_0}^{n+n_0} x_1(k - m)
\end{aligned}$$

Now, consider

$$\begin{aligned}
y_1(n) &= \sum_{k=n-n_0}^{n+n_0} x_1(k) \\
y_1(n - m) &= \sum_{k=n-m-n_0}^{n-m+n_0} x_1(k) \\
&= \sum_{k=n-n_0}^{n+n_0} x_1(k - m)
\end{aligned}$$

Clearly, $y_2(n) = y_1(n - m)$; thus, this system is time-invariant.

(c) $x(n)$ is bounded by a finite integer B_x , i.e.,

$$|x(n)| < B_x \quad \text{for all } n$$

We then find that

$$\begin{aligned}
|y(n)| &= \left| \sum_{k=n-n_0}^{n+n_0} x(k) \right| \\
&\leq \sum_{k=n-n_0}^{n+n_0} |x(k)| \\
&\leq \sum_{k=n-n_0}^{n+n_0} B_x \\
&\leq B_x [(n + n_0) - (n - n_0) + 1] \\
&\leq B_x (2n_0 + 1) = C
\end{aligned}$$

Example 1.35 Consider a continuous-time system with input $x(t)$ and output $y(t)$ related by

$$y(t) = x(\sin(t))$$

(a) Is this system causal?

(b) Is this system linear?

Solution

(a) Case I: Consider the output $y(t)$ at time $t = -\pi$

$$\begin{aligned}
y(t) \Big|_{t=-\pi} &= x(\sin(-\pi)) \\
y(-3.14) &= x(0)
\end{aligned}$$

In this case, the present output depends on the future input.

Case II: Consider the output at time $t = 0$

$$\begin{aligned}
y(t) \Big|_{t=0} &= x(\sin(0)) \\
y(0) &= x(0)
\end{aligned}$$

In this case, the present output depends on the present input.

Case III: Consider the output at time $t = \pi$

$$\begin{aligned}
y(t) \Big|_{t=\pi} &= x(\sin(\pi)) \\
y(3.14) &= x(0)
\end{aligned}$$

In this case, the present output depends on the past input.

The present output of this system depends on the present, past, and future inputs, and so this system is noncausal.

(b) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$x_1(t) \rightarrow y_1(t) = x_1(\sin(t))$$

$$x_2(t) \rightarrow y_2(t) = x_2(\sin(t))$$

$$x_3(t) \rightarrow y_3(t) = x_3(\sin(t))$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$. That is,

$$x_3(t) = ax_1(t) + bx_2(t)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(t) = ay_1(t) + by_2(t)$$

Consider the LHS of the above equation

$$\begin{aligned} y_3(t) &= x_3(\sin(t)) \\ &= ax_1(\sin(t)) + bx_2(\sin(t)) \\ &= ay_1(t) + by_2(t) = \text{RHS} \end{aligned}$$

Since LHS = RHS, we conclude that the system is linear.

Example 1.36 The system that follow have input $x(t)$ or $x(n)$ and output $y(t)$ or $y(n)$. For each system, determine whether it is (i) memoryless, (ii) stable, (iii) causal, (iv) linear, or (v) time-invariant.

$$(a) y(t) = \int_{-\infty}^{2t} x(\tau) d\tau$$

$$(b) y(t) = \frac{dx(t)}{dt}$$

$$(c) y(t) = \begin{cases} 0, & t < 0 \\ x(t) + x(t-2), & t \geq 0 \end{cases}$$

$$(d) y(t) = \begin{cases} 0, & x(t) < 0 \\ x(t) + x(t-2), & x(t) \geq 0 \end{cases}$$

$$(e) y(n) = \mathcal{E}\{x(n-1)\}$$

$$(f) y(n) = x(n^2)$$

Solution

(a) (i) This system integrates the input from $-\infty$ to $2t$. The present output depends on the past, present, and future inputs. We may conclude that this system is a *with-memory* system.

(ii) Assume that the input signal $x(t)$ satisfies the condition

$$|x(t)| \leq B_x < \infty \quad \text{for all } t.$$

We then find that

$$\begin{aligned} |y(t)| &= \left| \int_{-\infty}^{2t} x(\tau) d\tau \right| \\ &= \int_{-\infty}^{2t} |x(\tau)| d\tau \\ &= \int_{-\infty}^{2t} B_x d\tau \\ &= \infty \end{aligned}$$

We conclude that in this system if any input is bounded by an arbitrary positive number B_x , the corresponding output is unbounded. Thus, the given system is *unstable*.

(iii) In this system, the present output anticipates with the future inputs (the upper limit of the integration is $2t$). Consequently, the given system is *noncausal*.

(iv) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$x_1(t) \rightarrow y_1(t) = \int_{-\infty}^{2t} x_1(\tau) d\tau$$

$$x_2(t) \rightarrow y_2(t) = \int_{-\infty}^{2t} x_2(\tau) d\tau$$

$$x_3(t) \rightarrow y_3(t) = \int_{-\infty}^{2t} x_3(\tau) d\tau$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$. That is,

$$x_3(t) = ax_1(t) + bx_2(t)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(t) = ay_1(t) + by_2(t)$$

Consider the LHS of the above equation

$$\begin{aligned}
 y_3(t) &= \int_{-\infty}^{2t} x_3(\tau) d\tau \\
 &= \int_{-\infty}^{2t} (ax_1(\tau) + bx_2(\tau)) d\tau \\
 &= a \int_{-\infty}^{2t} x_1(\tau) d\tau + b \int_{-\infty}^{2t} x_2(\tau) d\tau \\
 &= ay_1(t) + by_2(t) = \text{RHS}
 \end{aligned}$$

Since LHS = RHS, we conclude that the system is *linear*.

(v) Let $x_1(t)$ be an arbitrary input to the system, and let

$$y_1(t) = \int_{-\infty}^{2t} x_1(\tau) d\tau$$

be the corresponding output. Then consider a second input obtained by shifting $x_1(t)$ in time:

$$x_2(t) = x_1(t - t_0)$$

The output corresponding to this input is

$$\begin{aligned}
 y_2(t) &= \int_{-\infty}^{2t} x_2(\tau) d\tau \\
 &= \int_{-\infty}^{2t} x_1(\tau - t_0) d\tau \\
 &= \int_{-\infty}^{2t-t_0} x_1(\tau) d\tau
 \end{aligned}$$

Now, consider

$$\begin{aligned}
 y_1(t) &= \int_{-\infty}^{2t} x_1(\tau) d\tau \\
 y_1(t - t_0) &= \int_{-\infty}^{2(t-t_0)} x_1(\tau) d\tau
 \end{aligned}$$

We see that $y_2(t) \neq y_1(t - t_0)$, and therefore this system is not time-invariant.

(b) (i)

$$\begin{aligned}
 y(t) &= \frac{dx(t)}{dt} \\
 y(t) &= \lim_{\Delta t \rightarrow 0} \frac{x(t) - x(t - \Delta t)}{\Delta t}
 \end{aligned}$$

The present output depends on the present and past inputs. Consequently, this system is a *with memory* system.

(ii) Consider the bounded input

$$x(t) = u(t) \quad \text{unit step function}$$

then, the corresponding output is

$$\begin{aligned}
 y(t) &= \frac{du(t)}{dt} \\
 &= \delta(t)
 \end{aligned}$$

The bounded input produces an unbounded output. We may conclude that the given system is *unstable*.

(iii) The present output depends on the present and past values of the input. Consequently, this system is *causal*.

(iv) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$\begin{aligned}
 x_1(t) &\rightarrow y_1(t) = \frac{dx_1(t)}{dt} \\
 x_2(t) &\rightarrow y_2(t) = \frac{dx_2(t)}{dt} \\
 x_3(t) &\rightarrow y_3(t) = \frac{dx_3(t)}{dt}
 \end{aligned}$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$. That is,

$$x_3(t) = ax_1(t) + bx_2(t)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(t) = ay_1(t) + by_2(t)$$

Consider the LHS of the above equation

$$\begin{aligned}
 y_3(t) &= \frac{dx_3(t)}{dt} \\
 &= \frac{d}{dt} (ax_1(t) + bx_2(t))
 \end{aligned}$$

$$\begin{aligned}
&= a \frac{dx_1(t)}{dt} + b \frac{dx_2(t)}{dt} \\
&= ay_1(t) + by_2(t) = \text{RHS}
\end{aligned}$$

Since LHS = RHS, we conclude that the system is *linear*.

(v) Let $x_1(t)$ be an arbitrary input to the system, and let

$$y_1(t) = \frac{dx_1(t)}{dt}$$

be the corresponding output. Then consider a second input obtained by shifting $x_1(t)$ in time:

$$x_2(t) = x_1(t - t_0)$$

The output corresponding to this input is

$$\begin{aligned}
y_2(t) &= \frac{dx_2(t)}{dt} \\
&= \frac{dx_1(t - t_0)}{dt}
\end{aligned}$$

Now, consider

$$\begin{aligned}
y_1(t) &= \frac{dx_1(t)}{dt} \\
y_1(t - t_0) &= \frac{dx_1(t - t_0)}{dt}
\end{aligned}$$

We see that $y_2(t) = y_1(t - t_0)$, and therefore this system is time-invariant.

(c) (i) This system has memory, since the value of the output signal $y(t)$ at time t depends on the present and one past input signal $x(t)$.

(ii) Assume that the input signal $x(t)$ satisfies the condition

$$|x(t)| \leq B_x < \infty, \quad \text{for all } t$$

We then find that

$$\begin{aligned}
|y(t)| &= \begin{cases} 0, & t < 0 \\ |x(t) + x(t-2)|, & t \geq 0 \end{cases} \\
&\leq \begin{cases} 0, & t < 0 \\ |x(t)| + |x(t-2)|, & t \geq 0 \end{cases} \\
&\leq \begin{cases} 0, & t < 0 \\ 2B_x, & t \geq 0 \end{cases}
\end{aligned}$$

We conclude that in this system if any input is bounded by an arbitrary positive number B_x , the corresponding output is guaranteed to be bounded. Thus, the given system is stable.

(iii) This system is causal, since the present output $y(t)$ at time t depends on the present and past values of the input signal $x(t)$.

(iv) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$x_1(t) \rightarrow y_1(t) = \begin{cases} 0, & t < 0 \\ x_1(t) + x_1(t-2), & t \geq 0 \end{cases}$$

$$x_2(t) \rightarrow y_2(t) = \begin{cases} 0, & t < 0 \\ x_2(t) + x_2(t-2), & t \geq 0 \end{cases}$$

$$x_3(t) \rightarrow y_3(t) = \begin{cases} 0, & t < 0 \\ x_3(t) + x_3(t-2), & t \geq 0 \end{cases}$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$. That is,

$$x_3(t) = ax_1(t) + bx_2(t)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(t) = ay_1(t) + by_2(t)$$

Consider the LHS of the above equation

$$\begin{aligned}
y_3(t) &= \begin{cases} 0, & t < 0 \\ x_3(t) + x_3(t-2), & t \geq 0 \end{cases} \\
&= \begin{cases} 0, & t < 0 \\ [ax_1(t) + bx_2(t)] \\ + [ax_1(t-2) + bx_2(t-2)], & t \geq 0 \end{cases} \\
&= \begin{cases} 0, & t < 0 \\ a[x_1(t) + x_1(t-2)] \\ + b[x_2(t) + x_2(t-2)], & t \geq 0 \end{cases} \\
&= ay_1(t) + by_2(t) = \text{RHS}
\end{aligned}$$

Since LHS = RHS, we conclude that the system is *linear*.

(v) Let $x_1(t)$ be an arbitrary input to the system, and let

$$y_1(t) = \begin{cases} 0, & t < 0 \\ x_1(t) + x_1(t-2), & t \geq 0 \end{cases}$$

be the corresponding output. Then consider a second input obtained by shifting $x_1(t)$ in time:

$$x_2(t) = x_1(t - t_0)$$

The output corresponding to this input is

$$\begin{aligned} y_2(t) &= \begin{cases} 0, & t < 0 \\ x_2(t) + x_2(t-2), & t \geq 0 \end{cases} \\ &= \begin{cases} 0, & t < 0 \\ x_1(t - t_0) + x_1(t - t_0 - 2), & t \geq 0 \end{cases} \end{aligned}$$

Now, consider

$$\begin{aligned} y_1(t) &= \begin{cases} 0, & t < 0 \\ x_1(t) + x_1(t-2), & t \geq 0 \end{cases} \\ y_1(t - t_0) &= \begin{cases} 0, & t - t_0 < 0 \\ x_1(t - t_0) + x_1(t - t_0 - 2), & t - t_0 \geq 0 \end{cases} \\ &= \begin{cases} 0, & t < t_0 \\ x_1(t - t_0) + x_1(t - t_0 - 2), & t \geq t_0 \end{cases} \end{aligned}$$

We see that $y_2(t) \neq y_1(t - t_0)$, and therefore this system is not time-invariant.

(d) (i) This system has memory, since the value of the output signal $y(t)$ at time t depends on the present and one past input signal $x(t)$.

(ii) Assume that the input signal $x(t)$ satisfies the condition

$$|x(t)| \leq B_x < \infty, \quad \text{for all } t$$

We then find that

$$\begin{aligned} |y(t)| &= \begin{cases} 0, & x(t) < 0 \\ |x(t) + x(t-2)|, & x(t) \geq 0 \end{cases} \\ &\leq \begin{cases} 0, & x(t) < 0 \\ |x(t)| + |x(t-2)|, & x(t) \geq 0 \end{cases} \\ &\leq \begin{cases} 0, & x(t) < 0 \\ 2B_x, & x(t) \geq 0 \end{cases} \end{aligned}$$

We conclude that in this system if any input is bounded by an arbitrary positive number B_x , the corresponding output is guaranteed to be bounded. Thus, the given system is stable.

(iii) This system is causal, since the present output $y(t)$ at time t depends on the present and past values of the input signal $x(t)$.

(iv) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$x_1(t) \rightarrow y_1(t) = \begin{cases} 0, & x_1(t) < 0 \\ x_1(t) + x_1(t-2), & x_1(t) \geq 0 \end{cases}$$

$$x_2(t) \rightarrow y_2(t) = \begin{cases} 0, & x_2(t) < 0 \\ x_2(t) + x_2(t-2), & x_2(t) \geq 0 \end{cases}$$

$$x_3(t) \rightarrow y_3(t) = \begin{cases} 0, & x_3(t) < 0 \\ x_3(t) + x_3(t-2), & x_3(t) \geq 0 \end{cases}$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$. That is,

$$x_3(t) = ax_1(t) + bx_2(t)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(t) = ay_1(t) + by_2(t)$$

Consider the LHS of the above equation

$$\begin{aligned} y_3(t) &= \begin{cases} 0, & x_3(t) < 0 \\ x_3(t) + x_3(t-2), & x_3(t) \geq 0 \end{cases} \\ &= \begin{cases} 0, & (ax_1(t) + bx_2(t)) < 0 \\ [ax_1(t) + bx_2(t)] \\ + [ax_1(t-2) + bx_2(t-2)], & (ax_1(t) + bx_2(t)) \geq 0 \end{cases} \\ &= \begin{cases} 0, & (ax_1(t) + bx_2(t)) < 0 \\ a[x_1(t) + x_1(t-2)] \\ + b[x_2(t) + x_2(t-2)], & (ax_1(t) + bx_2(t)) \geq 0 \end{cases} \\ &\neq \text{RHS} \end{aligned}$$

Since LHS $\neq$ RHS, we conclude that the system is *nonlinear*.

(v) Let $x_1(t)$ be an arbitrary input to the system, and let

$$y_1(t) = \begin{cases} 0, & x_1(t) < 0 \\ x_1(t) + x_1(t-2), & x_1(t) \geq 0 \end{cases}$$

be the corresponding output. Then consider a second input obtained by shifting $x_1(t)$ in time:

$$x_2(t) = x_1(t - t_0)$$

The output corresponding to this input is

$$\begin{aligned} y_2(t) &= \begin{cases} 0, & x_2(t) < 0 \\ x_2(t) + x_2(t-2), & x_2(t) \geq 0 \end{cases} \\ &= \begin{cases} 0, & x_1(t - t_0) < 0 \\ x_1(t - t_0) + x_1(t - t_0 - 2), & x_1(t - t_0) \geq 0 \end{cases} \end{aligned}$$

Now, consider

$$\begin{aligned} y_1(t) &= \begin{cases} 0, & x_1(t) < 0 \\ x_1(t) + x_1(t-2), & x_1(t) \geq 0 \end{cases} \\ y_1(t - t_0) &= \begin{cases} 0, & x_1(t - t_0) < 0 \\ x_1(t - t_0) + x_1(t - t_0 - 2), & x_1(t - t_0) \geq 0 \end{cases} \end{aligned}$$

We see that $y_2(t) = y_1(t - t_0)$, and therefore this system is time-invariant.

(e)

$$y(n) = \mathcal{E}\{x(n-1)\}$$

or, equivalently

$$y(n) = \frac{x(n-1) + x(-n-1)}{2}$$

(i) This system has memory, since the value of the output signal $y(n)$ at time n depends on the present and one past input signal $x(n)$.

(ii) Assume that the input signal $x(n)$ satisfies the condition

$$|x(n)| \leq B_x < \infty, \quad \text{for all } n$$

We then find that

$$\begin{aligned} |y(n)| &= \left| \frac{x(n-1) + x(-n-1)}{2} \right| \\ &\leq \left| \frac{x(n-1)}{2} \right| + \left| \frac{x(-n-1)}{2} \right| \\ &\leq B_x \end{aligned}$$

We conclude that in this system if any input is bounded by an arbitrary positive number B_x , the corresponding output is guaranteed to be bounded. Thus, the given system is stable.

(iii) This system is noncausal, since the present output $y(n)$ at time n depends on the future values of the input signal $x(n)$.

(iv) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(n)$ and $x_2(n)$.

$$x_1(n) \rightarrow y_1(n) = \frac{x_1(n-1) + x_1(-n-1)}{2}$$

$$x_2(n) \rightarrow y_2(n) = \frac{x_2(n-1) + x_2(-n-1)}{2}$$

$$x_3(n) \rightarrow y_3(n) = \frac{x_3(n-1) + x_3(-n-1)}{2}$$

Let $x_3(n)$ be a linear combination of $x_1(n)$ and $x_2(n)$. That is,

$$x_3(n) = ax_1(n) + bx_2(n)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(n) = ay_1(n) + by_2(n)$$

Consider the LHS of the above equation

$$\begin{aligned} y_3(n) &= \frac{x_3(n-1) + x_3(-n-1)}{2} \\ &= \frac{1}{2} [ax_1(n-1) + bx_2(n-1) \\ &\quad + [ax_1(-n-1) + bx_2(-n-1)]] \\ &= a[x_1(n-1) + x_1(-n-1)] \\ &\quad + b[x_2(n-1) + x_2(-n-1)] \\ &= ay_1(n) + by_2(n) = \text{RHS} \end{aligned}$$

Since LHS = RHS, we conclude that the system is linear.

(v) Let $x_1(n)$ be an arbitrary input to the system, and let

$$y_1(n) = \frac{x_1(n-1) + x_1(-n-1)}{2}$$

be the corresponding output. Then consider a second input obtained by shifting $x_1(n)$ in time:

$$x_2(n) = x_1(n - n_0)$$

The output corresponding to this input is

$$\begin{aligned} y_2(n) &= \frac{x_2(n-1) + x_2(-n-1)}{2} \\ &= \frac{x_1(n-n_0-1) + x_1(-n-n_0-1)}{2} \end{aligned}$$

Now, consider

$$\begin{aligned} y_1(n) &= \frac{x_1(n-1) + x_1(-n-1)}{2} \\ y_1(n-n_0) &= \frac{x_1(n-n_0-1) + x_1(-n+n_0-1)}{2} \end{aligned}$$

We see that $y_2(n) \neq y_1(n-n_0)$, and therefore this system is *not time-invariant*.

(f) (i) This system has memory, since the value of the output signal $y(n)$ at time n depends on the future input.

$$y(n)|_{n=n_0} = y(n_0) = x(n_0^2)$$

and $y(n)|_{n=-n_0} = y(-n_0) = x(n_0^2)$

(ii) Assume that the input signal $x(n)$ satisfies the condition

$$|x(n)| \leq B_x < \infty, \quad \text{for all } n$$

We then find that

$$\begin{aligned} |y(n)| &= |x(n^2)| \\ &= B_x \end{aligned}$$

We conclude that in this system if any input is bounded by an arbitrary positive number B_x , the corresponding output is guaranteed to be bounded. Thus, the given system is stable.

(iii) This system is noncausal, since the present output $y(n)$ at time n depends on the future values of the input signal $x(n)$.

(iv) To determine whether or not a system is linear, we consider two arbitrary inputs $x_1(n)$ and $x_2(n)$.

$$x_1(n) \rightarrow y_1(n) = x_1(n^2)$$

$$x_2(n) \rightarrow y_2(n) = x_2(n^2)$$

$$x_3(n) \rightarrow y_3(n) = x_3(n^2)$$

Let $x_3(n)$ be a linear combination of $x_1(n)$ and $x_2(n)$. That is,

$$x_3(n) = ax_1(n) + bx_2(n)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(n) = ay_1(n) + by_2(n)$$

Consider the LHS of the above equation

$$\begin{aligned} y_3(n) &= x_3(n^2) \\ &= ax_1(n^2) + bx_2(n^2) \\ &= ay_1(n) + by_2(n) = \text{RHS} \end{aligned}$$

Since LHS = RHS, we conclude that the system is *linear*.

(v) Let $x_1(n)$ be an arbitrary input to the system, and let

$$y_1(n) = x_1(n^2)$$

be the corresponding output. Then consider a second input obtained by shifting $x_1(n)$ in time:

$$x_2(n) = x_1(n-n_0)$$

The output corresponding to this input is

$$\begin{aligned} y_2(n) &= x_2(n^2) \\ &= x_1(n^2 - n_0) \end{aligned}$$

Now, consider

$$\begin{aligned} y_1(n) &= x_1(n^2) \\ y_1(n-n_0) &= x_1((n-n_0)^2) \end{aligned}$$

We see that $y_2(n) \neq y_1(n-n_0)$, and therefore this system is *not time-invariant*.

Example 1.37 (a) An arbitrary real-valued continuous-time signal $x(t)$ occupies the entire interval $-\infty < x(t) < \infty$. Show that the energy of the signal $x(t)$ is equal to the sum of the energy of the even component $x_e(t)$ and the energy of the odd component $x_o(t)$.

(b) Show that an arbitrary real-valued discrete-time signal $x(n)$ satisfies a relationship similar to that satisfied by the continuous signal in part (a).

Solution

(a) See Property 6 of continuous-time even and odd signals.

(b) See Property 4 of discrete-time even and odd signals.

Example 1.38 Let $x(t)$ be a continuous-time signal, and let

$$y_1(t) = x(2t) \quad \text{and} \quad y_2(t) = x(t/2)$$

The signal $y_1(t)$ represents a speeded up version of $x(t)$ in the sense that the duration of the signal is cut in half. Similarly, $y_2(t)$ represents a slowed down version of $x(t)$ in the sense that the duration of the signal is doubled. Consider the following statements:

- (1) If $x(t)$ is periodic, then $y_1(t)$ is periodic.
- (2) If $y_1(t)$ is periodic, then $x(t)$ is periodic.
- (3) If $x(t)$ is periodic, then $y_2(t)$ is periodic.
- (4) If $y_2(t)$ is periodic, then $x(t)$ is periodic.

For each of these statements, determine whether it is true and, if so, determine the relationship between the fundamental periods of the two signals considered in the statement.

Solution

(1) Let $x(t)$ be periodic with period T , i.e.,

$$x(t) = x(t + T)$$

$$x(2t) = x(2t + T)$$

and $y_1(t) = x(2t)$

$$y_1(t + T_0) = x(2(t + T_0))$$

$$= x(2t + 2T_0)$$

$$= x(2t + T) \quad \text{for } T = 2T_0$$

$$= x(2t) = y_1(t)$$

$y_1(t)$ is periodic with period $T_0 = T/2$.

(2) Let $y_1(t)$ be periodic with period T_0 .

$$y_1(t) = y_1(t + T_0)$$

Now, consider

$$x(2t) = y_1(t)$$

$$x(2(t + T/2)) = y_1(t + T/2)$$

$$x(2t + T) = y_1(t + T/2)$$

$$= y_1(t + T_0) \quad \text{for } T_0 = T/2$$

$$= y_1(t) = x(2t)$$

$x(t)$ is periodic with period $T = 2T_0$.

(3) Let $x(t)$ be periodic with period T , i.e.,

$$x(t) = x(t + T)$$

$$x(t/2) = x(t/2 + T)$$

and

$$y_2(t) = x(t/2)$$

$$y_2(t + T_0) = x((t + T_0)/2)$$

$$= x(t/2 + T_0/2)$$

$$= x(t/2 + T) \quad \text{for } T = T_0/2$$

$$= x(t/2) = y_2(t)$$

$y_2(t)$ is periodic with period $T_0 = 2T$.

(4) Let $y_2(t)$ be periodic with period T_0 .

$$y_2(t) = y_2(t + T_0)$$

Now, consider

$$x(t/2) = y_2(t)$$

$$x((t + 2T)/2) = y_2(t + 2T)$$

$$x(t/2 + T) = y_2(t + 2T)$$

$$= y_2(t + T_0) \quad \text{for } T_0 = 2T$$

$x(t)$ is periodic with period $T = T_0/2$.

Example 1.39 Let $x(n)$ be a discrete-time signal, and let

$$y_1(n) = x(2n)$$

$$\text{and } y_2(n) = \begin{cases} x(n/2), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases}$$

The signals $y_1(n)$ and $y_2(n)$ respectively represent in some sense the speeded up and slowed down versions of $x(n)$. Consider the following statements:

(1) If $x(n)$ is periodic, then $y_1(n)$ is periodic.

(2) If $y_1(n)$ is periodic, then $x(n)$ is periodic.

(3) If $x(n)$ is periodic, then $y_2(n)$ is periodic.

(4) If $y_2(n)$ is periodic, then $x(n)$ is periodic.

For each of these statements, determine whether it is true and, if so, determine the relationship between the fundamental periods of the two signals considered in the statement.

Solution

(1) True. Let $x(n)$ be periodic with period N .

$$x(n) = x(n + mN), \quad m = \text{an integer}$$

$$x(2n) = x(2n + mN)$$

Now, consider

$$y_1(n) = x(2n)$$

$$y_1(n + N_0) = x(2(n + N_0))$$

$$= x(2n + 2N_0)$$

$$= x(2n + mN), \quad \text{for } mN = 2N_0$$

$$= x(2n) = y_1(n)$$

$y_1(n)$ is periodic with period

$$\begin{aligned} N_0 &= \frac{mN}{2} \\ &= \begin{cases} \frac{N}{2}, & \text{if } N \text{ is even } (m = 1) \\ N, & \text{if } N \text{ is odd } (m = 2) \end{cases} \end{aligned}$$

(2) False. $y_1(n)$ being periodic does not imply $x(n)$ is periodic, i.e., let $x(n) = g(n) + h(n)$, where

$$g(n) = \begin{cases} 1, & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases}$$

$$\text{and } h(n) = \begin{cases} 0, & n = \text{an even number} \\ (1/2)^n, & n = \text{an odd number} \end{cases}$$

Then $y_1(n) = x(2n)$ is periodic but $x(n)$ is clearly not periodic (Fig. 1.32).

(3) True. Let $x(n)$ be periodic with period N .

$$x(n) = x(n + mN) \quad m = \text{an integer}$$

$$x(n/2) = x(n/2 + mN)$$

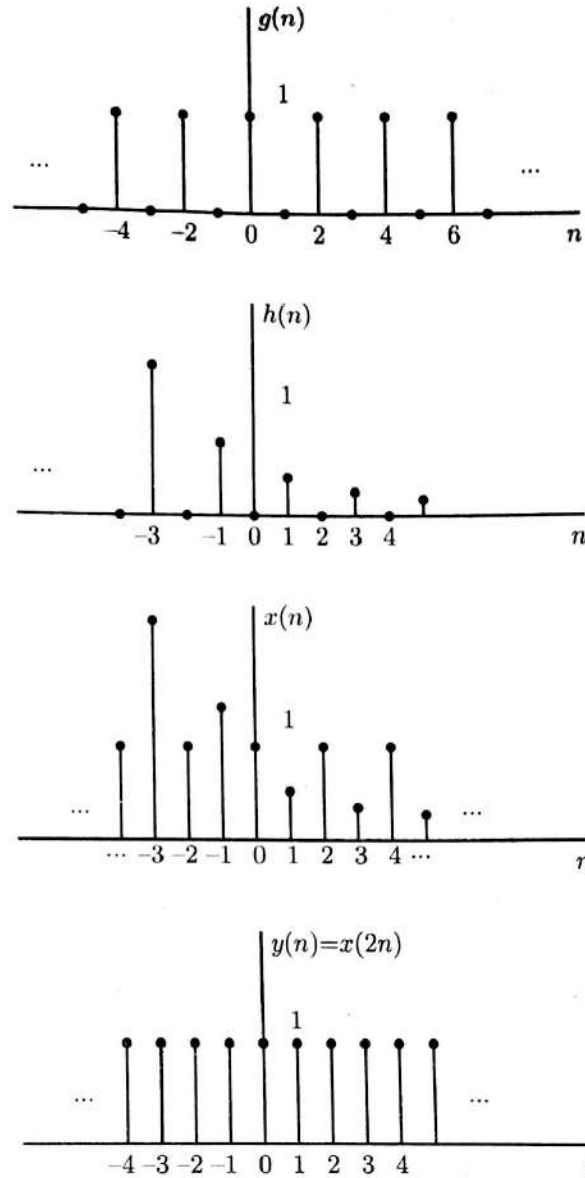


Fig. 1.32

Now, consider

$$y_2(n) = \begin{cases} x(n/2), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases}$$

$$y_2(n + N_0) = \begin{cases} x\left(\frac{1}{2}(n + N_0)\right), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases}$$

$$= \begin{cases} x\left(\frac{n}{2} + \frac{N_0}{2}\right), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases}$$

Assume $mN = N_0/2$, then we have

$$\begin{aligned} y_2(n + N_0) &= \begin{cases} x\left(\frac{n}{2} + mN\right), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases} \\ &= \begin{cases} x\left(\frac{n}{2}\right), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases} \\ &= y_2(n) \end{aligned}$$

$y_2(n)$ is periodic with period

$$\begin{aligned} N_0 &= 2mN \\ &= 2N \quad (m = 1) \end{aligned}$$

(4) True. Let $y_2(n)$ be periodic with period N .

$$y_2(n) = y_2(n + mN), \quad m = \text{an integer}$$

Now, consider

$$y_2(n) = \begin{cases} x\left(\frac{n}{2}\right), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases}$$

$$y_2(n + 2N_0) = \begin{cases} x\left(\frac{1}{2}(n + 2N_0)\right), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases}$$

$$y_2(n + 2N_0) = \begin{cases} x\left(\frac{n}{2} + N_0\right), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases}$$

Assume $N = 2N_0$, then we have

$$y_2(n + N) = \begin{cases} x\left(\frac{n}{2} + N_0\right), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases}$$

$$y_2(n) = \begin{cases} x\left(\frac{n}{2} + N_0\right), & n = \text{an even number} \\ 0, & n = \text{an odd number} \end{cases}$$

$x(n)$ is periodic with period $N_0 = N/2$.

Example 1.40 A discrete-time system is both linear and time-invariant. Suppose the output $y(n)$ due to an input $x(n) = \delta(n)$ is as shown in Fig. 1.33(a).

(a) Find the output $y_1(n)$ due to an input $x_1(n) = \delta(n - 1)$.

(b) Find the output $y_2(n)$ due to an input $x_2(n) = 2\delta(n) - \delta(n - 2)$.

(c) Find the output $y_3(n)$ due to the input depicted in Fig. 1.33(b).

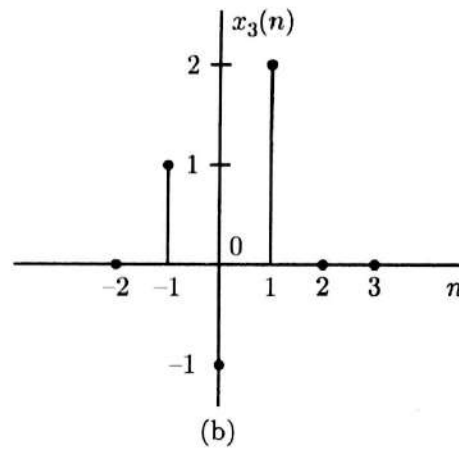
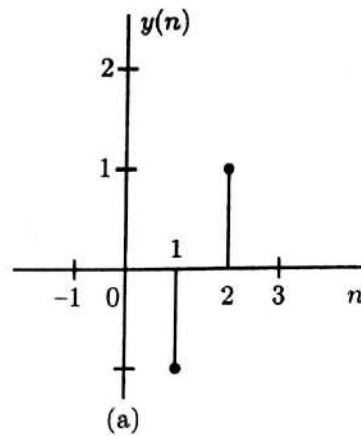


Fig. 1.33

Solution

(a) The given system is time-invariant. Therefore, a time shift in the input signal causes an identical time shift in the output signal [Fig. 1.34(a)]. The output corresponding to the input $x_1(n) = \delta(n - 1)$ is $y_1(n) = y(n - 1)$.

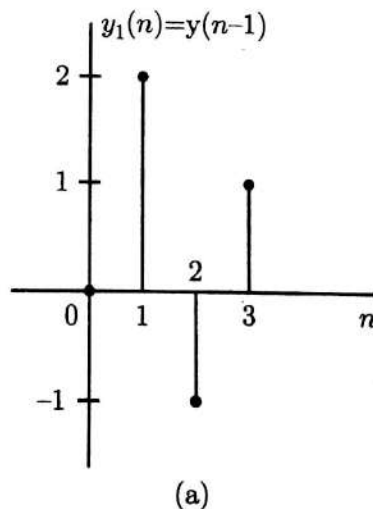


Fig. 1.34(a)

(b) The given system is both linear and time-invariant. The output [shown in Fig. 1.34(b)] corresponding to the input (i) $2\delta(n)$ is $2y(n)$, (ii) $\delta(n-2)$ is $y(n-2)$, and (iii) $2\delta(n) - \delta(n-2)$ is $y_2(n) = 2y(n) - y(n-2)$.

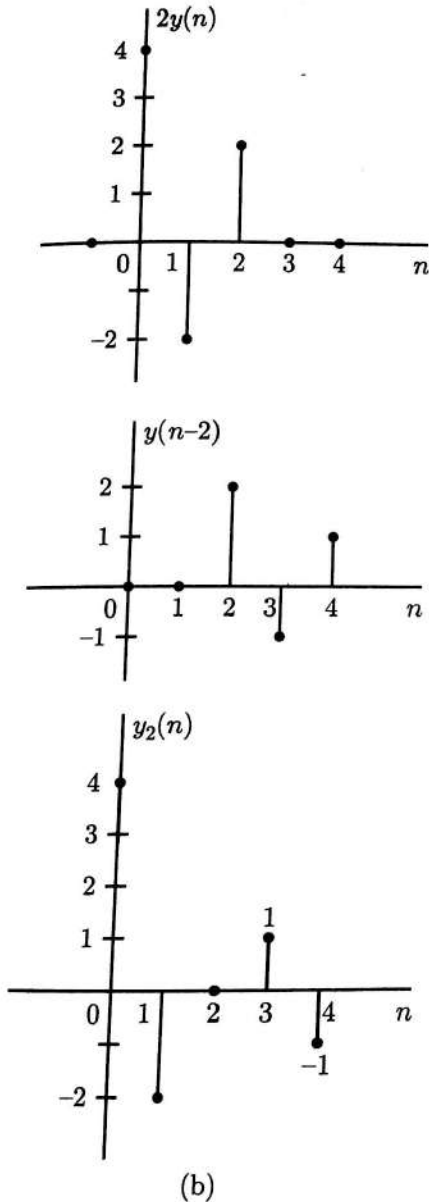


Fig. 1.34(b)

(c) From Fig. 1.33(b), the output [shown in Fig. 1.34(c)] corresponding to the input $x_3(n) = \delta(n+1) - \delta(n) + 2\delta(n-1)$ is $y_3(n) = y(n+1) - y(n) + 2y(n-1)$.

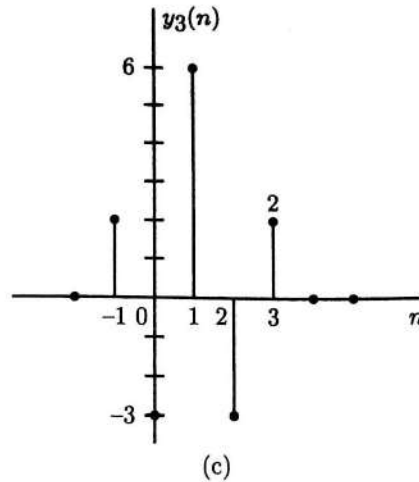


Fig. 1.34(c)

Example 1.41 (a) Find and sketch dx/dt for the signal $x(t)$ shown in Fig. 1.35(a).

(b) Find and sketch d^2x/dt^2 for the signal $x(t)$ depicted in Fig. 1.35(b).

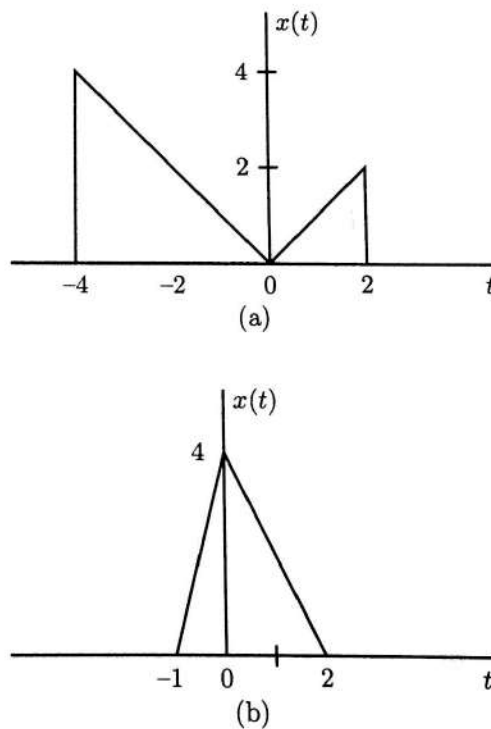


Fig. 1.35

Solution

(a) From Fig. 1.35(a)

$$x(t) = 4u(t+4) - r(t+4) + 2r(t) - r(t-2) - 2u(t-2)$$

$$\begin{aligned}\frac{d}{dt}x(t) &= 4\delta(t+4) - u(t+4) \\ &\quad + 2u(t) - u(t-2) - 2\delta(t-2)\end{aligned}$$

The derivative of $x(t)$ is shown in Fig. 1.36(a).

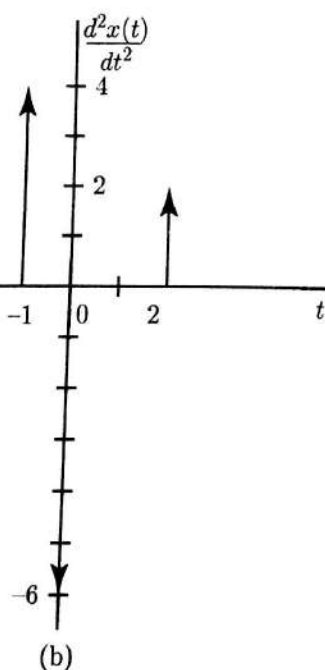
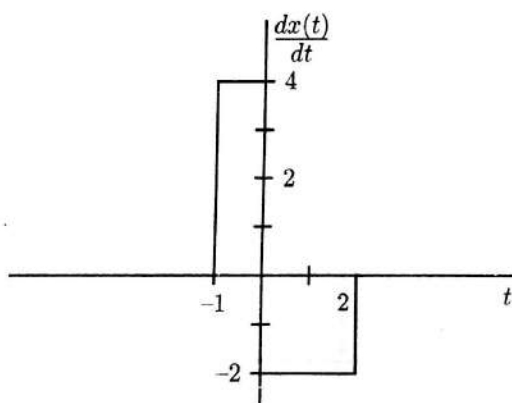
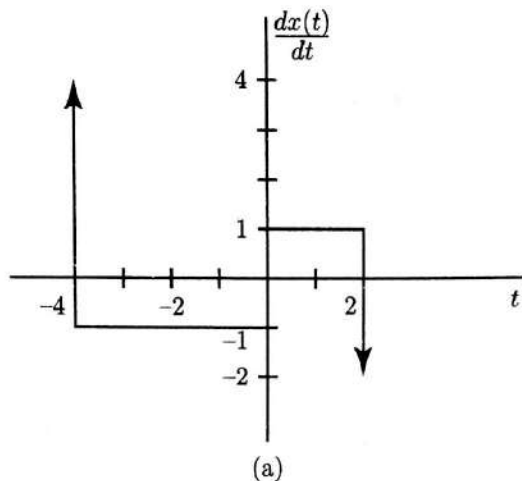


Fig. 1.36

(b) From Fig. 1.35(b)

$$x(t) = 4r(t+1) - 4r(t) - 2r(t) + 2r(t-2)$$

$$\frac{d}{dt}x(t) = 4u(t+1) - 4u(t) - 2u(t) + 2u(t-2)$$

$$\frac{d^2}{dt^2}x(t) = 4\delta(t+1) - 4\delta(t) - 2\delta(t) + 2\delta(t-2)$$

dx/dt and d^2x/dt^2 is shown in Fig. 1.36(b).

Example 1.42 Find and sketch $\int_{-\infty}^t x(t)dt$ for the signal $x(t)$ illustrated in Figs 1.37(a) and (b).

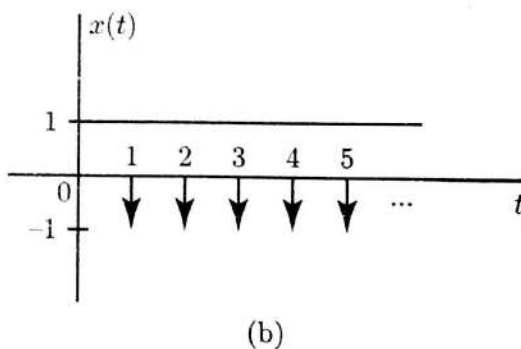
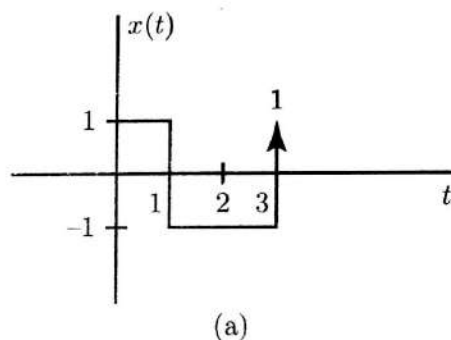


Fig. 1.37

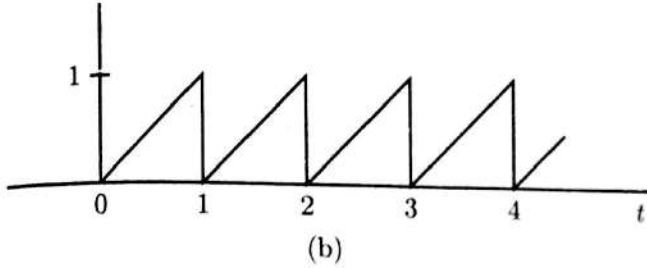
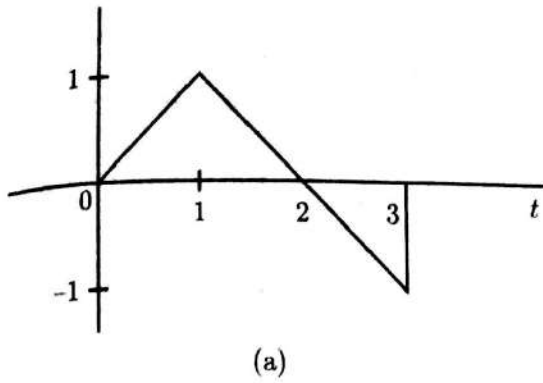
Solution

(a) From Fig. 1.37(a)

$$x(t) = u(t) - 2u(t-1) + u(t-3) + \delta(t-3)$$

$$\int_{-\infty}^t x(t)dt = r(t) - 2r(t-1) + r(t-3) + u(t-3)$$

The integrated version of Fig. 1.37(a) is shown in Fig. 1.38(a).


Fig. 1.38

From Fig. 1.37(b)

$$x(t) = u(t) - \delta(t-1) - \delta(t-2) - \delta(t-3)$$

$$\int_{-\infty}^t x(t) dt = r(t) - u(t-1) - u(t-2) - u(t-3)$$

The integrated version of Fig. 1.37(b) is shown in Fig. 1.38(b).

Example 1.43 Sketch the following signals:

(a) $x_1(t) = \delta(\cos t)$

(b) $x_2(t) = \operatorname{sgn}\left(\sin \frac{\pi}{T} t\right)$

(c) $x_3(t) = t \operatorname{sgn}(\cos t) \quad 0 \leq t \leq 2\pi$

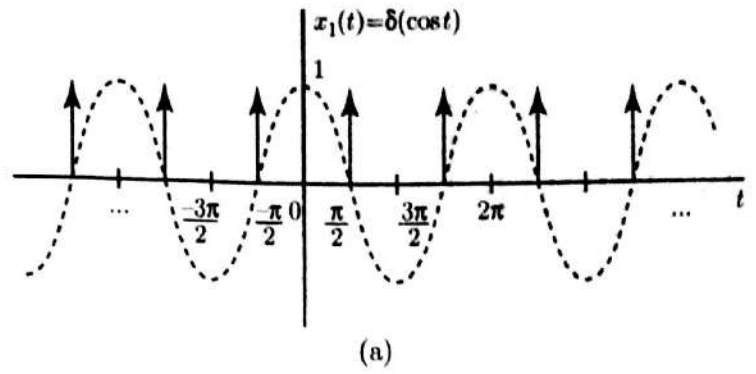
(d) $x_4(t) = u\left(\sin \frac{\pi}{T} t\right) - u\left(-\sin \frac{\pi}{T} t\right)$

Solution

(a) Since $\delta(t) = 0, \quad t \neq 0$

we get $x_1(t) = \delta(\cos t) = 0, \quad \cos t \neq 0$

$x_1(t)$ is shown in Fig. 1.39(a).

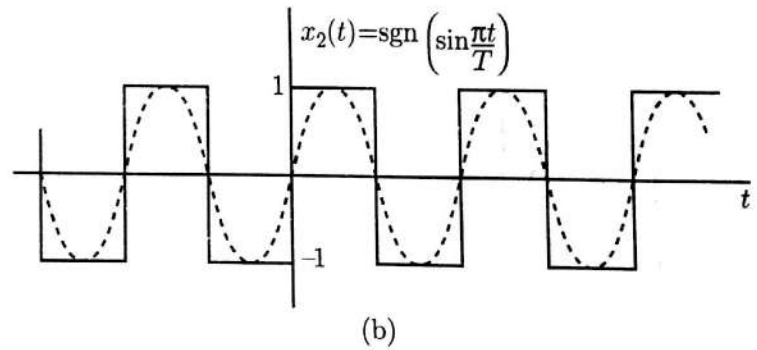

Fig. 1.39(a)

(b) Since
$$\operatorname{sgn}(t) = \begin{cases} 1, & t > 0 \\ -1, & t < 0 \end{cases}$$

we get

$$x_2(t) = \operatorname{sgn}\left(\sin \frac{\pi}{T} t\right) = \begin{cases} 1, & \sin \frac{\pi}{T} t > 0 \\ -1, & \sin \frac{\pi}{T} t < 0 \end{cases}$$

$x_2(t)$ is shown in Fig. 1.39(b)

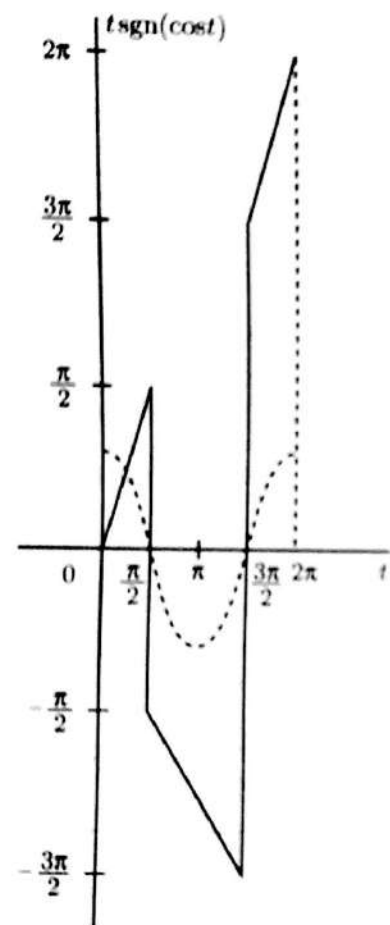
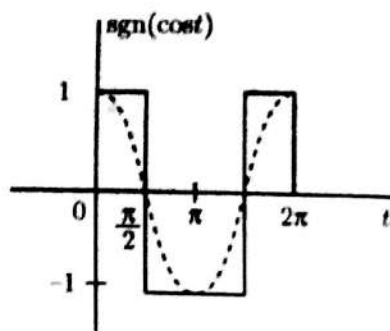

Fig. 1.39(b)

(c) Since
$$\operatorname{sgn}(t) = \begin{cases} 1, & t > 0 \\ -1, & t < 0 \end{cases}$$

we get

$$t \operatorname{sgn}(\cos t) = \begin{cases} t, & \cos t > 0 \\ -t, & \cos t < 0 \end{cases}$$

The signal $t \operatorname{sgn}(\cos t)$ is shown in Fig. 1.39(c).



(c)

Fig. 1.39(c)

(d) Since

$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

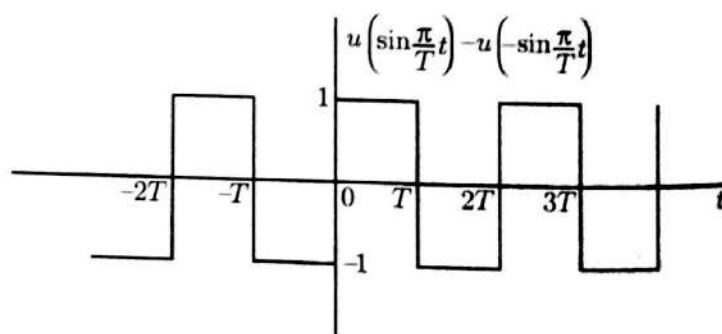
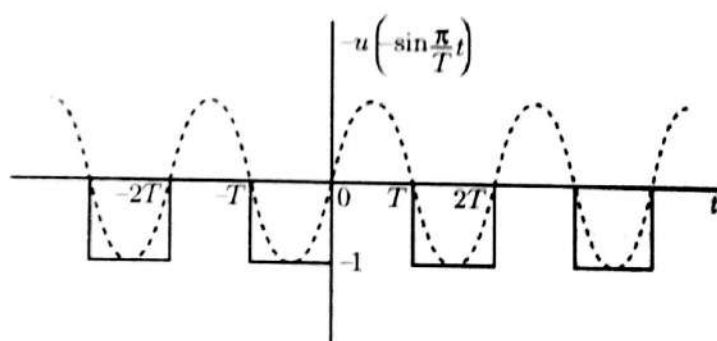
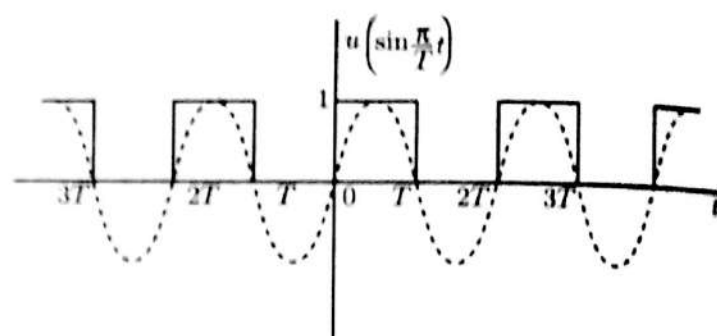
we get

$$u\left(\sin\frac{\pi}{T}t\right) = \begin{cases} 1, & \sin\frac{\pi}{T}t > 0 \\ 0, & \sin\frac{\pi}{T}t < 0 \end{cases}$$

$$u\left(-\sin\frac{\pi}{T}t\right) = \begin{cases} 1, & \sin\frac{\pi}{T}t < 0 \\ 0, & \sin\frac{\pi}{T}t > 0 \end{cases}$$

$$u\left(\sin\frac{\pi}{T}t\right) - u\left(-\sin\frac{\pi}{T}t\right) = \begin{cases} 1, & \sin\frac{\pi}{T}t > 0 \\ -1, & \sin\frac{\pi}{T}t < 0 \end{cases}$$

The signal $u\left(\sin\frac{\pi}{T}t\right) - u\left(-\sin\frac{\pi}{T}t\right)$ is shown in Fig. 1.39(d).



(d)

Fig. 1.39(d)

Example 1.44 Write the equation for the waveforms in Fig. 1.40 using shifted step functions.

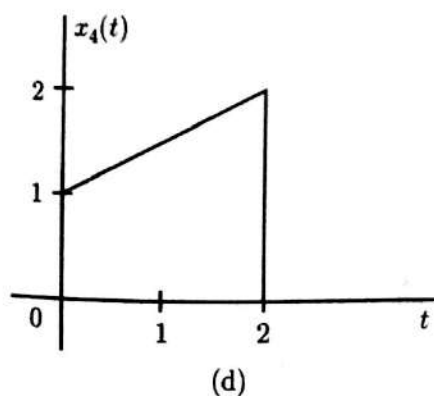
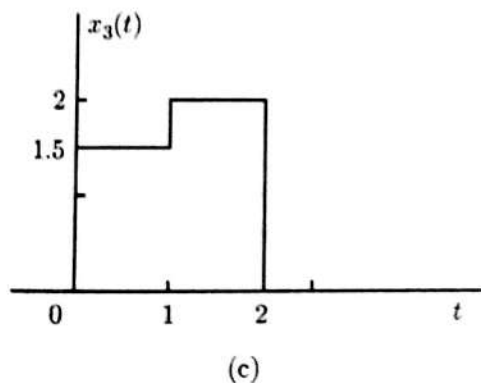
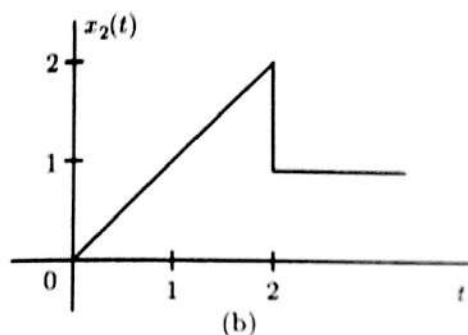
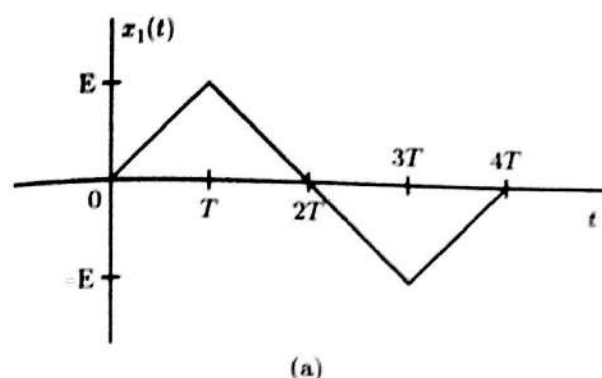


Fig. 1.40

Solution

(a)

$$\begin{aligned} x_1(t) &= \frac{E}{T}r(t) - \frac{2E}{T}r(t-T) + \frac{2E}{T}r(t-3T) \\ &\quad - \frac{E}{T}r(t-4T) \\ &= \frac{E}{T}tu(t) - \frac{2E}{T}(t-T)u(t-T) \\ &\quad + \frac{2E}{T}(t-3T)u(t-3T) \\ &\quad - \frac{E}{T}(t-4T)u(t-4T) \end{aligned}$$

(b)

$$\begin{aligned} x_2(t) &= r(t) - r(t-2) - u(t-2) \\ &= tu(t) - (t-2)u(t-2) - u(t-2) \\ &= tu(t) - (t-1)u(t-2) \end{aligned}$$

(c)

$$x_3(t) = 1.5u(t) + 0.5u(t-1) - 2u(t-2)$$

(d)

$$\begin{aligned} x_4(t) &= u(t) + \frac{1}{2}r(t) - \frac{1}{2}r(t-2) - 2u(t-2) \\ &= u(t) + \frac{1}{2}tu(t) - \frac{1}{2}(t-2)u(t-2) \\ &\quad - 2u(t-2) \\ &= \left(\frac{1}{2}t + 1\right)u(t) - \left(\frac{1}{2}t + 1\right)u(t-2) \end{aligned}$$

Example 1.45 The waveform $x(t)$ in Fig. 1.41 is defined as

$$x(t) = \begin{cases} \frac{3}{\epsilon^3}(t-\epsilon)^2, & 0 \leq t \leq \epsilon \\ 0, & \text{elsewhere} \end{cases}$$

Show that as $\epsilon \rightarrow 0$, $x(t)$ becomes a unit impulse.

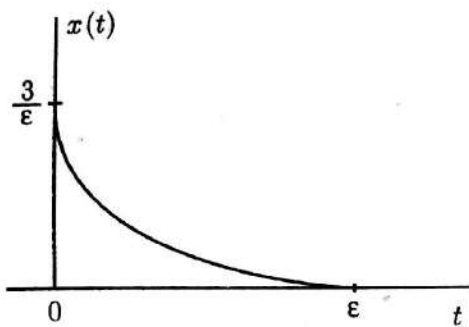


Fig. 1.41

Solution

Area under the unit impulse function is unity.

$$\begin{aligned} \int_{-\infty}^{\infty} x(t) dt &= \int_0^{\epsilon} \frac{3}{\epsilon^3} (t - \epsilon^2) dt \\ &= \frac{3}{\epsilon^3} \left[\frac{t^2}{2} - \epsilon^2 t \right]_0^{\epsilon} \\ &= 1 \end{aligned}$$

As $\epsilon \rightarrow 0$, the width of $x(t) \rightarrow 0$ and height approaches ∞ and its area remains at unity. Therefore, as $\epsilon \rightarrow 0$, $x(t)$ becomes a unit impulse function.

Example 1.46 Sketch and check whether the given signal

$$x(t) = \sum_{n=-\infty}^{\infty} e^{-(2t-n)} u(2t-n)$$

is periodic? If yes, compute its average power.

Solution

Consider the given signal

$$x(t) = \sum_{n=-\infty}^{\infty} e^{-(2t-n)} u(2t-n)$$

For $x(t)$ to be periodic, $x(t+T) = x(t)$; therefore,

$$\begin{aligned} x(t+T) &= \sum_{n=-\infty}^{\infty} e^{-(2(t+T)-n)} u(2(t+T)-n) \\ &= \sum_{n=-\infty}^{\infty} e^{-(2t+2T-n)} u(2t+2T-n) \end{aligned}$$

A change of variables is performed by letting $2T-n = -m$ (m -an integer), which also yields $m = -\infty$ as $n = -\infty$, and $m = \infty$ as $n = \infty$. Therefore,

$$\begin{aligned} x(t+T) &= \sum_{m=-\infty}^{\infty} e^{-(2t-m)} u(2t-m) \\ &= x(t) \end{aligned}$$

Thus, $x(t)$ is periodic if

$$2T - n = -m$$

$$T = \frac{n-m}{2}$$

Thus, the fundamental period (minimum value of T) is $T = 1/2$. Now, again consider the given signal

$$x(t) = \sum_{n=-\infty}^{\infty} e^{-(2t-n)} u(2t-n)$$

$$\text{Since } u(2t-n) = \begin{cases} 1, & (2t-n) > 0 \rightarrow t > \frac{n}{2} \\ 0, & \text{otherwise} \end{cases}$$

Therefore,

$$x(t) = \begin{cases} \sum_{n=-\infty}^{\infty} e^{-(2t-n)}, & t > \frac{n}{2} \\ 0, & \text{otherwise} \end{cases}$$

To determine $x(t)$ over the interval $0 < t < \frac{1}{2}$, $n = 0, -1, -2, \dots$. Therefore, we have

$$\begin{aligned} x(t) &= \sum_{n=-\infty}^0 e^{-(2t-n)}, \quad 0 < t < \frac{1}{2} \\ &= e^{-2t} \sum_{n=-\infty}^0 e^n \\ &= e^{-2t} \sum_{n=0}^{\infty} (e^{-1})^n \\ x(t) &= \frac{e^{-2t}}{1 - e^{-1}}, \quad 0 < t < \frac{1}{2} \end{aligned}$$

The average power contained in $x(t)$ is given by

$$\begin{aligned}
 P_x &= \frac{1}{T} \int_0^T |x(t)|^2 dt \\
 &= 2 \int_0^{\frac{1}{2}} \left[\frac{e^{-2t}}{1 - e^{-1}} \right]^2 dt \\
 &= \frac{2}{(1 - e^{-1})^2} \int_0^{\frac{1}{2}} e^{-4t} dt \\
 &= \frac{2}{(1 - e^{-1})^2} \left. \frac{e^{-4t}}{-4} \right|_0^{\frac{1}{2}} \\
 &= \frac{1}{2} \frac{1 - e^{-2}}{(1 - e^{-1})^2} \\
 &= \frac{1}{2} \frac{(1 - e^{-1})(1 + e^{-1})}{(1 - e^{-1})^2} \\
 &= \frac{1}{2} \frac{1 + e^{-1}}{1 - e^{-1}} \\
 P_x &= \frac{1}{2} \frac{e - 1}{e + 1}
 \end{aligned}$$

The signal $x(t)$ is shown in Fig. 1.29.

Example 1.47 Determine whether the following signals are periodic, and for those which are, find the fundamental period:

- (a) $x(n) = (-1)^n$
 (b) $x(n) = (-1)^{n^2}$
 (c) $x(t) = \sum_{k=-\infty}^{\infty} (-1)^k \delta(t - 2k)$

Solution

(a) Consider the given signal

$$x(n) = (-1)^n$$

For $x(n)$ to be periodic, $x(n + N) = x(n)$; therefore,

$$\begin{aligned}
 x(n + N) &= (-1)^{n+N} \\
 &= (-1)^n (-1)^N \\
 &= x(n) \underbrace{(-1)^N}_{=1}
 \end{aligned}$$

Thus, $x(n)$ is periodic if

$$\begin{aligned}
 (-1)^N &= 1 = (-1)^{2m} \\
 N &= 2m
 \end{aligned}$$

Therefore, the fundamental period (for minimum value of N , choose $m = 1$) of $x(n)$ is $N = 2$.

(b) Consider the given signal

$$x(n) = (-1)^{n^2}$$

For $x(n)$ to be periodic, $x(n + N) = x(n)$; therefore,

$$\begin{aligned}
 x(n + N) &= (-1)^{(n+N)^2} \\
 &= (-1)^{n^2 + N^2 + 2nN} \\
 &= (-1)^{n^2} (-1)^{N^2 + 2nN} \\
 &= x(n) \underbrace{(-1)^{N^2 + 2nN}}_{=1}
 \end{aligned}$$

Thus, $x(n)$ is periodic if

$$\begin{aligned}
 (-1)^{N^2 + 2nN} &= 1 = (-1)^{2m} \\
 N^2 + 2nN &= 2m
 \end{aligned}$$

That is, $x(n)$ is periodic if $N^2 + 2nN$ is an even integer. If $N = 1$, then

$$(-1)^{N^2 + 2nN} = (-1)^{1 + 2n} = -1 \neq 1$$

Now if $N = 2$, then

$$(-1)^{N^2 + 2nN} = (-1)^{4 + 4n} = 1$$

Therefore, the fundamental period (for minimum value of N , choose $m = 1$) of $x(n)$ is $N = 2$.

(c) Consider the given signal

$$x(t) = \sum_{k=-\infty}^{\infty} (-1)^k \delta(t - 2k)$$

For $x(t)$ to be periodic, $x(t + T) = x(t)$; therefore,

$$x(t + T) = \sum_{k=-\infty}^{\infty} (-1)^k \delta(t + T - 2k)$$

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A change of variables is performed by letting $T - 2k = -2m$ (m = an integer). Therefore,

$$\begin{aligned} x(t+T) &= \sum_{m=-\infty}^{\infty} (-1)^{T/2+m} \delta(t-2m) \\ &= \sum_{m=-\infty}^{\infty} \underbrace{(-1)^m \delta(t-2m)}_{=x(t)} (-1)^{T/2} \\ &= x(t) \underbrace{(-1)^{T/2}}_{=1} \end{aligned}$$

Thus, $x(t)$ is periodic if

$$\begin{aligned} (-1)^{T/2} &= 1 = (-1)^{2m} \\ \frac{T}{2} &= 2m \\ T &= 4m \end{aligned}$$

Therefore, the fundamental period of $x(t)$ is $T = 4$.

SUMMARY

1. Signals can be classified as continuous-time or discrete-time signals.
2. A discrete-time signal is defined only at discrete instants of time.
3. Continuous-time signals that satisfy the condition $x(t) = x(t+T)$ are periodic with fundamental period T .
4. A discrete-time signal $x(n)$ is periodic with period N if $x(n+N) = x(n)$ for some integer N .
5. The complex exponential $x(t) = e^{j\omega_0 t}$ is periodic with period $T = 2\pi/\omega_0$ for all ω_0 .
6. The complex signal $x(n) = e^{j\omega_0 n}$ is periodic only if $\omega_0/2\pi$ is a rational number.
7. Harmonically related continuous-time exponentials $x_k(t) = e^{jk\omega_0 t}$ are periodic with common period $T = 2\pi/\omega_0$.
8. The signal $x(t)$ is an energy signal if $0 < E < \infty$.
9. The signal $x(t)$ is a power signal if $0 < P < \infty$.
10. The signal $x(t-t_0)$ is a time-shifted version of $x(t)$. If $t_0 > 0$, then the signal is delayed by t_0 . If $t_0 < 0$, then $x(t-t_0)$ represents an advanced replica of $x(t)$.
11. The signal $x(-t)$ is obtained by reflecting $x(t)$ about $t = 0$.
12. The signal $x(\alpha t)$ is a scaled version of $x(t)$. If $\alpha > 1$, then $x(\alpha t)$ is a compressed version of $x(t)$, whereas if $0 < \alpha < 1$, then $x(\alpha t)$ is an expanded version of $x(t)$.
13. Time-scaling of discrete-time signals may yield a signal that is completely different from the original signal.
14. The signal $x(t)$ is even symmetric if $x(t) = x(-t)$.

15. The signal $x(t)$ is odd symmetric if $x(t) = -x(-t)$.
16. In the classical (Dirac) definition, the unit impulse function $\delta(t)$ is characterized by unit area and is concentrated at a single instant $t = 0$.
17. Unit impulse, unit step, and unit ramp functions are related as

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

$$r(t) = \int_{-\infty}^t u(\tau) d\tau$$

18. The discrete-time unit step and impulse functions are related as

$$u(n) = \sum_{k=-\infty}^n \delta(k)$$

$$\delta(n) = u(n) - u(n-1)$$

19. A signal with finite energy is an energy signal. Similarly, a signal with a finite and nonzero power is a power signal. A signal can be either an energy signal or a power signal, but not both. However, there are signals that are neither energy nor power signals.
20. A signal whose complete physical description is known in a mathematical or graphical form is a deterministic signal. A random signal is known only in terms of its probabilistic description such as mean value or mean square value, rather than by its mathematical or graphical form.

21. A continuous-time system is a transformation that operates on a continuous-time input signal to produce a continuous-time output signal.
22. A system is linear if it follows the principle of superposition.
23. A system is time-invariant if a time shift in the input signal causes an identical time shift in the output.
24. The systems satisfying both linearity and time-invariant properties are known as linear time-invariant or LTI systems.
25. A system is memoryless if the present value of the output $y(t)$ depends only on the present value of the input $x(t)$.
26. A system is causal if the output $y(t_0)$ depends only on values of the input $x(t)$ for $t \leq t_0$.
27. A system is invertible if the distinct inputs lead to the distinct outputs.
28. A system is BIBO stable if bounded inputs result in bounded output.
29. Concepts such as linearity, memory, time invariance, and causality in discrete-time systems are similar to those in continuous-time systems.

MULTIPLE-CHOICE QUESTIONS

1. Let $u(n)$ be a unit step sequence. The sequence $u(N - n)$ can be described as
 - (a) $x(n) = \begin{cases} 1, & n < N \\ 0, & \text{otherwise} \end{cases}$
 - (b) $x(n) = \begin{cases} 1, & n \leq N \\ 0, & \text{otherwise} \end{cases}$
 - (c) $x(n) = \begin{cases} 1, & n > N \\ 0, & \text{otherwise} \end{cases}$
 - (d) $x(n) = \begin{cases} 1, & n \geq N \\ 0, & \text{otherwise} \end{cases}$
2. Given a unit step function $u(t)$. Its time derivative is
 - (a) a unit impulse
 - (b) another step function
 - (c) a unit ramp function
 - (d) a sine function
3. The function $\sin(\pi u)/\pi u$ is denoted by
 - (a) $\text{sinc}(\pi u)$
 - (b) $\text{sinc}(u)$
 - (c) signum
 - (d) none of these
4. A periodic signal $x(n)$ of period N_1 is added to another periodic signal of period N_2 . Then the period of the resulting signal is always
 - (a) $N_1 + N_2$
 - (b) $N_1 N_2$
 - (c) LCM of N_1 and N_2
 - (d) GCD of N_1 and N_2
5. The system having input $x(n)$ related to output $y(n)$ as $y(n) = \log_{10}|x(n)|$ is
 - (a) nonlinear, causal, stable
 - (b) linear, noncausal, stable
 - (c) nonlinear, causal, not stable
 - (d) linear, noncausal, not stable
6. To obtain $x(4 - 2n)$ from the given signal $x(n)$, the following precedence (or priority) rule is used for operations on the independent variable n :
 - (a) Time scaling $\rightarrow$ Time shifting $\rightarrow$ Reflection
 - (b) Reflection $\rightarrow$ Time scaling $\rightarrow$ Time shifting
 - (c) Time scaling $\rightarrow$ Reflection $\rightarrow$ Time shifting
 - (d) Time shifting $\rightarrow$ Time scaling $\rightarrow$ Reflection
7. The system described by $y(n) = nx(n)$ is
 - (a) linear, time-varying, and stable
 - (b) nonlinear, time-invariant, and unstable
 - (c) nonlinear, time-varying, and stable
 - (d) linear, time-varying, and unstable
8. The period of the signal $x(t) = 10 \sin(12\pi t) + 4 \cos(18\pi t)$ is
 - (a) $\pi/4$
 - (b) $1/6$
 - (c) $1/9$
 - (d) $1/3$
9. The area under the curve $\int_{-\infty}^{\infty} \delta(t) dt$ is
 - (a) ∞
 - (b) unity
 - (c) 0
 - (d) undefined
10. Given $x(n) = a^{|n|}$, $|a| < 1$ is
 - (a) an energy signal
 - (b) a power signal
 - (c) neither an energy nor a power signal
 - (d) an energy as well as a power signal
11. The system characterized by the equation $y(t) = ax(t) + b$ is

- (a) linear for any value of b
 (b) linear if $b > 0$
 (c) linear if $b < 0$
 (d) nonlinear
12. The discrete-time signal $x(n) = (-1)^n$ is periodic with fundamental period
 (a) 6
 (b) 4
 (c) 2
 (d) 0
13. A useful property of the unit impulse $\delta(t)$ is that
 (a) $\delta(at) = a\delta(t)$
 (b) $\delta(at) = \delta(t)$
 (c) $\delta(at) = \frac{1}{|a|}\delta(t)$
 (d) $\delta(at) = [\delta(t)]^a$
14. Any signal $x(t)$ can be represented as
 (a) $x_e(t) + x_o(t)$
 (b) $x_e(t) - x_o(t)$
 (c) $\frac{x_e(t)}{x_o(t)}$
 (d) $x_e(t) \times x_o(t)$
15. Periodic signals are
 (a) $x(t+T) = x(t)$
 (b) $x(t-T) = x(t)$
 (c) $x(n=mN) = x(n)$
 (d) all the above
16. Invertible systems are systems in which
 (a) output can be uniquely obtained from the knowledge of input
 (b) input signal can be uniquely determined by observing the output signal
 (c) both (a) and (b)
 (d) system output is always constant
17. Choose the false statement:
 (a) $t\delta(t) = 0$
 (b) $\cos(t)\delta(t-\pi) = -\delta(t-\pi)$
 (c) $\delta(t) = \int u(t)dt$
 (d) $t\delta'(t) = -\delta(t)$

PROBLEMS

1.1 Prove each of the following statements. In each case, let energy signal $x_1(t)$ have energy $E[x_1(t)]$, let energy signal $x_2(t)$ have energy $E[x_2(t)]$, and let T be a nonzero, finite, real-valued constant.

(i) Prove $E[Tx_1(t)] = T^2E[x_1(t)]$. That is, amplitude scaling a signal by constant T scales the signal energy by T^2 .

(ii) Prove $E[x_1(t)] = E[x_1(t-T)]$. That is, shifting a signal does not affect its energy.

(iii) If $(x_1(t) \neq 0) \Rightarrow (x_2(t) = 0)$ and $(x_2(t) \neq 0) \Rightarrow (x_1(t) = 0)$, then prove $E[x_1(t) + x_2(t)] = E[x_1(t)] + E[x_2(t)]$. That is, the energy of the sum of two nonoverlapping signals is the sum of the two individual energies.

(iv) Prove $E[x_1(Tt)] = 1/|T|E[x_1(t)]$. That is, time-scaling a signal by T reciprocally scales the signal energy by $1/|T|$.

1.2 Show that the power of a signal

$$x(t) = \sum_{k=-m}^n C_k e^{j\omega_k t} \text{ is } P_x = \sum_{k=-m}^n |C_k|^2$$

assuming all frequencies to be distinct, that is, $\omega_i \neq \omega_k$ for all $i \neq k$.

1.3 For the signal $x(t)$ illustrated in Fig. 1.35(a), sketch

- (i) $x(t-4)$
 (ii) $x(t/1.5)$
 (iii) $x(-t)$
 (iv) $x(2t-4)$
 (v) $x(2-t)$

1.4 In Fig. 1.42, express signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ in terms of signal $x(t)$ and its time-shifted, time-scaled, or time-reversed versions.

1.5 Simplify the following expressions:

- (i) $\left(\frac{\sin t}{t^2 + 2}\right)\delta(t)$
 (ii) $\left(\frac{\omega + 2}{\omega^2 + 9}\right)\delta(\omega)$
 (iii) $[e^{-t}\cos(3t - 60^\circ)]\delta(t)$
 (iv) $\left(\frac{1}{j\omega + 2}\right)\delta(\omega + 3)$
 (v) $\left(\frac{\sin k\omega}{\omega}\right)\delta(\omega)$

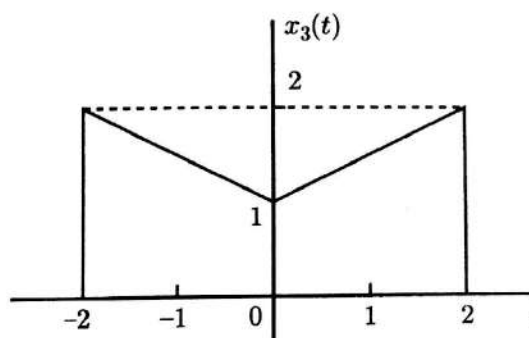
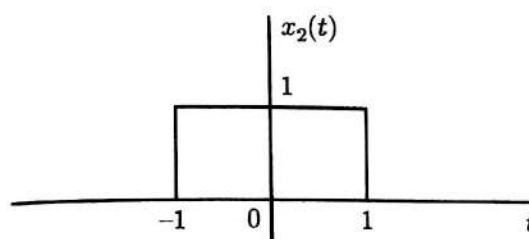
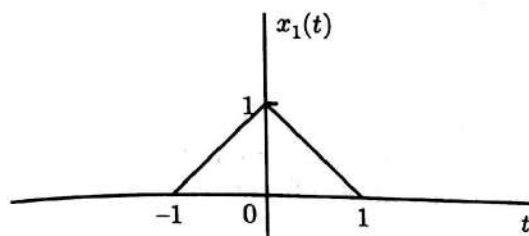
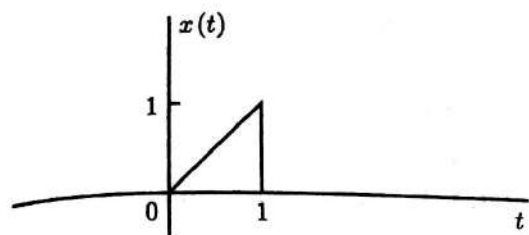


Fig. 1.42

1.6 Evaluate the following integrals:

(i) $\int_{-\infty}^{\infty} \left(\frac{2}{3}t - \frac{3}{2} \right) \delta(t-1) dt$

(ii) $\int_{-\infty}^{\infty} (t-1) \delta\left(\frac{2}{3}t - \frac{3}{2}\right) dt$

(iii) $\int_{-3}^{-2} \left[e^{-t+1} + \sin\left(\frac{2\pi}{3}t\right) \right] \delta\left(t - \frac{3}{2}\right) dt$

(iv) $\int_{-3}^{-2} \left[e^{-t+1} + \sin\left(\frac{2\pi}{3}t\right) \right] \delta\left(t - \frac{3}{2}\right) dt$

(v) $\int_{-\infty}^{\infty} e^{-5t+1} \delta'(t-5) dt$

1.7 (a) If $x_e(t)$ and $x_o(t)$ are even and odd components of a real signal $x(t)$, then show that

$$\int_{-\infty}^{\infty} x_e(t)x_o(t)dt = 0$$

(b) Show that

$$\int_{-\infty}^{\infty} x(t)dt = \int_{-\infty}^{\infty} x_e(t)dt$$

1.8 Find and sketch the even and odd components of the following:

- (i) $u(t)$
- (ii) $r(t) = tu(t)$
- (iii) $\sin(\omega_0 t)u(t)$
- (iv) $\cos(\omega_0 t)u(t)$

1.9 An aperiodic signal is defined as $x(t) = \sin(\pi t)u(t)$, where $u(t)$ is the continuous-time step function. Is the odd portion of this signal, $x_o(t)$, periodic? Justify your answer.

1.10 The output of the system $y(t)$ is related to the input $x(t)$ as

$$y(t) = \cos\left(2\pi f_c t + k \int_{-\infty}^t x(\tau) d\tau\right)$$

where k is a constant parameter.

- (a) Show that the system is nonlinear.
- (b) Show that the system is time-variant.

1.11 The systems that follow have input $x(n)$ and output $y(n)$. For each system, determine whether it is (i) memoryless, (ii) stable, (iii) causal, (iv) linear, or (v) time invariant.

- (a) $y(n) = \log_e[x(n)]$
- (b) $y(n) = \log_{10}[|x(n)|]$
- (c) $y(n) = \text{median}\{x(n-1), x(n), x(n+1)\}$
- (d) $y(n) = \text{Trun}[x(n)]$, where $\text{Trun}[x(n)]$ denotes the integer part of $x(n)$ obtained by truncation.
- (e) $y(n) = \text{Round}[x(n)]$, where $\text{Round}[x(n)]$ denotes the integer part of $x(n)$ obtained by rounding.
- (f) $y(n) = |x(n)|$
- (g) $y(n) = \text{sgn}[x(n)]$

1.12 Sketch the following signals:

(i) $x_1(t) = u(t) + 5u(t-1) - 2u(t-2)$

(ii) $x_2(t) = r(t) - r(t-1) - u(t-2)$

(iii) $x_3(t) = 2u(t) + \delta(t-1)$

(iv) $x_4(t) = u(t)u(t-2)$

(v) $x_1(t)x_2(t + \frac{1}{2})$

1.13 A discrete-time signal is shown in Fig. 1.43.

Sketch and label each of the following signals:

(i) $x(n-4)$

(ii) $x(3-n)$

(iii) $x(3n)$

(iv) $x(3n+1)$

(v) $x(n)u(3-n)$

(vi) $x(n-2)\delta(n-2)$

(vii) $\frac{1}{2}x(n) + \frac{1}{2}(-1)^n x(n)$

(viii) $x(n-1)^2$

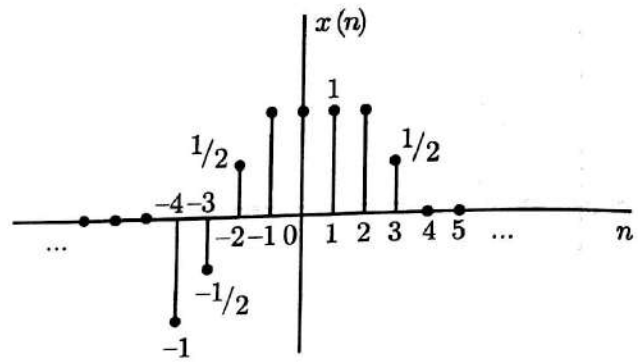


Fig. 1.43

1.14 Find the derivative of the waveforms in Fig. 1.40 and write the equations for the derivatives using shifted and/or impulse functions.

1.15 Show that if $x(n)$ is periodic with period N , then

$$\sum_{k=n_0}^n x(k) = \sum_{k=n_0+N}^{n+N} x(k)$$

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ANSWERS TO MULTIPLE-CHOICE QUESTIONS

- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|--------|
| 1. (b) | 2. (a) | 3. (b) | 4. (c) | 5. (c) | 6. (d) | 7. (d) | 8. (d) | 9. (b) |
| 10. (a) | 11. (d) | 12. (c) | 13. (c) | 14. (a) | 15. (d) | 16. (b) | 17. (c) | |

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